

Stationary Motion–Staticity Geometry: Schwarzschild and Kerr through Response Spectra, Weyl–Ernst Potentials, and Komar–Smarr Charges

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September 3, 2026

Abstract We develop the stationary and axisymmetric sector of motion–staticity (MS) geometry by applying the response-polar construction to the two-dimensional Killing plane generated by stationarity and axial symmetry. The starting point is the previously reconstructed local response norm

$$\mu_x(\xi) = \sqrt{-g_x(\xi, \xi)}$$

for future-timelike displacements. For commuting Killing fields k and m , we introduce the stationary response polynomial

$$\mathcal{R}_x(\Omega) := -g_x(k + \Omega m, k + \Omega m).$$

On every interval for which $k + \Omega m$ is future timelike, $\mathcal{R}_x(\Omega) = \mu_x(k + \Omega m)^2$. Since $\mathcal{R}_x$ is quadratic in Ω , its response spectrum reconstructs the complete Killing Gram matrix

$$G_{AB} = g(K_A, K_B), \quad K_A = (k, m),$$

and hence the local stationary redshift, frame-dragging, and axial geometry. In particular,

$$\rho^2 := -\det G = \frac{1}{4} \left[(\partial_\Omega \mathcal{R})^2 - 2\mathcal{R} \partial_\Omega^2 \mathcal{R} \right]$$

reconstructs the Weyl–Papapetrou radial invariant directly from massive response data.

This spectral formulation separates two geometrically different degeneracies. A static-limit surface is characterized by $\mathcal{R}(0) = 0$, whereas a regular non-axis Killing horizon is characterized by coalescence of the two null angular-velocity roots,

$$\mathcal{R}(\Omega_H) = \partial_\Omega \mathcal{R}(\Omega_H) = 0.$$

The unique maximum of $\mathcal{R}$ inside the timelike interval selects the locally nonrotating, zero-angular-momentum observer. Schwarzschild and Kerr provide the canonical examples: in Schwarzschild the static limit and horizon coincide, while in Kerr the response spectrum distinguishes the ergosurface, frame dragging, and the horizon angular velocity.

For stationary axisymmetric vacuum spacetimes, the zeroth-order response spectrum determines the real stationary metric data, including the Weyl radius and the Papapetrou dragging coefficient. The first derivative

hierarchy supplies the twist one-form of the stationary Killing field. Vacuum closure makes this form locally exact and therefore produces the complex Ernst potential

$$\mathcal{E} = f + i\psi, \quad f = -g(k, k).$$

The static reduction $\psi = 0$ yields the Weyl harmonic-potential description, with Schwarzschild represented by the standard finite rod, whereas Kerr gives the nontrivial rotating Ernst geometry.

At the global level, the same calibrated Killing fields define the standard Komar mass and angular momentum. In vacuum the corresponding Komar forms are closed, allowing the charge to be transported from spatial infinity to a regular Killing horizon. Evaluation on the horizon generator

$$\chi_H = k + \Omega_H m$$

then gives the Smarr relation. The Komar and Smarr identities themselves are classical general-relativistic results; the new structural content developed here is their organization within the MS response hierarchy

massive response spectrum $\longrightarrow$ Killing Gram geometry $\longrightarrow$ Weyl–Ernst structure $\longrightarrow$ horizon generator $\longrightarrow$ Komar–Smarr

No derivation of the vacuum Einstein equations, the Kerr uniqueness theorem, or black-hole thermodynamics from the compact MS state relation is claimed.

Keywords: motion–staticity geometry; stationary spacetime; Schwarzschild geometry; Kerr geometry; Weyl–Papapetrou coordinates; Ernst potential; Komar mass and angular momentum; Killing horizon; frame dragging; Smarr relation.

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1 Introduction

Stationary gravitational fields provide a particularly sharp setting in which to test the geometric content of the motion–staticity response formalism. They contain simultaneously local redshift, frame dragging, ergoregions, Killing horizons, global mass and angular momentum, and, in vacuum, the

integrable Weyl–Ernst structure. Schwarzschild and Kerr then supply two canonical geometries in which these structures can be displayed without approximation.

The purpose of the present paper is not to derive the Schwarzschild or Kerr solutions from the compact MS relation alone. Nor do we attempt to derive the classical Komar identities or the Smarr formula anew. Rather, we ask a more specific question:

Can the stationary Killing geometry of a spacetime be reconstructed and organized directly through the spectrum of massive MS responses, and can the familiar Weyl–Ernst and Komar–Smarr structures be placed into the same response hierarchy?

The answer developed below is affirmative within the stationary axisymmetric vacuum sector.

The local geometric input is the response-polar construction established in the preceding MS analysis. At each spacetime point x , a massive positive generator shell and the operational response

$$\Theta_x(P, \xi) = P(e_x \xi) \quad (1)$$

determine, for a future-timelike tangent vector ξ , the normalized polar functional

$$\mu_x(\xi) = \frac{1}{Q} \inf_{P \in \Sigma_{Q,x}^+} \Theta_x(P, \xi) = \sqrt{-g_x(\xi, \xi)} \quad (2)$$

The corresponding Lorentzian metric is recovered by

$$g_x = -\frac{1}{2} D^2(\mu_x^2). \quad (3)$$

These are recalled local results. The present paper introduces a new specialization of this geometry to stationary Killing families.

Let

$$k \quad (4)$$

be a stationary Killing field and, in the axisymmetric case, let

$$m \quad (5)$$

be an axial Killing field with closed spacelike orbits away from the rotation axis. We assume

$$[k, m] = 0. \quad (6)$$

The two fields span the local Killing plane

$$\mathcal{K}_x := \text{span}\{k_x, m_x\}. \quad (7)$$

Every constant-angular-velocity Killing combination is

$$\chi_\Omega := k + \Omega m \quad (8)$$

Whenever χ_Ω is future timelike, the MS response polar assigns to it the scalar

$$\mu_x(\chi_\Omega) = \sqrt{-g_x(\chi_\Omega, \chi_\Omega)}. \quad (9)$$

This motivates the central object of the paper.

Definition 1.1 (Stationary response polynomial). *For a stationary axisymmetric Killing pair (k, m) , define*

$$\boxed{\mathcal{R}_x(\Omega) := -g_x(k + \Omega m, k + \Omega m)} \quad (10)$$

On every interval of angular velocities for which χ_Ω is future timelike,

$$\boxed{\mathcal{R}_x(\Omega) = \mu_x(\chi_\Omega)^2} \quad (11)$$

The distinction in this definition is useful. The operational polar μ_x is defined on the future-timelike cone. The quadratic expression $\mathcal{R}_x(\Omega)$, once reconstructed from that timelike response interval, has a unique algebraic continuation to null and spacelike Killing combinations. We shall use $\mathcal{R}_x$ for this complete quadratic polynomial and reserve μ_x^2 for its directly massive-response-accessible timelike part.

Choose coordinates adapted to the Killing fields,

$$k = \partial_t, \quad m = \partial_\phi. \quad (12)$$

The Killing Gram matrix is

$$G := \begin{pmatrix} g_{tt} & g_{t\phi} \\ g_{t\phi} & g_{\phi\phi} \end{pmatrix}. \quad (13)$$

Then

$$\boxed{\mathcal{R}_x(\Omega) = -g_{tt} - 2\Omega g_{t\phi} - \Omega^2 g_{\phi\phi}} \quad (14)$$

This elementary quadratic identity is the key stationary spectral observation. The complete Killing Gram geometry can be reconstructed from the response spectrum. At any point for which an open interval of timelike χ_Ω exists,

$$g_{tt} = -\mathcal{R}(0) \quad (15)$$

$$g_{t\phi} = -\frac{1}{2} \left. \frac{d\mathcal{R}}{d\Omega} \right|_{\Omega=0} \quad (16)$$

$$g_{\phi\phi} = -\frac{1}{2} \frac{d^2\mathcal{R}}{d\Omega^2} \quad (17)$$

More generally, no distinguished value $\Omega = 0$ is required: three distinct timelike angular velocities suffice to determine the quadratic polynomial and hence the three independent entries of G .

Thus a finite set of counter-moving massive response measurements reconstructs the stationary Killing two-metric. This is the first new structural result developed in the present paper.

A particularly useful invariant is the determinant of the Killing Gram matrix. Define

$$\boxed{\rho^2 := -\det G = g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \quad (18)$$

In the stationary axisymmetric vacuum problem, ρ becomes the canonical Weyl–Papapetrou radial coordinate away from degeneracies. Using Eq. (14), one obtains the response formula

$$\boxed{\rho^2 = \frac{1}{4} \left[\left(\frac{d\mathcal{R}}{d\Omega} \right)^2 - 2\mathcal{R} \frac{d^2\mathcal{R}}{d\Omega^2} \right]} \quad (19)$$

The right-hand side is independent of Ω , as follows directly from the fact that $\mathcal{R}$ is quadratic.

Equation (19) gives a simple interpretation of the Weyl radial invariant: it is one quarter of the discriminant of the stationary massive-response polynomial.

Equivalently, the null Killing combinations satisfy

$$\mathcal{R}_x(\Omega) = 0. \quad (20)$$

Away from a degeneracy, the two roots are

$$\Omega_{\pm} = \frac{-g_{t\phi} \pm \sqrt{g_{t\phi}^2 - g_{tt}g_{\phi\phi}}}{g_{\phi\phi}} \quad (21)$$

If $g_{\phi\phi} > 0$, the future-timelike stationary observers lie between the two null roots.

This immediately gives two different spectral signatures that are central to the Schwarzschild–Kerr comparison.

The stationary Killing field itself becomes null when

$$\mathcal{R}(0) = 0 \quad (22)$$

This is the stationary-limit or ergosurface condition.

By contrast,

$$\rho^2 = 0 \quad (23)$$

means that the two roots of $\mathcal{R}(\Omega) = 0$ coalesce. On a regular non-axis component of this degeneracy, the coalesced null Killing combination has angular velocity

$$\Omega_H = -\frac{g_{t\phi}}{g_{\phi\phi}} \quad (24)$$

and satisfies

$$\mathcal{R}(\Omega_H) = 0, \quad \partial_{\Omega}\mathcal{R}(\Omega_H) = 0 \quad (25)$$

A double root of the Killing response polynomial is therefore the local algebraic signature of a degenerate Killing plane. Additional geometric and regularity conditions are required to identify such a component as an actual Killing horizon: the rotation axis also satisfies $\det G = 0$. We shall keep this distinction explicit throughout.

For a regular axisymmetric region with $g_{\phi\phi} > 0$, the response polynomial is concave,

$$\frac{d^2\mathcal{R}}{d\Omega^2} = -2g_{\phi\phi} < 0. \quad (26)$$

It therefore has a unique maximum at

$$\Omega_Z = -\frac{g_{t\phi}}{g_{\phi\phi}} \quad (27)$$

The corresponding Killing direction obeys

$$g(k + \Omega_Z m, m) = 0. \quad (28)$$

After normalization this is the locally nonrotating, zero-angular-momentum observer direction.

Its maximum response is

$$\boxed{\mathcal{R}_{\max} = \frac{\rho^2}{g_{\phi\phi}}} \quad (29)$$

Thus frame dragging is itself encoded spectrally: the displacement of the response maximum away from $\Omega = 0$ measures the dragging angular velocity.

The Schwarzschild geometry provides the static reference example. In geometrized units,

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (30)$$

Since

$$g_{t\phi} = 0, \quad (31)$$

the stationary response polynomial is

$$\mathcal{R}_{\text{Schw}}(\Omega) = 1 - \frac{2M}{r} - \Omega^2 r^2 \sin^2\theta. \quad (32)$$

The maximum occurs at $\Omega_Z = 0$. The stationary Killing response

$$\mu(k)^2 = 1 - \frac{2M}{r} \quad (33)$$

vanishes at $r = 2M$. In Schwarzschild the static-limit surface and the Killing horizon coincide.

Kerr separates these phenomena. In Boyer–Lindquist coordinates,

$$g_{t\phi} \neq 0, \quad (34)$$

so the response polynomial is displaced from $\Omega = 0$. Its stationary value $\mathcal{R}(0) = -g_{tt}$ vanishes on the ergosurface, while

$$-\det G = \Delta \sin^2\theta \quad (35)$$

vanishes both on the axis and on

$$\Delta = 0. \quad (36)$$

On the non-axis component $\Delta = 0$, the two null angular-velocity roots merge into the Kerr horizon generator. Thus the response polynomial distinguishes

ergosurface : $\mathcal{R}(0) = 0$, horizon : $\mathcal{R}(\Omega_H) = \mathcal{R}'(\Omega_H) = 0$, frame dragging : $\Omega_Z = \arg \max_{\Omega} \mathcal{R}(\Omega)$.	(37)
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This separation will be developed explicitly in the Kerr analysis.

The same Killing Gram data lead naturally to the Weyl–Papapetrou description. In a stationary axisymmetric vacuum region, the metric may be written locally in the form

$$ds^2 = -f(dt - \omega d\phi)^2 + f^{-1} \left[e^{2\gamma} (d\rho^2 + dz^2) + \rho^2 d\phi^2 \right], \quad (38)$$

subject to the usual regularity assumptions.

The real stationary potential is

$$f = -g(k, k) = \mathcal{R}(0) \quad (39)$$

where k is timelike, while

$$\omega = \frac{g_{t\phi}}{f} \quad (40)$$

is reconstructed algebraically from the first angular derivative of the response polynomial. The Weyl coordinate ρ is reconstructed by Eq. (19).

Thus the entire algebraic Killing-plane part of the Weyl–Papapetrou geometry is already contained in the zeroth-order stationary response spectrum.

The remaining rotating information is differential. Let

$$\varpi_\mu := \epsilon_{\mu\nu\rho\sigma} k^\nu \nabla^\rho k^\sigma \quad (41)$$

be the twist one-form of the stationary Killing field. In vacuum, the standard Killing and Ricci identities imply

$$d\varpi = 0. \quad (42)$$

Locally there is therefore a twist potential

$$d\psi = \varpi \quad (43)$$

The complex Ernst potential is

$$\mathcal{E} := f + i\psi \quad (44)$$

In canonical Weyl coordinates, the vacuum field equations reduce to the Ernst equation

$$(\Re \mathcal{E}) \left(\mathcal{E}_{,\rho\rho} + \frac{1}{\rho} \mathcal{E}_{,\rho} + \mathcal{E}_{,zz} \right) = \mathcal{E}_{,\rho}^2 + \mathcal{E}_{,z}^2 \quad (45)$$

The logical status of these quantities should be stated carefully. The algebraic dragging coefficient ω is determined pointwise by the Killing Gram matrix and hence by the stationary response polynomial. The Ernst twist potential ψ , by contrast, is not a pointwise algebraic function of the response spectrum. It requires first-derivative information and the vacuum integrability condition $d\varpi = 0$.

This gives a natural response hierarchy:

zeroth-order massive response $\longrightarrow G_{AB} \longrightarrow (f, \omega, \rho)$, first derivative of stationary geometry $\longrightarrow \varpi$, $R_{\mu\nu} = 0 \longrightarrow d\varpi = 0 \longrightarrow \psi$, $(f, \psi) \longrightarrow \mathcal{E}$.	(46)
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In the static sector,

$$\varpi = 0, \quad \psi = \text{const.}, \quad (47)$$

and one may set

$$\psi = 0. \quad (48)$$

Writing

$$f = e^{2U}, \quad (49)$$

the vacuum Ernst equation reduces to the axisymmetric flat-space Laplace equation

$$\boxed{U_{,\rho\rho} + \frac{1}{\rho}U_{,\rho} + U_{,zz} = 0} \quad (50)$$

This is the classical Weyl description of static axisymmetric vacuum spacetimes [?].

In the present interpretation,

$$\boxed{U = \frac{1}{2} \ln \mathcal{R}(0)} \quad (51)$$

where the stationary Killing field is timelike. The Weyl harmonic potential is therefore the logarithm of the stationary MS response gap. For Schwarzschild the corresponding Weyl source is the familiar finite rod. This gives a particularly direct bridge between the local response spectrum and the classical Weyl representation.

Kerr occupies the rotating branch:

$$\psi \neq 0. \quad (52)$$

Its Ernst potential combines the stationary redshift potential and the twist produced by frame dragging into a single complex object. The relation between the Kerr response polynomial, its Weyl discriminant, the twist potential, and the horizon double root will be made explicit below. The Kerr metric itself is not derived from MS response alone; it is used as the exact stationary vacuum solution on which the response hierarchy can be evaluated and tested.

The second global structure studied in this paper is supplied by the Komar forms. For a stationary vacuum spacetime, define in geometrized units

$$M_K[S] := -\frac{1}{8\pi} \int_S \star dk^b, \quad (53)$$

$$J_K[S] := \frac{1}{16\pi} \int_S \star dm^b. \quad (54)$$

These are standard general-relativistic quantities [?]. In vacuum the Killing identity gives

$$d \star dk^b = 0, \quad d \star dm^b = 0, \quad (55)$$

so their integrals are invariant under homologous deformation of the integration surface.

The significance for the present paper is structural. The local response spectrum reconstructs the normalized stationary and axial Killing geometry. The Komar forms then turn those calibrated Killing fields into global conserved surface charges. This is a different mechanism from the canonical spatial-infinity boundary completion studied in the preceding paper, although the two give the same mass and angular momentum in the common stationary asymptotically flat Einstein sector when their standard hypotheses are satisfied.

For a regular rotating Killing horizon, the horizon generator is

$$\boxed{\chi_H = k + \Omega_H m} \quad (56)$$

The response spectrum identifies Ω_H through the double-root condition Eq. (25). The first transverse derivative of the norm of χ_H then carries the surface-gravity scale. In standard Killing-horizon geometry,

$$\nabla_\mu (\chi_H^\nu \chi_\nu^H) = -2\kappa_H \chi_\mu^H \quad (57)$$

on the horizon, with the normalization of χ_H fixed at infinity.

Transporting the Komar identity from infinity to the horizon yields

$$M_K - 2\Omega_H J_K = \frac{\kappa_H A_H}{4\pi} \quad (58)$$

in geometrized units, and therefore

$$M_K = \frac{\kappa_H A_H}{4\pi} + 2\Omega_H J_K \quad (59)$$

For asymptotically flat stationary vacuum black holes this is the classical Smarr relation [?].

Equation (59) is not claimed as a new GR identity. What is new in the present organization is that the quantities entering it occur at successive levels of the stationary MS response hierarchy:

$$\begin{aligned} \mathcal{R}_x(\Omega) &\longrightarrow \text{Killing Gram matrix,} \\ \mathcal{R}_x(0) &\longrightarrow \text{stationary redshift potential,} \\ \arg \max \mathcal{R}_x &\longrightarrow \text{frame-dragging observer,} \\ \text{disc } \mathcal{R}_x &\longrightarrow \rho^2 = -\det G, \\ \text{double root of } \mathcal{R}_x &\longrightarrow \Omega_H, \\ \nabla(-\chi_H^2) &\longrightarrow \kappa_H, \\ (k, m) &\longrightarrow (M_K, J_K), \\ \text{vacuum horizon transport} &\longrightarrow \text{Smarr balance.} \end{aligned} \quad (60)$$

There is a complementary Ernst hierarchy:

$$\mathcal{R}(\Omega) \longrightarrow G_{AB} \longrightarrow (f, \rho, \omega) \longrightarrow \varpi \longrightarrow \psi \longrightarrow \mathcal{E} \longrightarrow \text{Weyl-Ernst vacuum structure} \quad (61)$$

The two chains meet at the horizon. The Ernst variables encode the stationary vacuum geometry between infinity, axis, and horizon, whereas the Komar identities convert the normalized Killing fields into global surface charges. The horizon generator selected by the response double root is the same linear combination that enters the Komar horizon balance. This is the central unifying structure investigated in the paper.

Several limitations should be stated at the outset.

First, the vacuum Einstein equations are not derived here from the stationary response spectrum. The response reconstructs the stationary Killing geometry once the soldered Lorentzian spacetime is given. Vacuum Einstein dynamics is an additional input when the Weyl and Ernst equations are invoked.

Second, the closure of the twist one-form,

$$d\varpi = 0, \quad (62)$$

is a vacuum general-relativistic consequence, not a consequence of the compact MS relation alone. The imaginary Ernst potential therefore belongs to a derivative-plus-dynamics level of the construction rather than to the zeroth-order local response spectrum.

Third, a local degeneracy

$$\det G = 0 \quad (63)$$

is not by itself sufficient to establish the existence of a global event horizon. The axis also has $\det G = 0$, and global causal and regularity conditions remain necessary. The response double-root criterion identifies the local non-axis Killing degeneracy associated with a horizon generator in the regular stationary black-hole sector.

Fourth, we do not prove the Kerr uniqueness theorem. Schwarzschild and Kerr are exact reference solutions used to expose the response structure. The classification of all regular stationary black holes requires the usual additional global hypotheses and uniqueness machinery.

Fifth, the Komar and Smarr formulas are standard GR results. The purpose of including them is to demonstrate that the stationary response variables selected locally by MS geometry connect naturally to the calibrated Killing fields whose global surface integrals control black-hole mechanics.

Finally, no derivation of black-hole entropy or of the quantum Bekenstein–Hawking relation is attempted. The appearance of surface gravity and horizon area in the Smarr balance is entirely classical in the present work.

Subject to these qualifications, the paper develops a stationary extension of the MS programme in which local massive response measurements acquire a direct relation to the classical invariant structures of black-hole geometry.

The organization is as follows.

Section ?? develops the response polynomial $\mathcal{R}_x(\Omega)$, proves reconstruction of the Killing Gram matrix, derives the Weyl discriminant, and identifies the locally nonrotating observer as the unique response maximum.

Section 2 specializes to static axisymmetric vacuum geometry. The response gap $-g(k, k)$ is converted into the Weyl harmonic potential, and Schwarzschild is analyzed in both standard and Weyl coordinates.

Section 3 treats Kerr. The response polynomial is evaluated explicitly, and the ergosurface, dragging angular velocity, horizon roots, and the coalescence at Ω_H are separated spectrally.

Section 4 introduces the twist hierarchy and the Ernst potential. We distinguish the pointwise Papapetrou dragging coefficient from the derivative-defined twist potential and show how the Schwarzschild and Kerr sectors occupy respectively the real and genuinely complex Ernst branches.

Section 5 studies the Komar mass and angular momentum associated with the response-calibrated stationary and axial Killing fields and relates them to the corresponding asymptotic charges.

Section 6 transports the Komar identities to a regular Killing horizon. The horizon angular velocity is identified with the double root of the response polynomial, the surface-gravity scale is read from the first transverse norm derivative, and the resulting horizon identity is organized into the Smarr relation.

The final section discusses what this stationary construction does and does not add to the classical theory, and outlines extensions to electrovacuum Ernst systems, multipole moments, horizon spectroscopy, and more general stationary response geometries.

2 Static response geometry, Weyl potentials, and the Schwarzschild rod

The stationary response spectrum becomes especially transparent in the static sector. The off-diagonal Killing-plane coefficient vanishes in a static Killing representation, the response polynomial becomes even in the angular parameter, and the stationary response gap itself supplies the real Weyl potential.

The purpose of this section is to make this relation precise and then to apply it to Schwarzschild. The resulting description connects three structures that are usually introduced separately:

$$\boxed{\text{static massive response} \longrightarrow \text{Weyl harmonic potential} \longrightarrow \text{Schwarzschild finite rod}} \quad (64)$$

The first arrow is the response interpretation developed here. The second is the classical Weyl representation of static axisymmetric vacuum spacetimes.

Throughout this section we use geometrized units

$$G = c = 1. \quad (65)$$

Accordingly, the Schwarzschild parameter M has dimensions of length. Its identification with the asymptotic mass charge will be discussed in the global Komar analysis below.

2.1 Static reduction of the stationary response spectrum

Let k be a hypersurface-orthogonal timelike Killing field,

$$k^b \wedge dk^b = 0, \quad (66)$$

and let m be the commuting axial Killing field,

$$[k, m] = 0. \quad (67)$$

In the standard static axisymmetric Killing representation one may choose coordinates

$$k = \partial_t, \quad m = \partial_\phi, \quad (68)$$

such that

$$g(k, m) = 0. \quad (69)$$

The stationary response polynomial of Sec. ?? then reduces to

$$\boxed{\mathcal{R}(\Omega) = -g(k, k) - g(m, m)\Omega^2} \quad (70)$$

It is even:

$$\boxed{\mathcal{R}(\Omega) = \mathcal{R}(-\Omega)} \quad (71)$$

Its unique maximum occurs at

$$\boxed{\Omega_Z = 0} \quad (72)$$

Thus the locally nonrotating direction coincides with the stationary Killing direction itself.

Define

$$\boxed{f := -g(k, k) = \mathcal{R}(0)} \quad (73)$$

Where k is future timelike,

$$f > 0, \quad (74)$$

and the response-polar construction gives

$$\mu(k) = \sqrt{f}. \quad (75)$$

It is therefore natural to define the static response potential

$$U := \ln \mu(k) = \frac{1}{2} \ln f = \frac{1}{2} \ln \mathcal{R}(0) \quad (76)$$

Equivalently,

$$f = e^{2U} \quad (77)$$

This relation is purely kinematical. At this stage no vacuum equation has been imposed.

2.2 Vacuum closure and the Weyl equation

We now add the standard static axisymmetric vacuum Einstein condition

$$R_{\mu\nu} = 0. \quad (78)$$

Under the usual regularity assumptions, the metric may locally be written in Weyl canonical form

$$ds^2 = -e^{2U} dt^2 + e^{-2U} \left[e^{2\gamma} (d\rho^2 + dz^2) + \rho^2 d\phi^2 \right] \quad (79)$$

Here

$$k = \partial_t, \quad m = \partial_\phi, \quad (80)$$

and

$$g(k, k) = -e^{2U}, \quad (81)$$

$$g(m, m) = \rho^2 e^{-2U}. \quad (82)$$

The determinant of the Killing Gram matrix is therefore

$$\det G = -\rho^2, \quad (83)$$

so the Weyl coordinate ρ is precisely the response discriminant introduced in the preceding section:

$$\rho^2 = -\det G = \frac{1}{4} \left[(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}'' \right] \quad (84)$$

The stationary response polynomial assumes the canonical static form

$$\mathcal{R}(\Omega) = e^{2U} - \rho^2 e^{-2U} \Omega^2 \quad (85)$$

Thus the entire Killing-plane geometry is encoded by the pair (U, ρ) .

The two stationary-null angular velocities are

$$\Omega_{\pm} = \pm \frac{e^{2U}}{\rho} \quad (86)$$

Consequently, at any regular point with $\rho > 0$, the timelike stationary-observer interval is

$$|\Omega| < \frac{e^{2U}}{\rho} \quad (87)$$

The vacuum Einstein equations reduce to

$$U_{,\rho\rho} + \frac{1}{\rho} U_{,\rho} + U_{,zz} = 0 \quad (88)$$

This is the axisymmetric Laplace equation in an auxiliary Euclidean three-space.

The remaining metric function is determined by quadratures:

$$\gamma_{,\rho} = \rho \left[(U_{,\rho})^2 - (U_{,z})^2 \right] \quad (89)$$

$$\gamma_{,z} = 2\rho U_{,\rho} U_{,z} \quad (90)$$

The integrability condition for these equations follows from Eq. (88).

Combining the Weyl reduction with Eq. (76) gives the central static response statement.

Theorem 2.1 (Static response–Weyl correspondence). *Let a region of a static axisymmetric spacetime be vacuum and let the stationary Killing field k be future timelike there. Then the normalized massive MS response amplitude*

$$\mu(k) = \sqrt{-g(k, k)} \quad (91)$$

determines the Weyl potential through

$$U = \ln \mu(k) = \frac{1}{2} \ln \mathcal{R}(0) \quad (92)$$

In canonical Weyl coordinates, vacuum Einstein dynamics implies

$$\left(\partial_\rho^2 + \frac{1}{\rho} \partial_\rho + \partial_z^2 \right) \ln \mu(k) = 0 \quad (93)$$

The theorem separates the response input from the dynamical input. The identity

$$U = \ln \mu(k)$$

follows from the stationary response geometry. The harmonic equation for U follows from the vacuum Einstein equations. The compact MS state relation alone does not imply Eq. (93).

Equation (93) nevertheless gives a useful interpretation of static vacuum geometry:

$$\text{the logarithm of the stationary massive-response gap is harmonic in Weyl auxiliary space} \quad (94)$$

2.3 Schwarzschild response spectrum in spherical coordinates

The canonical static example is Schwarzschild. In geometrized units,

$$ds^2 = - \left(1 - \frac{2M}{r} \right) dt^2 + \left(1 - \frac{2M}{r} \right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (95)$$

The Killing fields are

$$k = \partial_t, \quad m = \partial_\phi. \quad (96)$$

Their Gram coefficients are

$$g(k, k) = -\left(1 - \frac{2M}{r}\right), \quad (97)$$

$$g(k, m) = 0, \quad (98)$$

$$g(m, m) = r^2 \sin^2 \theta. \quad (99)$$

Hence the Schwarzschild stationary response polynomial is

$$\mathcal{R}_{\text{Schw}}(r, \theta; \Omega) = 1 - \frac{2M}{r} - r^2 \sin^2 \theta \Omega^2 \quad (100)$$

Several familiar Schwarzschild properties are immediate response statements.

First,

$$\mathcal{R}_{\text{Schw}}(0) = 1 - \frac{2M}{r} \quad (101)$$

Thus the stationary MS response amplitude is

$$\mu_{\text{Schw}}(k) = \sqrt{1 - \frac{2M}{r}} \quad (102)$$

It vanishes at

$$r = 2M. \quad (103)$$

Second, the response maximum occurs at

$$\Omega_Z = 0, \quad (104)$$

as expected for a nonrotating spacetime.

Third, the two stationary-null angular velocities are

$$\Omega_{\pm} = \pm \frac{\sqrt{1 - 2M/r}}{r \sin \theta} \quad (105)$$

Thus the allowed stationary timelike interval shrinks continuously as $r \rightarrow 2M^+$:

$$\Omega_+ - \Omega_- \rightarrow 0 \quad (106)$$

At the Schwarzschild horizon the common root is

$$\Omega_H = 0 \quad (107)$$

Unlike Kerr, Schwarzschild has no distinct ergoregion. The stationary Killing field becomes null precisely where the Killing-plane causal interval collapses:

$$\text{Schwarzschild:} \quad \mathcal{R}(0) = 0 \iff \rho^2 = -\det G = 0 \quad (108)$$

on the non-axis horizon component.

This coincidence will be broken by rotation in Sec. 3.

2.4 Schwarzschild in Weyl canonical coordinates

Introduce Weyl coordinates

$$\rho = \sqrt{r(r-2M)} \sin \theta \quad (109)$$

$$z = (r-M) \cos \theta \quad (110)$$

These coordinates directly reproduce the response discriminant:

$$\begin{aligned} -\det G &= \left(1 - \frac{2M}{r}\right) r^2 \sin^2 \theta \\ &= r(r-2M) \sin^2 \theta \\ &= \boxed{\rho^2}. \end{aligned} \quad (111)$$

Define the Euclidean distances from the two points $z = \mp M$ on the Weyl axis:

$$R_{\pm} := \sqrt{\rho^2 + (z \pm M)^2} \quad (112)$$

Using Eqs. (109) and (110), one finds in the exterior region

$$R_+ = r - M + M \cos \theta, \quad (113)$$

$$R_- = r - M - M \cos \theta. \quad (114)$$

Therefore

$$r = M + \frac{R_+ + R_-}{2} \quad (115)$$

$$\cos \theta = \frac{R_+ - R_-}{2M} \quad (116)$$

The Schwarzschild response gap becomes

$$\begin{aligned} f_{\text{Schw}} &= 1 - \frac{2M}{r} \\ &= \boxed{\frac{R_+ + R_- - 2M}{R_+ + R_- + 2M}} \end{aligned} \quad (117)$$

Hence the response/Weyl potential is

$$U_{\text{Schw}} = \ln \mu_{\text{Schw}}(k) = \frac{1}{2} \ln \left(\frac{R_+ + R_- - 2M}{R_+ + R_- + 2M} \right) \quad (118)$$

The second Weyl metric function is

$$e^{2\gamma_{\text{Schw}}} = \frac{(R_+ + R_-)^2 - 4M^2}{4R_+R_-} \quad (119)$$

Substitution of Eqs. (118) and (119) into the Weyl metric (79) reproduces the Schwarzschild exterior geometry.

2.5 The finite-rod representation

The potential Eq. (118) has the classical auxiliary Newtonian representation

$$U_{\text{Schw}}(\rho, z) = -\frac{1}{2} \int_{-M}^M \frac{d\zeta}{\sqrt{\rho^2 + (z - \zeta)^2}} \quad (120)$$

Thus, in the auxiliary Euclidean space of the Weyl reduction, U_{Schw} is the potential of a finite rod occupying

$$\rho = 0, \quad -M \leq z \leq M \quad (121)$$

with the standard Weyl linear density $1/2$.

This “rod” is not physical matter. Schwarzschild is vacuum outside and on the regular horizon in the appropriate spacetime extension. The rod is an auxiliary source representation of the harmonic Weyl potential after the nonlinear gravitational problem has been reduced to the flat axisymmetric Laplace equation.

In the present response interpretation, Eq. (120) says

$$\ln \mu_{\text{Schw}}(k) = -\frac{1}{2} \int_{-M}^M \frac{d\zeta}{\sqrt{\rho^2 + (z - \zeta)^2}} \quad (122)$$

The logarithm of the stationary massive-response amplitude therefore has the harmonic potential of the Schwarzschild Weyl rod.

This gives the static chain

$$\mu_{\text{Schw}}(k) \longrightarrow \ln \mu_{\text{Schw}}(k) = U_{\text{Schw}} \longrightarrow \text{finite Weyl rod of length } 2M \quad (123)$$

The length of the rod is

$$L_{\text{rod}} = 2M. \quad (124)$$

The fact that the same parameter M later equals the global Komar and ADM mass in geometrized units is a standard property of the Schwarzschild solution. That global charge interpretation is deliberately deferred to Sec. 5.

2.6 Axis and horizon as distinct Weyl degeneracies

The Weyl coordinate ρ vanishes both on the rotation axis and on the Schwarzschild horizon rod. Since

$$\rho^2 = -\det G, \quad (125)$$

both are degeneracies of the Killing Gram determinant. The response spectrum distinguishes them.

Consider first an exterior axis segment,

$$\rho = 0, \quad |z| > M. \quad (126)$$

There,

$$f_{\text{Schw}} > 0, \quad (127)$$

while

$$g(m, m) = \rho^2 f^{-1} \longrightarrow 0. \quad (128)$$

Hence

$$\boxed{\mathcal{R}(0) > 0, \quad \mathcal{R}'' \longrightarrow 0} \quad (129)$$

The degeneracy occurs because the axial Killing field itself vanishes.

Now consider the interior of the finite rod,

$$\rho = 0, \quad |z| < M. \quad (130)$$

This is the Schwarzschild horizon in Weyl coordinates. Here

$$f_{\text{Schw}} \longrightarrow 0, \quad (131)$$

but the axial orbit size remains finite away from the rod endpoints.

The distinction can be seen explicitly by expanding at small ρ with fixed $|z| < M$. One finds

$$R_+ + R_- = 2M + \frac{M\rho^2}{M^2 - z^2} + O(\rho^4). \quad (132)$$

Therefore

$$\boxed{f_{\text{Schw}} = \frac{\rho^2}{4(M^2 - z^2)} + O(\rho^4)} \quad (133)$$

The stationary response amplitude behaves as

$$\boxed{\mu_{\text{Schw}}(k) = \frac{\rho}{2\sqrt{M^2 - z^2}} + O(\rho^3)} \quad (134)$$

Meanwhile,

$$g_{\phi\phi} = \frac{\rho^2}{f_{\text{Schw}}} \quad (135)$$

has the finite limit

$$\boxed{g_{\phi\phi} \longrightarrow 4(M^2 - z^2)} \quad (|z| < M). \quad (136)$$

Indeed, using

$$z = M \cos \theta \quad (137)$$

on $r = 2M$, this becomes

$$g_{\phi\phi} = 4M^2 \sin^2 \theta, \quad (138)$$

as required by the spherical form of the Schwarzschild horizon.

The response polynomial near the rod is consequently

$$\mathcal{R}_{\text{Schw}}(\Omega) = \frac{\rho^2}{4(M^2 - z^2)} - 4(M^2 - z^2)\Omega^2 + O(\rho^2\Omega^2) + O(\rho^4). \quad (139)$$

At the rod itself,

$$\boxed{\mathcal{R}_{\text{Schw}}(\Omega) = -4(M^2 - z^2)\Omega^2, \quad |z| < M} \quad (140)$$

Thus

$$\boxed{\mathcal{R}_{\text{Schw}}(0) = 0, \quad \mathcal{R}'_{\text{Schw}}(0) = 0} \quad (141)$$

and

$$\mathcal{R}''_{\text{Schw}}(0) = -8(M^2 - z^2) < 0. \quad (142)$$

The Schwarzschild horizon therefore appears as a genuine double root of the stationary response polynomial with

$$\Omega_H = 0. \quad (143)$$

The axis and horizon rod are consequently distinguished by

	$\mathcal{R}(0)$	$g(m, m)$
regular exterior axis	> 0	0
horizon rod interior	0	> 0

(144)

even though both satisfy

$$\rho^2 = -\det G = 0. \quad (145)$$

This explicitly demonstrates why the response discriminant alone cannot distinguish the rotation axis from a horizon. The full polynomial does.

2.7 Collapse of the stationary causal interval

The static Weyl response roots are

$$\Omega_{\pm} = \pm \frac{f}{\rho}. \quad (146)$$

For Schwarzschild,

$$\Omega_{\pm} = \pm \frac{f_{\text{Schw}}}{\rho} \quad (147)$$

Near the interior of the horizon rod, Eq. (133) gives

$$\Omega_{\pm} = \pm \frac{\rho}{4(M^2 - z^2)} + O(\rho^3) \quad (148)$$

Thus

$$\Omega_{\pm} \longrightarrow 0 \quad (149)$$

and the complete stationary timelike interval collapses to the horizon generator k .

By contrast, approaching a regular exterior axis segment with $f > 0$,

$$\frac{f}{\rho} \longrightarrow \infty. \quad (150)$$

The apparent root separation diverges because the axial generator m itself shrinks to zero. This is another way to distinguish the two components of the Weyl boundary.

We therefore obtain the static spectral picture

axis:	$g(m, m) \rightarrow 0, \quad \Omega_{\pm} \rightarrow \infty,$
horizon rod:	$f \rightarrow 0, \quad \Omega_{\pm} \rightarrow 0.$

(151)

2.8 Asymptotic normalization and the rod parameter

Let

$$R := \sqrt{\rho^2 + z^2}. \quad (152)$$

At large Weyl radius,

$$U_{\text{Schw}} = -\frac{M}{R} + O(R^{-3}). \quad (153)$$

Consequently,

$$f_{\text{Schw}} = e^{2U_{\text{Schw}}} = 1 - \frac{2M}{R} + O(R^{-2}). \quad (154)$$

The normalization

$$U \longrightarrow 0 \quad (R \rightarrow \infty) \quad (155)$$

is equivalent to fixing

$$g(k, k) \longrightarrow -1, \quad (156)$$

and therefore fixes the normalization of the stationary Killing generator.

In response language,

$$\boxed{\mu(k) \longrightarrow 1} \quad (157)$$

at spatial infinity.

The leading deviation is

$$\ln \mu(k) = -\frac{M}{R} + O(R^{-3}). \quad (158)$$

Thus the Schwarzschild parameter appears simultaneously as

$$\boxed{\begin{aligned} M &= \text{coefficient of the leading asymptotic response potential,} \\ 2M &= \text{length of the finite Weyl rod.} \end{aligned}} \quad (159)$$

The equality of this parameter with the global mass charge is not inferred from the harmonic representation alone. In the Schwarzschild solution it is established by the standard asymptotic or Komar charge calculation, which will be revisited in Sec. 5.

2.9 Response interpretation of the Weyl rod

The Schwarzschild construction can now be summarized without treating the Weyl rod as a physical source.

The local stationary response gives

$$\mu(k)^2 = -g(k, k). \quad (160)$$

Static vacuum Einstein dynamics implies

$$\Delta_{\text{ax}} \ln \mu(k) = 0. \quad (161)$$

For the asymptotically flat Schwarzschild branch, the resulting harmonic function is

$$\ln \mu(k) = U_{\text{Schw}}, \quad (162)$$

whose auxiliary source representation is the finite segment

$$\rho = 0, \quad -M \leq z \leq M. \quad (163)$$

Thus

stationary MS response amplitude	$\mu(k),$	(164)
static response potential	$U = \ln \mu(k),$	
vacuum closure	$\Delta_{\text{ax}} U = 0,$	
Schwarzschild solution	$U = U_{\text{rod}},$	
horizon	finite response-gap rod.	

The phrase “response-gap rod” should be interpreted geometrically. Along the rod interior,

$$\mu(k) \rightarrow 0, \quad (165)$$

so the stationary Killing response gap closes. At the same time the axial orbit remains nondegenerate. The rod therefore represents the locus at which the static stationary generator becomes null, not a distribution of massive MS particles.

This distinction is especially important in view of the local positive generator picture. Nothing in Eq. (122) means that the Schwarzschild mass is obtained by integrating local one-particle shell parameters Q along the rod. The rod belongs to the auxiliary Weyl representation of the global vacuum geometry.

2.10 Why Schwarzschild is the static benchmark for Kerr

The Schwarzschild response spectrum has three properties that rotation will modify:

$\begin{aligned} \mathcal{R}_{\text{Schw}}(\Omega) &= \mathcal{R}_{\text{Schw}}(-\Omega), \\ \Omega_Z &= 0, \\ \mathcal{R}_{\text{Schw}}(0) = 0 &\iff \Omega_+ = \Omega_- = 0 \quad \text{on the horizon.} \end{aligned}$	(166)
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Kerr changes all three.

The response polynomial is no longer even because

$$g(k, m) \neq 0. \quad (167)$$

Its maximum shifts to

$$\Omega_Z \neq 0, \quad (168)$$

encoding local frame dragging. Most importantly,

$$\mathcal{R}(0) = 0 \quad (169)$$

occurs on the stationary-limit surface, whereas collapse of the two null roots occurs at a different surface,

$$\Delta = 0. \quad (170)$$

On the latter, the double root occurs not at zero angular velocity but at the nonzero horizon value

$$\Omega_H. \quad (171)$$

The Schwarzschild coincidence

$$\boxed{\text{static limit} = \text{horizon} = \text{zero-angular-velocity double root}} \quad (172)$$

is therefore a special nonrotating limit of a richer stationary response geometry.

The next section develops this rotating structure explicitly for Kerr. We shall show that the exact Kerr response polynomial admits the canonical form

$$\mathcal{R}_{\text{Kerr}}(\Omega) = \mathcal{R}_{\text{max}} - g_{\phi\phi} (\Omega - \Omega_Z)^2, \quad (173)$$

with

$$-\det G = \Delta \sin^2 \theta. \quad (174)$$

The ergosurface will be identified by

$$\mathcal{R}_{\text{Kerr}}(0) = 0, \quad (175)$$

while the horizon will be identified by the genuinely different condition

$$\mathcal{R}_{\text{Kerr}}(\Omega_H) = \partial_\Omega \mathcal{R}_{\text{Kerr}}(\Omega_H) = 0. \quad (176)$$

This will provide the rotating counterpart of the Schwarzschild response-gap rod.

3 Kerr stationary response spectrum

The Schwarzschild geometry provides the static benchmark for the stationary response construction. Its response spectrum is even in the angular parameter, the response maximum occurs at zero angular velocity, and the vanishing of the stationary response $\mathcal{R}(0)$ coincides with collapse of the entire stationary timelike interval.

Kerr separates these structures.

The purpose of this section is to show that the exact Kerr geometry admits a particularly simple stationary response representation in which redshift, frame dragging, the ergosurface, the horizon, and the Weyl–Papapetrou discriminant are read from different features of a single quadratic polynomial.

Throughout this section we use geometrized units

$$G = c = 1. \quad (177)$$

We take

$$M > 0, \quad |a| \leq M, \quad (178)$$

and initially work in the domain of outer communications of the Kerr solution.

3.1 Kerr Killing geometry

In Boyer–Lindquist coordinates (t, r, θ, ϕ) , define

$$\Sigma := r^2 + a^2 \cos^2 \theta \quad (179)$$

$$\Delta := r^2 - 2Mr + a^2 \quad (180)$$

$$\mathcal{A} := (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta \quad (181)$$

Equivalently,

$$\mathcal{A} = (r^2 + a^2)\Sigma + 2Ma^2r \sin^2 \theta. \quad (182)$$

The Kerr metric is

$$ds^2 = -\left(1 - \frac{2Mr}{\Sigma}\right)dt^2 - \frac{4Mar \sin^2 \theta}{\Sigma}dt d\phi \\ + \frac{\Sigma}{\Delta}dr^2 + \Sigma d\theta^2 + \frac{\mathcal{A} \sin^2 \theta}{\Sigma}d\phi^2. \quad (183)$$

The stationary and axial Killing fields are

$$k = \partial_t, \quad m = \partial_\phi. \quad (184)$$

Their Gram coefficients are

$$g_{tt} = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} \quad (185)$$

$$g_{t\phi} = -\frac{2Mar \sin^2 \theta}{\Sigma} \quad (186)$$

$$g_{\phi\phi} = \frac{\mathcal{A} \sin^2 \theta}{\Sigma} \quad (187)$$

The determinant of the Killing Gram matrix simplifies drastically:

$$-\det G = g_{t\phi}^2 - g_{tt}g_{\phi\phi} = \Delta \sin^2 \theta \quad (188)$$

Thus the response discriminant introduced in Sec. ?? is

$$\rho^2 = \Delta \sin^2 \theta \quad (189)$$

This already identifies the two geometrically different ways in which the Killing Gram determinant can vanish:

$$\sin \theta = 0 \quad (190)$$

is the rotation axis, whereas

$$\Delta = 0 \quad (191)$$

gives the Kerr Killing-horizon candidates.

The horizon radii are

$$r_{\pm} = M \pm \sqrt{M^2 - a^2} \quad (192)$$

In the black-hole exterior,

$$r > r_+, \quad \Delta > 0. \quad (193)$$

3.2 The exact Kerr response polynomial

For

$$\chi_\Omega = k + \Omega m, \quad (194)$$

the stationary response polynomial is

$$\mathcal{R}_{\text{Kerr}}(r, \theta; \Omega) := -g(\chi_\Omega, \chi_\Omega). \quad (195)$$

Using Eqs. (185)–(187), one finds

$$\mathcal{R}_{\text{Kerr}}(\Omega) = \frac{\Delta - a^2 \sin^2 \theta + 4Mar \sin^2 \theta \Omega - \mathcal{A} \sin^2 \theta \Omega^2}{\Sigma} \quad (196)$$

The linear term in Ω is the response signature of rotation. In particular,

$$\mathcal{R}_{\text{Kerr}}(\Omega) - \mathcal{R}_{\text{Kerr}}(-\Omega) = \frac{8Mar \sin^2 \theta}{\Sigma} \Omega \quad (197)$$

Thus co-rotating and counter-rotating massive responses differ already at first order in angular velocity.

For $a = 0$, this odd contribution disappears and Eq. (196) reduces exactly to the Schwarzschild response polynomial of Sec. 2.

3.3 Response optimization, Kerr lapse, and frame dragging

The Kerr metric admits the equivalent decomposition

$$ds^2 = -\alpha^2 dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + g_{\phi\phi} (d\phi - \omega dt)^2, \quad (198)$$

where

$$\alpha^2 = \frac{\Delta \Sigma}{\mathcal{A}} \quad (199)$$

$$\omega = \frac{2Mar}{\mathcal{A}} = -\frac{g_{t\phi}}{g_{\phi\phi}} \quad (200)$$

The quantity ω is the standard Kerr frame-dragging angular velocity of the locally nonrotating observer.

Substituting these expressions into the response polynomial gives the canonical spectral form

$$\mathcal{R}_{\text{Kerr}}(\Omega) = \alpha^2 - g_{\phi\phi} (\Omega - \omega)^2 \quad (201)$$

This equation gives a direct MS response interpretation of two standard Kerr quantities.

Theorem 3.1 (Kerr lapse and dragging from response optimization). *Away from the rotation axis in the domain of outer communications, the stationary Kerr response polynomial has a unique maximum. Its location is the local frame-dragging angular velocity,*

$$\arg \max_{\Omega} \mathcal{R}_{\text{Kerr}}(\Omega) = \omega = \frac{2Mar}{\mathcal{A}} \quad (202)$$

and its maximum value is the square of the Kerr lapse,

$$\max_{\Omega} \mathcal{R}_{\text{Kerr}}(\Omega) = \alpha^2 = \frac{\Delta \Sigma}{\mathcal{A}} \quad (203)$$

Thus the locally nonrotating Killing direction is

$$\tilde{k} := k + \omega m, \quad (204)$$

and its response amplitude is

$$\mu(k + \omega m) = \alpha \quad (205)$$

After normalization,

$$n = \alpha^{-1} (k + \omega m) \quad (206)$$

is the standard ZAMO four-velocity.

In response language, the lapse is therefore the *optimized stationary massive-response gap*, while frame dragging is the angular velocity at which that gap is maximal:

$$\begin{aligned} \omega &= \arg \max_{\Omega} \mathcal{R}_{\text{Kerr}}, \\ \alpha &= \sqrt{\max_{\Omega} \mathcal{R}_{\text{Kerr}}}. \end{aligned} \quad (207)$$

This interpretation is distinct from the ordinary stationary response

$$\mu(k)^2 = \mathcal{R}(0), \quad (208)$$

which measures the norm of the asymptotically normalized stationary Killing field rather than the optimized locally nonrotating direction.

3.4 Null roots and the Kerr stationary-observer interval

The null stationary directions satisfy

$$\mathcal{R}_{\text{Kerr}}(\Omega) = 0. \quad (209)$$

Using Eq. (201), their angular velocities are

$$\Omega_{\pm} = \omega \pm \frac{\alpha}{\sqrt{g_{\phi\phi}}} \quad (210)$$

Explicitly,

$$\Omega_{\pm} = \frac{2Mar}{\mathcal{A}} \pm \frac{\Sigma \sqrt{\Delta}}{\mathcal{A} \sin \theta} \quad (211)$$

for

$$0 < \theta < \pi. \quad (212)$$

The stationary timelike observer interval is therefore

$$\Omega_- < \Omega < \Omega_+ \quad (213)$$

Its center is

$$\frac{\Omega_+ + \Omega_-}{2} = \omega \quad (214)$$

and its width is

$$\Omega_+ - \Omega_- = \frac{2\Sigma\sqrt{\Delta}}{\mathcal{A}\sin\theta} \quad (215)$$

Thus Kerr frame dragging shifts the center of the stationary causal interval, while the Weyl discriminant controls its width.

Indeed,

$$\rho = \sqrt{\Delta} \sin \theta, \quad (216)$$

and the general relation

$$\Omega_+ - \Omega_- = \frac{2\rho}{g_{\phi\phi}} \quad (217)$$

is recovered immediately.

3.5 Two different response gaps

The Kerr geometry contains two natural response quantities:

$$\mathcal{R}_{\text{stat}} := \mathcal{R}_{\text{Kerr}}(0) = -g(k, k) \quad (218)$$

$$\mathcal{R}_{\text{opt}} := \max_{\Omega} \mathcal{R}_{\text{Kerr}}(\Omega) = \alpha^2 \quad (219)$$

Explicitly,

$$\mathcal{R}_{\text{stat}} = \frac{\Delta - a^2 \sin^2 \theta}{\Sigma} = 1 - \frac{2Mr}{\Sigma} \quad (220)$$

whereas

$$\mathcal{R}_{\text{opt}} = \frac{\Delta\Sigma}{\mathcal{A}} \quad (221)$$

They obey

$$\mathcal{R}_{\text{stat}} = \mathcal{R}_{\text{opt}} - g_{\phi\phi}\omega^2 \quad (222)$$

This single identity explains the separation of the ergosurface and the horizon.

For Schwarzschild,

$$\omega = 0, \quad (223)$$

so

$$\mathcal{R}_{\text{stat}} = \mathcal{R}_{\text{opt}}. \quad (224)$$

For Kerr,

$$\omega \neq 0, \quad (225)$$

and the two gaps close on different surfaces.

3.6 The ergosurface as loss of the zero-angular-velocity observer

The stationary-limit surface is determined by

$$g(k, k) = 0, \quad (226)$$

or

$$\boxed{\mathcal{R}_{\text{Kerr}}(0) = 0} \quad (227)$$

Using Eq. (220), this gives

$$r^2 - 2Mr + a^2 \cos^2 \theta = 0. \quad (228)$$

The two stationary-limit radii are

$$\boxed{r_{\text{E}\pm}(\theta) = M \pm \sqrt{M^2 - a^2 \cos^2 \theta}} \quad (229)$$

The outer ergosurface is

$$r = r_{\text{E}+}(\theta). \quad (230)$$

At a generic non-axis point of the outer ergosurface,

$$\alpha^2 > 0. \quad (231)$$

Thus the optimized response gap remains open even though

$$\mathcal{R}(0) = 0. \quad (232)$$

Stationary timelike observers still exist, but $\Omega = 0$ has reached the boundary of their allowed interval.

Since

$$\mathcal{R}(0) = \alpha^2 - g_{\phi\phi}\omega^2, \quad (233)$$

the ergosurface condition is

$$\boxed{\alpha^2 = g_{\phi\phi}\omega^2} \quad (234)$$

The two null roots there are therefore

$$\boxed{\{\Omega_-, \Omega_+\} = \{0, 2\omega\}} \quad (235)$$

up to their ordering under reversal of the rotational orientation.

Hence the response interpretation of the ergosurface is

$$\boxed{\text{ergosurface} \iff \text{one null stationary root passes through } \Omega = 0} \quad (236)$$

Crossing into the ergoregion removes the asymptotically stationary direction k from the timelike cone:

$$\mathcal{R}(0) < 0. \quad (237)$$

Nevertheless,

$$\mathcal{R}_{\text{max}} = \alpha^2 > 0 \quad (238)$$

until the horizon is reached.

This is the local response statement behind the familiar fact that no observer can remain static inside the Kerr ergoregion, while appropriately co-rotating stationary timelike observers continue to exist outside the horizon.

3.7 The horizon as collapse of the optimized response gap

The outer Kerr horizon is

$$r = r_+, \quad (239)$$

where

$$\Delta(r_+) = 0. \quad (240)$$

Consequently,

$$\alpha^2 = \mathcal{R}_{\max} \longrightarrow 0 \quad (241)$$

At the same time,

$$\Omega_+ - \Omega_- \longrightarrow 0. \quad (242)$$

Thus the entire stationary timelike interval collapses.

At $\Delta = 0$,

$$\mathcal{A}_+ = (r_+^2 + a^2)^2. \quad (243)$$

Using

$$2Mr_+ = r_+^2 + a^2, \quad (244)$$

the limiting dragging angular velocity becomes

$$\begin{aligned} \omega(r_+) &= \frac{2Mar_+}{(r_+^2 + a^2)^2} \\ &= \boxed{\frac{a}{r_+^2 + a^2}} \end{aligned} \quad (245)$$

Define

$$\Omega_H := \frac{a}{r_+^2 + a^2} \quad (246)$$

The two null roots satisfy

$$\lim_{r \rightarrow r_+^+} \Omega_{\pm} = \Omega_H \quad (247)$$

Thus the horizon generator is

$$\chi_H = k + \Omega_H m \quad (248)$$

The response polynomial itself takes an especially simple form directly on the horizon. Let

$$\Sigma_+ := r_+^2 + a^2 \cos^2 \theta. \quad (249)$$

Away from the axis,

$$g_{\phi\phi}|_H = \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+}. \quad (250)$$

Since the optimized response vanishes, Eq. (201) becomes

$$\mathcal{R}_{\text{Kerr}}(\Omega)|_H = -\frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+} (\Omega - \Omega_H)^2 \quad (251)$$

We therefore obtain the central spectral characterization.

Theorem 3.2 (Kerr horizon as a response double root). *At every non-axis point of the outer Kerr horizon, the stationary response polynomial has the unique double root Ω_H :*

$$\boxed{\mathcal{R}_{\text{Kerr}}(\Omega_H) = 0, \quad \partial_{\Omega} \mathcal{R}_{\text{Kerr}}(\Omega_H) = 0} \quad (252)$$

Equivalently,

$$\boxed{\Omega_+ = \Omega_- = \Omega_H} \quad (253)$$

This is stronger than the ergosurface condition. The ergosurface makes one particular Killing direction null. The horizon collapses the complete stationary timelike interval to a single null Killing generator.

In response language,

$$\boxed{\begin{array}{ll} \text{ergosurface} & : \mathcal{R}(0) = 0, \quad \mathcal{R}_{\text{max}} > 0, \\ \text{horizon} & : \mathcal{R}_{\text{max}} = 0, \quad \mathcal{R}(\Omega_H) = \mathcal{R}'(\Omega_H) = 0. \end{array}} \quad (254)$$

This distinction is absent in Schwarzschild because $\omega = 0$.

3.8 Static response on the Kerr horizon

The asymptotically normalized stationary Killing field k does not itself become the horizon generator when $a \neq 0$. Indeed, Eq. (251) evaluated at $\Omega = 0$ gives

$$\boxed{\mathcal{R}_{\text{Kerr}}(0)|_H = -\frac{a^2 \sin^2 \theta}{\Sigma_+}} \quad (255)$$

Thus, away from the rotation axis,

$$\mathcal{R}_{\text{Kerr}}(0) < 0 \quad (256)$$

on the horizon.

The stationary Killing field is therefore spacelike there. Only the co-rotating combination

$$k + \Omega_H m \quad (257)$$

is null.

This gives a compact response interpretation of the Kerr ergoregion:

$$\boxed{\text{between the ergosurface and horizon, } \mathcal{R}(0) < 0 \quad \text{but} \quad \max_{\Omega} \mathcal{R}(\Omega) > 0} \quad (258)$$

The distinction between these two response gaps is one of the main reasons the rotating problem is structurally richer than the static one.

3.9 Weyl coordinates and the Kerr horizon rod

The response discriminant also leads directly to canonical Weyl coordinates.

In the Kerr exterior define

$$\boxed{\rho = \sqrt{\Delta} \sin \theta} \quad (259)$$

$$\boxed{z = (r - M) \cos \theta} \quad (260)$$

The first equation is not an independent coordinate ansatz: it is exactly the response discriminant

$$\rho^2 = -\det G. \quad (261)$$

Introduce

$$\ell := \sqrt{M^2 - a^2} \quad (262)$$

Then

$$r_{\pm} = M \pm \ell. \quad (263)$$

On the outer horizon,

$$r = r_+ = M + \ell, \quad (264)$$

so

$$\rho = 0 \quad (265)$$

and

$$z = \ell \cos \theta. \quad (266)$$

Hence the complete outer horizon maps to the finite Weyl rod

$$\rho = 0, \quad -\ell \leq z \leq \ell \quad (267)$$

Its coordinate length in Weyl orbit space is

$$L_H^{\text{Weyl}} = 2 \sqrt{M^2 - a^2} \quad (268)$$

The two exterior axis components are

$$\rho = 0, \quad |z| > \ell. \quad (269)$$

Thus, exactly as in Schwarzschild, the full response polynomial rather than $\rho = 0$ alone is required to distinguish the axis from the horizon rod.

The rod formula immediately reproduces the Schwarzschild limit:

$$a \rightarrow 0 \quad \implies \quad \ell \rightarrow M, \quad (270)$$

so

$$L_H^{\text{Weyl}} \longrightarrow 2M. \quad (271)$$

This is precisely the Schwarzschild rod obtained in Sec. 2.

Rotation therefore shortens the horizon rod in Weyl orbit space:

$$2M \longrightarrow 2 \sqrt{M^2 - a^2} \quad (272)$$

This is a classical property of the Kerr Weyl representation. Its role here is to show that the same quantity arises directly from the response discriminant $-\det G$.

3.10 The extremal limit

The extremal limit is

$$|a| \rightarrow M. \quad (273)$$

Then

$$r_+ = r_- = M, \quad (274)$$

and

$$\ell = \sqrt{M^2 - a^2} \rightarrow 0. \quad (275)$$

Consequently,

$$\boxed{L_H^{\text{Weyl}} \rightarrow 0} \quad (276)$$

Thus the nonextremal finite horizon rod contracts to a point in canonical Weyl orbit space in the extremal limit.

The stationary response roots also exhibit a change in their radial coalescence behavior. Since

$$\Delta = (r - r_+)(r - r_-), \quad (277)$$

Eq. (215) gives, for a nonextremal black hole,

$$\Omega_+ - \Omega_- = O(\sqrt{r - r_+}) \quad (278)$$

in Boyer–Lindquist radial coordinate.

For extremal Kerr,

$$\Delta = (r - M)^2, \quad (279)$$

and therefore

$$\Omega_+ - \Omega_- = O(r - M). \quad (280)$$

These particular powers refer to the Boyer–Lindquist radial parameter and should not themselves be regarded as invariant statements. What is invariant within the stationary Killing plane is the root coalescence

$$\Omega_+ - \Omega_- \rightarrow 0 \quad (281)$$

and the simultaneous vanishing

$$-\det G \rightarrow 0. \quad (282)$$

The change from a finite Weyl horizon rod to a pointlike extremal degeneracy is the corresponding orbit-space statement.

3.11 Schwarzschild as the zero-dragging limit

The complete Schwarzschild response structure is recovered smoothly by taking

$$a \rightarrow 0. \quad (283)$$

Then

$$\Sigma \longrightarrow r^2, \quad (284)$$

$$\Delta \longrightarrow r(r - 2M), \quad (285)$$

$$\mathcal{A} \longrightarrow r^4, \quad (286)$$

$$\omega \longrightarrow 0, \quad (287)$$

$$\alpha^2 \longrightarrow 1 - \frac{2M}{r}. \quad (288)$$

Equation (201) becomes

$$\mathcal{R}(\Omega) \longrightarrow 1 - \frac{2M}{r} - r^2 \sin^2 \theta \Omega^2 \quad (289)$$

and

$$\Omega_H \longrightarrow 0. \quad (290)$$

Moreover,

$$\mathcal{R}_{\text{stat}} = \mathcal{R}_{\text{opt}} \quad (291)$$

in this limit. Therefore the two Kerr response-gap closures collapse into the single Schwarzschild condition

$$r = 2M. \quad (292)$$

We may summarize the comparison as

	Schwarzschild	Kerr
response center	0	ω
stationary gap	$\mathcal{R}(0)$	$\mathcal{R}(0)$
optimized gap	$\mathcal{R}(0)$	α^2
static-limit closure	$r = 2M$	$r = r_{\text{E}+}(\theta)$
root coalescence	$r = 2M, \Omega_H = 0$	$r = r_+, \Omega_H \neq 0$

(293)

Rotation does not merely perturb the Schwarzschild response spectrum. It splits one static response-gap closure into two geometrically distinct phenomena:

$$\boxed{\text{loss of the static observer} \quad \text{and} \quad \text{collapse of all stationary timelike observers}} \quad (294)$$

3.12 A response hierarchy for Kerr

The Kerr calculation may now be organized entirely in terms of the stationary response polynomial:

$$\begin{aligned}
\mathcal{R}_{\text{Kerr}}(\Omega) &\longrightarrow (g_{tt}, g_{t\phi}, g_{\phi\phi}), \\
\arg \max_{\Omega} \mathcal{R}_{\text{Kerr}} &\longrightarrow \omega, \\
\max_{\Omega} \mathcal{R}_{\text{Kerr}} &\longrightarrow \alpha^2, \\
\mathcal{R}_{\text{Kerr}}(0) = 0 &\longrightarrow \text{ergosurface}, \\
\frac{1}{4} [(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}''] &\longrightarrow \rho^2 = \Delta \sin^2 \theta, \\
\Omega_+ - \Omega_- &\longrightarrow \text{width of the stationary causal interval}, \\
\Omega_+ = \Omega_- &\longrightarrow \text{Killing-plane degeneracy}, \\
\mathcal{R}(\Omega_H) = \mathcal{R}'(\Omega_H) = 0 &\longrightarrow \text{outer horizon generator}.
\end{aligned} \tag{295}$$

None of the Kerr metric coefficients, ergosurface, ZAMO angular velocity, or horizon angular velocity is new as a general-relativistic quantity. The structural result developed here is that all of them are different spectral features of the same massive-response quadratic form.

In particular, the distinction

$$\boxed{\mathcal{R}(0) \quad \text{versus} \quad \max_{\Omega} \mathcal{R}(\Omega)} \tag{296}$$

provides a compact response-theoretic separation of the ergosurface and the horizon.

3.13 From algebraic dragging to differential twist

The response polynomial reconstructs the complete pointwise Killing Gram geometry. For Kerr this includes the dragging coefficient

$$\omega = -\frac{g(k, m)}{g(m, m)}. \tag{297}$$

This is a zeroth-order algebraic quantity.

It does not yet determine the full differential rotational structure of the stationary Killing field. That structure is encoded by its twist one-form,

$$\varpi = \star (k^b \wedge dk^b), \tag{298}$$

which depends on first spacetime derivatives of the stationary geometry.

In vacuum,

$$d\varpi = 0, \tag{299}$$

so locally

$$\varpi = d\psi. \tag{300}$$

The real stationary response quantity

$$f = -g(k, k) = \mathcal{R}(0) \tag{301}$$

and the twist potential ψ then combine into the complex Ernst potential

$$\mathcal{E} = f + i\psi. \quad (302)$$

The next section therefore moves from the zeroth-order stationary response spectrum to its first-derivative vacuum completion:

$$\boxed{\mathcal{R}(\Omega) \longrightarrow G_{AB} \longrightarrow \text{twist one-form} \longrightarrow \psi \longrightarrow \mathcal{E}} \quad (303)$$

Schwarzschild will appear as the real, twist-free branch of this construction, while Kerr supplies its canonical complex rotating branch.

4 Twist completion and the Ernst potential

The stationary response polynomial developed in the preceding sections reconstructs the complete pointwise metric on the Killing plane. In particular, it determines the stationary norm

$$f := -g(k, k) = \mathcal{R}(0), \quad (304)$$

the mixed Killing coefficient

$$g(k, m) = -\frac{1}{2} \partial_\Omega \mathcal{R}(\Omega)|_{\Omega=0}, \quad (305)$$

and the Weyl discriminant

$$\rho^2 = -\det G. \quad (306)$$

These are zeroth-order quantities: they require the response spectrum at a single spacetime point but no spacetime derivatives.

Rotation contains additional information. A stationary Killing field can fail to be hypersurface orthogonal even after its pointwise norm and its algebraic mixing with the axial Killing field have been specified. The missing information is the twist.

The purpose of this section is to show how the zeroth-order stationary MS response geometry admits a natural first-derivative vacuum completion:

$$\boxed{\mathcal{R}(\Omega) \longrightarrow G_{AB} \longrightarrow \varpi \xrightarrow{R_{\mu\nu}=0} \psi \longrightarrow \mathcal{E} = f + i\psi} \quad (307)$$

Here ϖ is the twist one-form of the stationary Killing field, ψ its vacuum twist potential, and $\mathcal{E}$ the complex Ernst potential.

The Ernst equations themselves are standard consequences of the stationary axisymmetric vacuum Einstein equations [?]. The new role assigned to them here is structural: their real part is the stationary response gap reconstructed directly from massive MS responses, while their imaginary part is the integrable first-derivative rotational completion of that response geometry.

Throughout this section we use geometrized units

$$G = c = 1. \quad (308)$$

4.1 Algebraic rotation versus differential twist

It is important first to distinguish two notions of rotational structure.
The response polynomial determines the pointwise algebraic mixing

$$g(k, m), \quad (309)$$

or equivalently the displacement of the response maximum,

$$\Omega_Z = -\frac{g(k, m)}{g(m, m)} = \arg \max_{\Omega} \mathcal{R}(\Omega) \quad (310)$$

This is the locally nonrotating angular velocity.

It is a zeroth-order metric quantity.

Staticity, however, is not determined by $g(k, m)$ at a point. The invariant local criterion for the stationary Killing field k to be hypersurface orthogonal is

$$k^b \wedge dk^b = 0 \quad (311)$$

This condition depends on the first spacetime derivative of the stationary geometry.

Define the twist one-form of k by

$$\varpi := \star (k^b \wedge dk^b) \quad (312)$$

In components,

$$\varpi_\mu = \epsilon_{\mu\nu\rho\sigma} k^\nu \nabla^\rho k^\sigma \quad (313)$$

Then, wherever k is nonzero,

$$\varpi = 0 \iff k \text{ is locally hypersurface orthogonal} \quad (314)$$

Thus the stationary response hierarchy contains two distinct rotational levels:

$$\begin{array}{ll} g(k, m) \text{ or } \Omega_Z & \text{pointwise algebraic dragging,} \\ \varpi & \text{differential stationary twist.} \end{array} \quad (315)$$

The two are related by the stationary field equations in an axisymmetric vacuum spacetime, but they should not be identified.

4.2 Vacuum integrability of the twist

The Killing equation and Ricci identity imply the standard relation

$$\nabla_{[\mu} \varpi_{\nu]} = \epsilon_{\mu\nu\rho\sigma} k^\rho R^\sigma{}_\lambda k^\lambda \quad (316)$$

Therefore, in vacuum,

$$R_{\mu\nu} = 0, \quad (317)$$

we have

$$\boxed{d\varpi = 0} \quad (318)$$

On a simply connected stationary vacuum domain there consequently exists a scalar twist potential

$$\psi \quad (319)$$

such that

$$\boxed{d\psi = \varpi} \quad (320)$$

The potential is defined up to an additive constant. For an asymptotically flat stationary spacetime we fix this freedom by

$$\boxed{\psi \longrightarrow 0 \quad \text{at spatial infinity}} \quad (321)$$

The normalization of k is likewise fixed asymptotically:

$$\boxed{g(k, k) \longrightarrow -1} \quad (322)$$

Thus

$$f \longrightarrow 1. \quad (323)$$

These two conditions remove the multiplicative normalization ambiguity in the stationary Killing field and the additive ambiguity in the twist potential. They will also be essential when the same Killing fields are used in the Komar integrals below.

4.3 The Ernst potential as complex response completion

Define the complex Ernst potential by

$$\boxed{\mathcal{E} := f + i\psi} \quad (324)$$

Since

$$f = \mathcal{R}(0), \quad (325)$$

the real part is directly determined by the stationary response spectrum:

$$\boxed{\Re \mathcal{E} = \mathcal{R}(0)} \quad (326)$$

The imaginary part is determined by the first-derivative twist:

$$\boxed{d \Im \mathcal{E} = \star (k^b \wedge dk^b)} \quad (327)$$

Thus the Ernst potential admits the response interpretation

$$\boxed{\mathcal{E} = \underbrace{\mathcal{R}(0)}_{\text{stationary response gap}} + i \underbrace{\psi}_{\text{integrated stationary twist}}} \quad (328)$$

The first term is accessible from the zeroth-order massive-response spectrum wherever k is time-like. The second is not a pointwise algebraic response quantity. It requires first spacetime derivatives and the vacuum integrability condition Eq. (318).

This distinction is central to the present paper.

4.4 Weyl–Papapetrou variables from the response spectrum

In a regular orthogonally transitive stationary axisymmetric vacuum region, the metric can be written locally in Weyl–Papapetrou form

$$ds^2 = -f(dt - \omega_P d\phi)^2 + f^{-1} \left[e^{2\gamma} (d\rho^2 + dz^2) + \rho^2 d\phi^2 \right] \quad (329)$$

We use the symbol ω_P for the Papapetrou metric coefficient in order to distinguish it from the locally nonrotating angular velocity Ω_Z of Eq. (310).

The Killing-plane coefficients are

$$g_{tt} = -f, \quad (330)$$

$$g_{t\phi} = f\omega_P, \quad (331)$$

$$g_{\phi\phi} = f^{-1}\rho^2 - f\omega_P^2. \quad (332)$$

Therefore

$$-\det G = \rho^2. \quad (333)$$

The stationary response polynomial reconstructs f , ω_P , and ρ algebraically:

$$f = \mathcal{R}(0) \quad (334)$$

$$\omega_P = \frac{g_{t\phi}}{f} = -\frac{\partial_\Omega \mathcal{R}(0)}{2\mathcal{R}(0)} \quad (335)$$

$$\rho^2 = \frac{1}{4} \left[(\partial_\Omega \mathcal{R})^2 - 2\mathcal{R}\partial_\Omega^2 \mathcal{R} \right] \quad (336)$$

The Papapetrou coefficient should not be confused with Ω_Z . Indeed,

$$\Omega_Z = -\frac{f\omega_P}{f^{-1}\rho^2 - f\omega_P^2} \quad (337)$$

so in general

$$\omega_P \neq \Omega_Z \quad (338)$$

The former is a coefficient in the Weyl–Papapetrou metric parametrization; the latter is the angular velocity of the locally nonrotating observer.

4.5 Twist from the first jet of the response spectrum

With the orientation convention adopted here, the stationary twist potential satisfies

$$\psi_{,\rho} = \frac{f^2}{\rho} \omega_{P,z} \quad (339)$$

$$\psi_{,z} = -\frac{f^2}{\rho} \omega_{P,\rho} \quad (340)$$

Reversal of the spacetime orientation reverses ψ and complex conjugates the Ernst potential; no physical metric quantity is affected.

Using Eq. (335), these relations can be written entirely in terms of the first spacetime jet of the stationary response polynomial:

$$\boxed{\psi_{,\rho} = \frac{\mathcal{R}(0)^2}{\rho} \partial_z \left[-\frac{\partial_\Omega \mathcal{R}(0)}{2\mathcal{R}(0)} \right]} \quad (341)$$

$$\boxed{\psi_{,z} = -\frac{\mathcal{R}(0)^2}{\rho} \partial_\rho \left[-\frac{\partial_\Omega \mathcal{R}(0)}{2\mathcal{R}(0)} \right]} \quad (342)$$

This gives a direct response interpretation of the twist hierarchy.

Theorem 4.1 (Vacuum Ernst completion of stationary response data). *Let a simply connected region be stationary, axisymmetric, orthogonally transitive, and vacuum. Suppose the stationary and axial Killing fields have been fixed by the asymptotic calibration used in the preceding sections.*

Then the stationary massive-response polynomial $\mathcal{R}_x(\Omega)$ determines pointwise

$$f, \quad \omega_P, \quad \rho. \quad (343)$$

Its first spacetime jet determines the closed twist one-form $d\psi$. Hence, after fixing $\psi \rightarrow 0$ at infinity, the response data determine the complex Ernst potential

$$\boxed{\mathcal{E} = \mathcal{R}(0) + i\psi} \quad (344)$$

The word “determine” here includes one integration step for the imaginary part. The twist potential is not an algebraic function of the response polynomial at a single point. Vacuum closure makes its line integral path-independent in the stated simply connected domain.

4.6 The Ernst equation

The stationary axisymmetric vacuum Einstein equations combine into the complex Ernst equation

$$\boxed{(\Re \mathcal{E}) \left(\mathcal{E}_{,\rho\rho} + \frac{1}{\rho} \mathcal{E}_{,\rho} + \mathcal{E}_{,zz} \right) = \mathcal{E}_{,\rho}^2 + \mathcal{E}_{,z}^2} \quad (345)$$

Writing

$$\mathcal{E} = f + i\psi, \quad (346)$$

the real and imaginary parts are

$$\boxed{f \left(f_{,\rho\rho} + \frac{1}{\rho} f_{,\rho} + f_{,zz} \right) = f_{,\rho}^2 + f_{,z}^2 - \psi_{,\rho}^2 - \psi_{,z}^2} \quad (347)$$

$$\boxed{f \left(\psi_{,\rho\rho} + \frac{1}{\rho} \psi_{,\rho} + \psi_{,zz} \right) = 2(f_{,\rho} \psi_{,\rho} + f_{,z} \psi_{,z})} \quad (348)$$

The remaining orbit-space metric coefficient γ is obtained by quadratures:

$$\gamma_{,\rho} = \frac{\rho}{4f^2} \left[f_{,\rho}^2 - f_{,z}^2 + \psi_{,\rho}^2 - \psi_{,z}^2 \right] \quad (349)$$

$$\gamma_{,z} = \frac{\rho}{2f^2} \left[f_{,\rho} f_{,z} + \psi_{,\rho} \psi_{,z} \right] \quad (350)$$

Thus the hierarchy of reconstruction is

$$\mathcal{R} \longrightarrow (f, \omega_P, \rho) \longrightarrow \psi \longrightarrow \mathcal{E} \longrightarrow \gamma \quad (351)$$

The first arrow is algebraic and response-theoretic. The second uses first spacetime derivatives and vacuum twist integrability. The last two use the stationary vacuum Einstein equations.

4.7 The static branch as a real Ernst sector

The static case is characterized by

$$\varpi = 0. \quad (352)$$

With the asymptotic normalization $\psi \rightarrow 0$, this gives

$$\psi = 0 \quad (353)$$

Hence the Ernst potential becomes purely real:

$$\mathcal{E}_{\text{static}} = f = \mathcal{R}(0) \quad (354)$$

The Ernst equation reduces to

$$f \Delta_{\text{ax}} f = |\nabla f|^2, \quad (355)$$

where

$$\Delta_{\text{ax}} := \partial_\rho^2 + \frac{1}{\rho} \partial_\rho + \partial_z^2. \quad (356)$$

Therefore

$$\Delta_{\text{ax}} \ln f = 0. \quad (357)$$

Since

$$f = e^{2U}, \quad (358)$$

we recover

$$\Delta_{\text{ax}} U = 0 \quad (359)$$

exactly as in Sec. 2.

Thus the Weyl harmonic-potential construction is the real, twist-free subsector of the Ernst response completion:

$$\text{Weyl static vacuum} \subset \text{Ernst stationary vacuum} \quad (360)$$

For Schwarzschild,

$$\mathcal{E}_{\text{Schw}} = 1 - \frac{2M}{r} \quad (361)$$

Equivalently,

$$\mathcal{E}_{\text{Schw}} = \frac{R_+ + R_- - 2M}{R_+ + R_- + 2M}. \quad (362)$$

The finite Weyl rod of the preceding section is therefore the real Ernst branch corresponding to a vanishing twist potential.

4.8 The Kerr Ernst potential

Kerr supplies the canonical complex rotating branch.

With the orientation convention of Eqs. (339) and (340), the Kerr Ernst potential is

$$\mathcal{E}_{\text{Kerr}} = 1 - \frac{2M}{r - ia \cos \theta} \quad (363)$$

Equivalently,

$$\mathcal{E}_{\text{Kerr}} = \frac{r - 2M - ia \cos \theta}{r - ia \cos \theta} \quad (364)$$

Its real part is

$$\begin{aligned} \Re \mathcal{E}_{\text{Kerr}} &= 1 - \frac{2Mr}{r^2 + a^2 \cos^2 \theta} \\ &= \boxed{1 - \frac{2Mr}{\Sigma}} \\ &= \boxed{-g_{tt} = \mathcal{R}_{\text{Kerr}}(0)} \end{aligned} \quad (365)$$

The imaginary part is

$$\psi_{\text{Kerr}} = -\frac{2Ma \cos \theta}{\Sigma} \quad (366)$$

Thus

$$\mathcal{E}_{\text{Kerr}} = \underbrace{\left(1 - \frac{2Mr}{\Sigma}\right)}_{\mathcal{R}_{\text{Kerr}}(0)} - i \underbrace{\frac{2Ma \cos \theta}{\Sigma}}_{\text{stationary twist potential}} \quad (367)$$

This formula exhibits directly the two levels of Kerr stationary geometry:

$$\begin{aligned} \Re \mathcal{E}_{\text{Kerr}} &= \text{stationary response gap,} \\ \Im \mathcal{E}_{\text{Kerr}} &= \text{integrated rotational twist.} \end{aligned} \quad (368)$$

The Schwarzschild limit is immediate:

$$a \longrightarrow 0 \quad \implies \quad \psi_{\text{Kerr}} \longrightarrow 0, \quad (369)$$

and

$$\mathcal{E}_{\text{Kerr}} \longrightarrow 1 - \frac{2M}{r} = \mathcal{E}_{\text{Schw}}. \quad (370)$$

4.9 Asymptotic twist and angular momentum

At large r ,

$$f = 1 - \frac{2M}{r} + O(r^{-2}), \quad (371)$$

$$\psi_{\text{Kerr}} = -\frac{2Ma \cos \theta}{r^2} + O(r^{-4}). \quad (372)$$

Since

$$J = Ma \quad (373)$$

for Kerr,

$$\psi_{\text{Kerr}} = -\frac{2J \cos \theta}{r^2} + O(r^{-4}) \quad (374)$$

Thus the leading real Ernst asymptotic coefficient carries the mass monopole, while the leading imaginary coefficient carries the angular momentum dipole:

$$\begin{aligned} \Re \mathcal{E} &= 1 - \frac{2M}{r} + \dots, \\ \Im \mathcal{E} &= -\frac{2J \cos \theta}{r^2} + \dots. \end{aligned} \quad (375)$$

At this stage the symbols M and J are the standard Kerr parameters. Their interpretation as global Komar charges will be established in Sec. 5. The asymptotic Ernst expansion already anticipates that identification.

4.10 Prolate coordinates and the Kerr horizon rod

For subextremal Kerr,

$$|a| < M, \quad (376)$$

define

$$\ell := \sqrt{M^2 - a^2}. \quad (377)$$

Introduce prolate spheroidal coordinates

$$x := \frac{r - M}{\ell}, \quad y := \cos \theta \quad (378)$$

Then

$$\rho = \ell \sqrt{x^2 - 1} \sqrt{1 - y^2} \quad (379)$$

$$z = \ell xy \quad (380)$$

The outer horizon corresponds to

$$x = 1, \quad (381)$$

and hence

$$\rho = 0, \quad -\ell \leq z \leq \ell. \quad (382)$$

Thus the prolate Ernst coordinates reproduce the Kerr horizon rod derived from the response discriminant in Sec. 3.

In these variables, Eq. (364) becomes

$$\mathcal{E}_{\text{Kerr}} = \frac{\ell x - M - iay}{\ell x + M - iay} \quad (383)$$

This representation makes the orbit-space role of

$$\ell = \sqrt{M^2 - a^2} \quad (384)$$

explicit. The same quantity controls the horizon-rod half-length and the prolate focal separation.

The extremal limit

$$\ell \rightarrow 0 \quad (385)$$

degenerates this particular prolate coordinate representation, consistently with the collapse of the finite Weyl horizon rod found in the preceding section.

4.11 A Cayley transform of the Ernst potential

A further useful variable is obtained by the fractional-linear transformation

$$\xi := \frac{1 - \mathcal{E}}{1 + \mathcal{E}} \quad (386)$$

For Kerr, Eq. (363) gives the remarkably simple expression

$$\xi_{\text{Kerr}} = \frac{M}{r - M - ia \cos \theta} \quad (387)$$

In prolate coordinates,

$$\xi_{\text{Kerr}} = \frac{M}{\ell x - iay} \quad (388)$$

For Schwarzschild,

$$a = 0, \quad (389)$$

so

$$\xi_{\text{Schw}} = \frac{M}{r - M} \quad (390)$$

The Kerr expression is especially revealing on the north symmetry axis. There

$$\rho = 0, \quad y = 1, \quad z = r - M \quad (391)$$

in the exterior component, and therefore

$$\xi_{\text{Kerr}}^{\text{axis}}(z) = \frac{M}{z - ia} \quad (392)$$

At large z ,

$$\xi_{\text{Kerr}}^{\text{axis}}(z) = \frac{M}{z} + \frac{iMa}{z^2} - \frac{Ma^2}{z^3} - \frac{iMa^3}{z^4} + \dots \quad (393)$$

This alternating real–imaginary pattern anticipates the well-known Kerr mass-current multipole structure. A rigorous identification with the Geroch–Hansen multipole moments requires the standard conformal completion and multipole definitions and is not performed here. Nevertheless, the compact form Eq. (392) shows how the two basic parameters M and a seed the complete stationary Ernst hierarchy.

This observation suggests a natural extension of the MS response programme to stationary multipole spectroscopy, but that problem lies beyond the scope of the present paper.

4.12 Ergosurface crossing and the response interpretation

A subtle point concerns the ergoregion.

Where the asymptotically normalized stationary Killing field k is future timelike,

$$f = \mathcal{R}(0) > 0, \quad (394)$$

and

$$\mu(k) = \sqrt{f} \quad (395)$$

has its direct massive-response interpretation.

At the stationary-limit surface,

$$f = 0, \quad (396)$$

and inside the ergoregion,

$$f < 0. \quad (397)$$

The local massive response polar $\mu(k)$ is therefore no longer defined there as a future-timelike norm of k .

The polynomial

$$\mathcal{R}(\Omega) \quad (398)$$

itself, however, remains a well-defined algebraic quadratic form on the Killing plane. The Ernst real part

$$f = \mathcal{R}(0) \quad (399)$$

is therefore understood inside the ergoregion as the algebraic continuation of the stationary response coefficient reconstructed from the Killing geometry, not as the square of a timelike massive response amplitude.

This distinction mirrors the one already made in Sec. ??: the direct MS polar is defined on the future-timelike sector, while the reconstructed Lorentzian quadratic form extends uniquely to null and spacelike directions.

The rotating response hierarchy is therefore

$\mu(k + \Omega m)^2$	direct timelike response,	(400)
$\mathcal{R}(\Omega)$	reconstructed Killing quadratic form,	
$\mathcal{R}(0) = f$	real Ernst field,	
$d\psi$	first-derivative twist,	
$\mathcal{E} = f + i\psi$	complex stationary vacuum potential.	

4.13 What the Ernst completion adds

The response polynomial by itself determines the Killing-plane metric but does not distinguish all spacetimes with the same pointwise Gram data. The twist completion introduces precisely the first-derivative information required to encode local stationary rotation.

This gives the hierarchy

zeroth-order response $\longrightarrow (f, \omega_p, \rho),$	(401)
first spacetime derivative $\longrightarrow \varpi,$	
vacuum integrability $\longrightarrow \psi,$	
complex completion $\longrightarrow \mathcal{E} = f + i\psi,$	
vacuum Ernst equation $\longrightarrow$ stationary axisymmetric geometry.	

For Schwarzschild,

$$\psi = 0, \quad (402)$$

so the response hierarchy remains real and reduces to Weyl harmonic geometry.

For Kerr,

$$\psi \neq 0, \quad (403)$$

and the stationary response gap and rotational twist combine into the simple complex field

$\mathcal{E}_{\text{Kerr}} = 1 - \frac{2M}{r - ia \cos \theta}$	(404)
---	-------

This compact expression already contains the mass scale, frame-rotation parameter, asymptotic angular momentum dipole, static Schwarzschild limit, and the orbit-space structure underlying the Kerr horizon rod.

None of these facts constitutes a derivation of Kerr from the compact MS state relation. The Ernst equation is the stationary-axisymmetric form of the vacuum Einstein equations, and the Kerr solution is a classical exact solution of that equation. The response contribution is the preceding organization:

$\Re \mathcal{E} = \mathcal{R}(0), \quad d \Im \mathcal{E} = \star(k^b \wedge dk^b)$	(405)
--	-------

The next step is global rather than local. The stationary and axial Killing fields have now been normalized by the same asymptotic reference used to define the response spectrum. In vacuum their Killing two-forms generate closed Komar fluxes. The following section therefore asks how the local response-calibrated generators

$$k, \quad m \quad (406)$$

produce the global mass and angular momentum:

response-calibrated Killing geometry $\longrightarrow$ Komar surface charges	(407)
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5 Komar charges from response-calibrated Killing generators

The preceding sections have remained primarily local. The stationary response polynomial reconstructs the pointwise Killing-plane geometry, while its first spacetime jet, together with the vacuum Einstein equations, produces the twist potential and the Ernst field.

We now turn to global conserved quantities.

The stationary and axial Killing fields

$$k, \quad m \quad (408)$$

have already been calibrated by the asymptotic structure used throughout the paper. The stationary field is normalized by

$$g(k, k) \longrightarrow -1 \quad \text{at spatial infinity,} \quad (409)$$

while the axial field is normalized so that its regular orbits have period

$$2\pi. \quad (410)$$

These normalizations are essential. Komar charges are linear in the Killing generator, so a rescaling of k or m would rescale the corresponding surface integral. The asymptotic response calibration therefore fixes not only the stationary response spectrum but also the normalization of the global Killing generators used below.

Throughout this section we use geometrized units

$$G = c = 1. \quad (411)$$

5.1 The Komar functional on the Killing algebra

Let ξ be a Killing vector field and let

$$\xi^b := g(\xi, \cdot) \quad (412)$$

be its metric dual.

For an oriented closed two-surface S , define the Komar functional

$$\mathcal{K}_S[\xi] := -\frac{1}{8\pi} \int_S \star d\xi^b. \quad (413)$$

The sign convention is chosen so that the asymptotically normalized stationary Killing field gives positive mass for Schwarzschild and Kerr.

For the stationary Killing generator,

$$M_K[S] := \mathcal{K}_S[k] = -\frac{1}{8\pi} \int_S \star dk^b. \quad (414)$$

The standard Komar angular momentum is

$$J_K[S] := \frac{1}{16\pi} \int_S \star dm^b. \quad (415)$$

Hence

$$\mathcal{K}_S[m] = -2J_K[S]. \quad (416)$$

The factor of two is the familiar relative normalization between Komar mass and angular momentum.

Since $\mathcal{K}_S$ is linear in its Killing argument, the one-parameter stationary Killing family

$$\chi_\Omega = k + \Omega m \quad (417)$$

has the Komar profile

$$\begin{aligned} \mathcal{K}_S[\chi_\Omega] &= \mathcal{K}_S[k] + \Omega \mathcal{K}_S[m] \\ &= M_K[S] - 2\Omega J_K[S]. \end{aligned} \quad (418)$$

It is useful to denote

$$\mathcal{K}_S(\Omega) := \mathcal{K}_S[k + \Omega m] = M_K[S] - 2\Omega J_K[S]. \quad (419)$$

This is an affine function of Ω .

The same Killing family therefore carries two distinct structures:

$$\begin{aligned} \mathcal{R}_x(\Omega) &= -g_x(k + \Omega m, k + \Omega m), \\ \mathcal{K}_S(\Omega) &= -\frac{1}{8\pi} \int_S \star d(k + \Omega m)^b. \end{aligned} \quad (420)$$

The first is quadratic and local; the second is linear and global.

This distinction will be central to the horizon analysis.

5.2 Vacuum surface independence

For any Killing field ξ , the standard Killing identity implies

$$\nabla_\mu \nabla^\mu \xi_\nu = -R_{\nu\mu} \xi^\mu. \quad (421)$$

In differential-form language this gives

$$d \star d\xi^b = 2 \star (R_{\mu\nu} \xi^\nu dx^\mu), \quad (422)$$

with the orientation convention used in Eq. (413).

Hence in vacuum,

$$R_{\mu\nu} = 0, \quad (423)$$

we have

$$d \star d\xi^b = 0. \quad (424)$$

Let S_1 and S_2 be homologous closed two-surfaces bounding a three-dimensional region $\mathcal{V}$ contained in the vacuum domain. Stokes' theorem gives

$$\begin{aligned} \mathcal{K}_{S_2}[\xi] - \mathcal{K}_{S_1}[\xi] &= -\frac{1}{8\pi} \int_{\mathcal{V}} d \star d\xi^b \\ &= 0. \end{aligned} \quad (425)$$

Thus

$$\mathcal{K}_{S_2}[\xi] = \mathcal{K}_{S_1}[\xi] \quad (426)$$

for homologous surfaces in stationary vacuum.

In particular,

$$M_K[S] = \text{constant}, \quad J_K[S] = \text{constant} \quad (427)$$

throughout a connected vacuum region admitting the corresponding Killing fields.

This is the mechanism that will allow the Komar charge to be transported from spatial infinity to the horizon.

Proposition 5.1 (Vacuum conservation of the Komar profile). *In a stationary axisymmetric vacuum region,*

$$\mathcal{K}_S(\Omega) = M_K - 2\Omega J_K \quad (428)$$

is independent of the homologous integration surface S for every fixed constant Ω .

The qualifier “constant” is important. If Ω were allowed to vary over spacetime, then $k + \Omega m$ would not in general be a Killing field and the closure statement would no longer follow.

5.3 Asymptotic extraction of mass and angular momentum

The Komar charges are directly reflected in the large-distance stationary geometry. For a stationary asymptotically flat vacuum spacetime in its asymptotic rest frame,

$$g_{tt} = -1 + \frac{2M}{r} + O(r^{-2}), \quad (429)$$

$$g_{t\phi} = -\frac{2J \sin^2 \theta}{r} + O(r^{-2}). \quad (430)$$

Hence the stationary response gap satisfies

$$\mathcal{R}(0) = 1 - \frac{2M}{r} + O(r^{-2}). \quad (431)$$

Similarly, the response-spectral odd coefficient behaves as

$$\begin{aligned} \partial_\Omega \mathcal{R}(\Omega)|_{\Omega=0} &= -2g_{t\phi} \\ &= \frac{4J \sin^2 \theta}{r} + O(r^{-2}). \end{aligned} \quad (432)$$

Thus the leading asymptotic even and odd sectors of the stationary response spectrum encode respectively mass and angular momentum.

In Ernst language,

$$\Re \mathcal{E} = 1 - \frac{2M}{r} + O(r^{-2}), \quad (433)$$

$$\Im \mathcal{E} = -\frac{2J \cos \theta}{r^2} + O(r^{-3}), \quad (434)$$

with the twist orientation convention fixed in Sec. 4.

Standard asymptotic evaluation of the Komar integrals then gives

$$M_K = M, \quad J_K = J. \quad (435)$$

For Kerr,

$$M_K = M, \quad J_K = Ma. \quad (436)$$

For Schwarzschild,

$$M_K = M, \quad J_K = 0. \quad (437)$$

The quantities M and J that appeared in the exact metrics and Ernst potentials are therefore the corresponding global Killing charges once asymptotic normalization has been fixed.

5.4 Relation to the preceding asymptotic MS charge

The Komar construction should be distinguished from the canonical Hamiltonian boundary construction developed in the preceding paper.

There, an asymptotic translation a acquires a differentiable Hamiltonian through boundary completion:

$$G[a] = G_0[a] + B_\infty[a], \quad (438)$$

and the translation boundary functional defines

$$P_{\infty}^{\text{MS}} \in t_{\infty}^*. \quad (439)$$

After Einstein normalization, the corresponding positive-energy four-vector agrees with the ADM four-momentum.

The Komar construction begins instead with an exact Killing field ξ of a stationary spacetime and integrates its derivative:

$$\mathcal{K}[\xi] = -\frac{1}{8\pi} \int \star d\xi^{\flat}. \quad (440)$$

Thus

$$\begin{aligned} \text{canonical construction} &: \text{bulk constraint} + \text{boundary differentiability}, \\ \text{Komar construction} &: \text{exact Killing symmetry} + \text{closed Killing flux}. \end{aligned} \quad (441)$$

They are different mechanisms.

In a stationary asymptotically flat vacuum spacetime, however, the asymptotically normalized stationary Killing field approaches the asymptotic unit time translation used in the canonical construction. Standard general relativity then gives

$$M_K = M_{\text{ADM}}. \quad (442)$$

Combining this with the preceding MS–ADM identification yields

$$M_K = M_{\text{ADM}} = M_{\infty}^{\text{MS}} \quad (443)$$

in the stationary asymptotically flat Einstein sector and in its rest frame.

Equivalently, for the normalized stationary Killing translation,

$$B_{\infty}[k] = E_{\infty}^{\text{MS}} = M_{\text{ADM}} = M_K \quad (444)$$

in geometrized units.

This equality should not obscure the different logical constructions on its two sides.

The angular-momentum sector requires an additional qualification. The preceding MS boundary paper constructed only asymptotic translations. It did not construct the linearly growing rotational generator or its boundary completion. Therefore we do not write an “MS angular momentum” identity derived from that work.

Standard stationary general relativity gives, under the usual asymptotically flat parity and axisymmetry assumptions,

$$J_K = J_{\text{ADM}}. \quad (445)$$

But the construction of a corresponding rotational MS Hamiltonian charge is a separate problem.

Thus the precise status is

$$\begin{aligned} M_K &= M_{\text{ADM}} = M_{\infty}^{\text{MS}}, \\ J_K &= J_{\text{ADM}}, \quad \text{while no rotational MS boundary charge is constructed here.} \end{aligned} \quad (446)$$

5.5 The local quadratic spectrum and the global Komar line

The same family

$$\chi_\Omega = k + \Omega m \quad (447)$$

now carries two complementary functions:

$$\mathcal{R}_x(\Omega) = -g_x(\chi_\Omega, \chi_\Omega), \quad (448)$$

$$\mathcal{K}(\Omega) = M_K - 2\Omega J_K. \quad (449)$$

Their mathematical structures are different:

$$\deg_\Omega \mathcal{R} = 2, \quad \deg_\Omega \mathcal{K} = 1. \quad (450)$$

The response spectrum answers a local causal question:

$$\mathcal{R}_x(\Omega) \geq 0 \iff \chi_\Omega \text{ timelike/null/spacelike at } x. \quad (451)$$

The Komar profile answers a global charge question:

$$\mathcal{K}(\Omega) = \text{conserved Komar charge associated with } \chi_\Omega. \quad (452)$$

This gives a useful two-level structure:

$$\begin{aligned} \text{local Killing spectrum :} & \quad \mathcal{R}_x(\Omega), \\ \text{global Killing charge line :} & \quad \mathcal{K}(\Omega). \end{aligned} \quad (453)$$

A black-hole horizon links them by selecting a distinguished value of Ω .

5.6 The horizon-selected Komar combination

For a regular stationary rotating horizon, the local response spectrum selects the horizon angular velocity through

$$\mathcal{R}_H(\Omega_H) = 0, \quad \partial_\Omega \mathcal{R}_H(\Omega_H) = 0. \quad (454)$$

The corresponding Killing generator is

$$\chi_H = k + \Omega_H m. \quad (455)$$

The global Komar functional evaluated on exactly this locally selected generator is

$$\mathcal{K}[\chi_H] = M_K - 2\Omega_H J_K. \quad (456)$$

Thus

$$\underbrace{\mathcal{R}'_H(\Omega_H) = 0}_{\text{local response selection}} \implies \underbrace{\mathcal{K}[\chi_H] = M_K - 2\Omega_H J_K}_{\text{global horizon-generator charge}}. \quad (457)$$

This is the point at which the local response spectrum and the global Komar construction meet.

5.7 Schwarzschild

For Schwarzschild,

$$J_K = 0, \quad \Omega_H = 0. \quad (458)$$

Hence

$$\mathcal{K}[\chi_H] = M. \quad (459)$$

The Weyl horizon rod has length

$$L_H^{\text{Weyl}} = 2M. \quad (460)$$

Therefore

$$\mathcal{K}[\chi_H] = \frac{1}{2}L_H^{\text{Weyl}} \quad \text{for Schwarzschild.} \quad (461)$$

The next section will show that the same quantity is also

$$\frac{\kappa_H A_H}{4\pi}. \quad (462)$$

5.8 Kerr and the horizon-rod half-length

For Kerr,

$$M_K = M, \quad J_K = Ma, \quad (463)$$

while

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \quad (464)$$

Using the horizon identity

$$r_+^2 + a^2 = 2Mr_+, \quad (465)$$

we obtain

$$\begin{aligned} M - 2\Omega_H J &= M - 2 \frac{a}{r_+^2 + a^2} Ma \\ &= M - \frac{a^2}{r_+} \\ &= r_+ - M. \end{aligned} \quad (466)$$

Since

$$r_+ - M = \sqrt{M^2 - a^2}, \quad (467)$$

we find

$$M - 2\Omega_H J = \sqrt{M^2 - a^2}. \quad (468)$$

Recall from Sec. 3 that the Kerr horizon rod in Weyl orbit space has length

$$L_H^{\text{Weyl}} = 2\sqrt{M^2 - a^2}. \quad (469)$$

Therefore

$$M - 2\Omega_H J = \frac{1}{2}L_H^{\text{Weyl}}. \quad (470)$$

This identity gives a striking connection between three structures that entered the analysis independently:

$$\begin{aligned}
&\text{response double root} \longrightarrow \Omega_H, \\
&\text{global Komar functional} \longrightarrow M - 2\Omega_H J, \\
&\text{Weyl orbit geometry} \longrightarrow \frac{1}{2}L_H^{\text{Weyl}}.
\end{aligned} \tag{471}$$

For Kerr, all three combine into

$$M - 2\Omega_H J = \sqrt{M^2 - a^2} = \frac{1}{2}L_H^{\text{Weyl}}. \tag{472}$$

The first equality is the Komar charge of the horizon Killing generator; the second is the orbit-space rod relation.

Neither equality is a new Kerr identity. Their role here is to expose the common quantity selected by the response, Komar, and Weyl descriptions.

5.9 Extremality as vanishing horizon-generator Komar charge

Equation (472) has an immediate extremal limit.

For

$$|a| \longrightarrow M, \tag{473}$$

we have

$$\sqrt{M^2 - a^2} \longrightarrow 0. \tag{474}$$

Therefore

$$M - 2\Omega_H J \longrightarrow 0. \tag{475}$$

At the same time,

$$L_H^{\text{Weyl}} \longrightarrow 0. \tag{476}$$

Thus extremal Kerr simultaneously exhibits

$$\begin{aligned}
&\text{collapsed Weyl horizon rod,} \\
&\text{vanishing horizon-generator Komar charge.}
\end{aligned} \tag{477}$$

The next section will show that these are accompanied by

$$\kappa_H \longrightarrow 0. \tag{478}$$

For Kerr the common nonextremality scale is therefore

$$\sqrt{M^2 - a^2}. \tag{479}$$

This quantity already appeared as the Weyl rod half-length; it now appears as the Komar charge of the horizon generator. It will appear once more as the surface-gravity–area contribution in the Smarr identity.

5.10 Komar versus Ernst asymptotics

The Ernst and Komar descriptions provide complementary ways of reading the same asymptotic charges.

The Ernst field has asymptotic expansion

$$\mathcal{E} = 1 - \frac{2M}{r} - i \frac{2J \cos \theta}{r^2} + \dots, \quad (480)$$

while the Komar integrals return

$$M_K = M, \quad J_K = J. \quad (481)$$

Thus

$$\begin{aligned} \text{Ernst asymptotics} &\longrightarrow \text{multipolar coefficients,} \\ \text{Komar fluxes} &\longrightarrow \text{global conserved charges.} \end{aligned} \quad (482)$$

For the monopole and axial current dipole, the two descriptions identify the same M and J .

This agreement should not be interpreted as saying that all Ernst multipoles possess independent Komar surface integrals. Komar's construction applies directly to exact Killing generators; higher multipoles require different asymptotic definitions.

5.11 What has been added by the response organization

The Komar formulas themselves are classical [?]. Their vacuum surface independence and their equality with the standard stationary asymptotic charges are likewise standard general-relativistic results.

The response contribution developed in this paper is the chain

$$\begin{aligned} \text{massive response spectrum} &\longrightarrow \text{normalized Killing-plane geometry,} \\ \text{asymptotic response calibration} &\longrightarrow \text{normalization of } k \text{ and } m, \\ (k, m) &\longrightarrow (M_K, J_K), \\ \mathcal{R}'_H(\Omega_H) = 0 &\longrightarrow \chi_H = k + \Omega_H m, \\ \chi_H &\longrightarrow M_K - 2\Omega_H J_K. \end{aligned} \quad (483)$$

The local response spectrum therefore selects the Killing combination on which the global Komar functional must be evaluated at the horizon.

For Kerr this produces the particularly compact identity

$$\mathcal{K}[\chi_H] = M - 2\Omega_H J = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (484)$$

The remaining question is geometric: what does this conserved horizon-generator Komar charge become when evaluated directly on a regular Killing horizon?

For a horizon generator χ_H , surface gravity is defined by

$$\chi_H^\nu \nabla_\nu \chi_H^\mu = \kappa_H \chi_H^\mu \quad (485)$$

on the horizon. Equivalently,

$$\nabla_\mu (\chi_H^2) = -2\kappa_H \chi_H^\mu \quad (486)$$

there.

The next section evaluates the same Komar functional on a horizon cross-section and establishes

$$\mathcal{K}[\chi_H] = \frac{\kappa_H A_H}{4\pi}. \quad (487)$$

Combined with Eq. (456), this gives

$$M = 2\Omega_H J + \frac{\kappa_H A_H}{4\pi}, \quad (488)$$

the vacuum Smarr relation.

The horizon calculation will also show that, for Kerr,

$$\frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (489)$$

Thus response degeneracy, Komar conservation, Weyl rod geometry, and horizon mechanics converge on the same nonextremality scale.

6 Horizon response, surface gravity, and the Smarr identity

The preceding sections have produced two structures associated with the same stationary Killing family

$$\chi_\Omega = k + \Omega m. \quad (490)$$

Locally, its causal geometry is encoded by the quadratic response spectrum

$$\mathcal{R}_x(\Omega) = -g_x(\chi_\Omega, \chi_\Omega), \quad (491)$$

whereas globally its vacuum Komar charge is the affine function

$$\mathcal{K}(\Omega) = M_K - 2\Omega J_K. \quad (492)$$

A regular rotating Killing horizon connects these two objects. The response spectrum selects a distinguished angular velocity Ω_H , while the Komar functional evaluated on the corresponding Killing generator gives the combination

$$M_K - 2\Omega_H J_K. \quad (493)$$

The horizon geometry then converts this same quantity into a surface-gravity–area term.

The resulting chain is

$$\text{response double root} \longrightarrow \chi_H \longrightarrow \text{horizon Komar charge} \longrightarrow \kappa_H A_H \longrightarrow \text{Smarr balance}. \quad (494)$$

The Komar and Smarr identities used in this section are standard results of stationary vacuum general relativity [?, ?]. The response contribution is the preceding selection and interpretation of the horizon generator and its norm.

Throughout this section we use geometrized units

$$G = c = 1. \quad (495)$$

6.1 The horizon generator from response-root coalescence

Let $\mathcal{H}$ be a regular non-axis stationary axisymmetric Killing horizon. The stationary and axial Killing fields are normalized as in the preceding sections:

$$g(k, k) \longrightarrow -1 \quad \text{at spatial infinity,} \quad (496)$$

and the regular orbits of m have period 2π .

On the horizon there exists a Killing generator

$$\chi_H = k + \Omega_H m. \quad (497)$$

For a regular stationary axisymmetric black hole, Ω_H is constant over each connected horizon component.

From the response point of view, the same quantity is selected by the degenerate Killing-plane spectrum. Away from the rotation axis,

$$-\det G \longrightarrow 0 \quad (498)$$

on the horizon, so the two stationary-null angular velocities coalesce:

$$\Omega_+ = \Omega_- = \Omega_H. \quad (499)$$

Equivalently,

$$\mathcal{R}_H(\Omega_H) = 0, \quad \partial_\Omega \mathcal{R}_H(\Omega_H) = 0. \quad (500)$$

Since

$$\partial_\Omega \mathcal{R} = -2g(k + \Omega m, m), \quad (501)$$

the second condition gives

$$g(\chi_H, m) = 0 \quad (502)$$

on the horizon.

At a non-axis horizon point,

$$g(m, m) > 0, \quad (503)$$

and therefore

$$\Omega_H = - \frac{g(k, m)}{g(m, m)} \Big|_{\mathcal{H}}. \quad (504)$$

Thus the angular velocity of the horizon is already fixed at the zeroth-order response level.

This statement has a precise limitation. The local double-root condition identifies the null direction of the degenerate Killing plane. Its extension to a globally regular Killing horizon and the constancy of Ω_H require the usual stationary-horizon structure; they are not consequences of the compact MS state relation alone.

6.2 The horizon response gap

Once Ω_H has been selected, hold it fixed and define the spacetime scalar

$$\mathcal{G}_H(x) := \mathcal{R}_x(\Omega_H) = -g_x(\chi_H, \chi_H). \quad (505)$$

On the horizon,

$$\mathcal{G}_H = 0. \quad (506)$$

Where χ_H is future timelike,

$$\mathcal{G}_H = \mu(\chi_H)^2. \quad (507)$$

The quantity $\mathcal{G}_H$ should be distinguished from the optimized response

$$\max_{\Omega} \mathcal{R}_x(\Omega). \quad (508)$$

The latter permits the maximizing angular velocity to vary from point to point. By contrast, $\mathcal{G}_H$ evaluates the response of one globally fixed Killing field,

$$\chi_H = k + \Omega_H m. \quad (509)$$

Both vanish on a regular stationary horizon, but their off-horizon extensions are different. This distinction becomes important when defining surface gravity.

6.3 Surface gravity as a transverse response derivative

Surface gravity is defined by the standard Killing-horizon relation

$$\chi_H^\nu \nabla_\nu \chi_H^\mu = \kappa_H \chi_H^\mu \quad (510)$$

on $\mathcal{H}$.

Equivalently,

$$\nabla_\mu (\chi_H^\nu \chi_\nu^H) = -2\kappa_H \chi_\mu^H \quad (511)$$

on the horizon.

Since

$$\mathcal{G}_H = -\chi_H^2, \quad (512)$$

this becomes

$$\nabla_\mu \mathcal{G}_H = 2\kappa_H \chi_\mu^H \quad (513)$$

on $\mathcal{H}$.

Let n^μ be an auxiliary transverse null vector field on the horizon normalized by

$$g(\chi_H, n) = -1. \quad (514)$$

Contracting Eq. (513) with n^μ gives

$$\kappa_H = -\frac{1}{2} n^\mu \nabla_\mu \mathcal{G}_H \Big|_{\mathcal{H}}. \quad (515)$$

Thus surface gravity is the normalized transverse first derivative of the response gap belonging to the horizon-selected Killing generator.

This produces a useful two-directional hierarchy:

$$\begin{aligned}\partial_{\Omega}\mathcal{R}_H &\longrightarrow \Omega_H, \\ \nabla_{\perp}\mathcal{R}(\Omega_H) &\longrightarrow \kappa_H.\end{aligned}\tag{516}$$

The first derivative is taken in the one-dimensional Killing-family parameter space. The second is taken in spacetime transverse to the horizon.

Accordingly,

$$\begin{aligned}\Omega_H &= \text{angular spectral degeneracy scale}, \\ \kappa_H &= \text{transverse response-gradient scale}.\end{aligned}\tag{517}$$

The surface gravity is therefore not determined by the zeroth-order response polynomial at a single horizon point. It requires the first spacetime jet of the selected Killing norm.

6.4 Nonextremal and extremal response degeneracy

Equation (515) also separates nonextremal and extremal horizons.

For a nonextremal Killing horizon,

$$\kappa_H \neq 0,\tag{518}$$

so the selected horizon response gap has a nonvanishing normalized transverse first derivative:

$$n^{\mu}\nabla_{\mu}\mathcal{G}_H \neq 0.\tag{519}$$

For an extremal horizon,

$$\kappa_H = 0,\tag{520}$$

and hence

$$n^{\mu}\nabla_{\mu}\mathcal{G}_H = 0.\tag{521}$$

Thus an extremal horizon is more degenerate than a nonextremal one in two different senses:

$$\begin{aligned}\text{both:} \quad & \mathcal{R}_H(\Omega_H) = \partial_{\Omega}\mathcal{R}_H(\Omega_H) = 0, \\ \text{extremal only:} \quad & n^{\mu}\nabla_{\mu}\mathcal{R}(\Omega_H) = 0.\end{aligned}\tag{522}$$

The first line concerns degeneracy inside the Killing plane. The second adds degeneracy in the transverse spacetime direction.

This provides a response-theoretic characterization of the distinction between ordinary and extremal stationary horizons.

6.5 Komar evaluation on a nonextremal horizon

Let $\mathcal{B}$ be a bifurcation two-surface of a regular nonextremal Killing horizon generated by χ_H . On $\mathcal{B}$,

$$\chi_H = 0,\tag{523}$$

and its derivative is related to the binormal $\epsilon_{\mu\nu}^{\perp}$ of $\mathcal{B}$ by

$$\nabla_{\mu}\chi_H^H = \kappa_H\epsilon_{\mu\nu}^{\perp}.\tag{524}$$

With the orientation convention used in Sec. 5, the Komar functional is

$$\mathcal{K}_{\mathcal{B}}[\chi_H] = -\frac{1}{8\pi} \int_{\mathcal{B}} \star d\chi_H^{\flat}. \quad (525)$$

Using Eq. (524), one obtains

$$\mathcal{K}_{\mathcal{B}}[\chi_H] = \frac{\kappa_H A_H}{4\pi}, \quad (526)$$

where

$$A_H := \int_{\mathcal{B}} dA \quad (527)$$

is the horizon area.

The factor $1/(4\pi)$ arises from the factor of two between $d\chi^{\flat}$ and $\nabla_{\mu}\chi_{\nu}$.

This is a classical Killing-horizon identity. It does not depend on MS geometry.

6.6 Vacuum transport from infinity to the horizon

Because χ_H is a Killing field and the exterior region is vacuum,

$$d \star d\chi_H^{\flat} = 0. \quad (528)$$

Therefore its Komar integral is independent of the homologous integration surface.

Let S_{∞} be a sufficiently large sphere at spatial infinity. Then

$$\mathcal{K}_{S_{\infty}}[\chi_H] = \mathcal{K}_{\mathcal{B}}[\chi_H]. \quad (529)$$

At infinity,

$$\begin{aligned} \mathcal{K}_{S_{\infty}}[\chi_H] &= \mathcal{K}[k] + \Omega_H \mathcal{K}[m] \\ &= M_K - 2\Omega_H J_K. \end{aligned} \quad (530)$$

At the horizon,

$$\mathcal{K}_{\mathcal{B}}[\chi_H] = \frac{\kappa_H A_H}{4\pi}. \quad (531)$$

Hence

$$M_K - 2\Omega_H J_K = \frac{\kappa_H A_H}{4\pi}. \quad (532)$$

Rearranging gives the vacuum Smarr relation.

Theorem 6.1 (Response-selected Komar–Smarr balance). *Let an asymptotically flat stationary axisymmetric vacuum black hole possess a connected regular nonextremal Killing horizon. Let the stationary and axial Killing fields be normalized at infinity as above, and let Ω_H be the horizon angular velocity selected by the response double-root condition*

$$\mathcal{R}_H(\Omega_H) = \partial_{\Omega} \mathcal{R}_H(\Omega_H) = 0. \quad (533)$$

Then

$$M_K = 2\Omega_H J_K + \frac{\kappa_H A_H}{4\pi}. \quad (534)$$

The Smarr identity is classical. The response content of the theorem lies in the preceding selection of the Killing generator:

$$\mathcal{R}_H \longrightarrow \Omega_H \longrightarrow \chi_H \longrightarrow \mathcal{K}[\chi_H]. \quad (535)$$

The horizon geometry then supplies $\kappa_H A_H/(4\pi)$.

6.7 Three levels of the horizon response structure

The stationary black-hole construction can now be separated into three levels.

At zeroth order in spacetime derivatives,

$$\mathcal{R}_H(\Omega) \quad (536)$$

determines the causal structure of the Killing plane and selects the horizon angular velocity through its double root.

At first order transverse to the horizon,

$$\nabla_\perp \mathcal{R}(\Omega_H) \quad (537)$$

determines the surface gravity after the normalization of χ_H has been fixed.

Globally,

$$-\frac{1}{8\pi} \int \star d\chi_H^b \quad (538)$$

transports the selected Killing generator between infinity and the horizon.

Thus

$$\begin{aligned} \text{response 0-jet} &\longrightarrow \Omega_H, \\ \text{response transverse 1-jet} &\longrightarrow \kappa_H, \\ \text{Killing flux} &\longrightarrow M - 2\Omega_H J, \\ \text{horizon integration} &\longrightarrow \kappa_H A_H / (4\pi). \end{aligned} \quad (539)$$

This is the horizon counterpart of the Ernst hierarchy of the preceding section.

6.8 Schwarzschild

For Schwarzschild,

$$a = 0, \quad J = 0, \quad \Omega_H = 0. \quad (540)$$

The horizon radius is

$$r_H = 2M. \quad (541)$$

The horizon generator is simply

$$\chi_H = k. \quad (542)$$

The surface gravity is

$$\kappa_H = \frac{1}{4M}. \quad (543)$$

The horizon area is

$$A_H = 4\pi(2M)^2 = 16\pi M^2. \quad (544)$$

Therefore

$$\frac{\kappa_H A_H}{4\pi} = M. \quad (545)$$

The Smarr formula reduces to

$$M = \frac{\kappa_H A_H}{4\pi}. \quad (546)$$

From Sec. 2, the Schwarzschild horizon rod has Weyl length

$$L_H^{\text{Weyl}} = 2M. \quad (547)$$

Hence

$$M = \frac{\kappa_H A_H}{4\pi} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (548)$$

Thus the static response-gap rod, the horizon Komar charge, and the surface-gravity–area term all carry the same Schwarzschild scale.

6.9 Kerr surface gravity

We now evaluate the same structure for Kerr.

Recall

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}. \quad (549)$$

The horizon angular velocity is

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \quad (550)$$

Evaluation of the Killing-horizon definition gives the standard Kerr surface gravity

$$\kappa_H = \frac{r_+ - r_-}{2(r_+^2 + a^2)}. \quad (551)$$

Since

$$r_+ - r_- = 2\sqrt{M^2 - a^2}, \quad (552)$$

this is

$$\kappa_H = \frac{\sqrt{M^2 - a^2}}{r_+^2 + a^2}. \quad (553)$$

This makes explicit that the transverse first jet of the horizon response gap is controlled by the same nonextremality scale

$$\sqrt{M^2 - a^2} \quad (554)$$

that appeared in the Weyl rod and horizon-generator Komar charge.

6.10 Kerr horizon area

The induced metric on a spatial cross-section $r = r_+$ of the Kerr horizon has

$$q_{\theta\theta} = \Sigma_+, \quad (555)$$

$$q_{\phi\phi} = \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+}, \quad (556)$$

where

$$\Sigma_+ = r_+^2 + a^2 \cos^2 \theta. \quad (557)$$

Therefore

$$\sqrt{\det q} = (r_+^2 + a^2) \sin \theta. \quad (558)$$

Integration gives

$$A_H = 4\pi(r_+^2 + a^2). \quad (559)$$

Using

$$r_+^2 + a^2 = 2Mr_+, \quad (560)$$

one may also write

$$A_H = 8\pi Mr_+. \quad (561)$$

6.11 The Kerr nonextremality identity

Combining Eqs. (551) and (559), we obtain

$$\begin{aligned} \frac{\kappa_H A_H}{4\pi} &= \frac{r_+ - r_-}{2(r_+^2 + a^2)} \frac{4\pi(r_+^2 + a^2)}{4\pi} \\ &= \frac{r_+ - r_-}{2} \\ &= \sqrt{M^2 - a^2}. \end{aligned} \quad (562)$$

Section 5 gave independently

$$M - 2\Omega_H J = \sqrt{M^2 - a^2}, \quad (563)$$

while Sec. 3 gave

$$\frac{1}{2}L_H^{\text{Weyl}} = \sqrt{M^2 - a^2}. \quad (564)$$

We therefore obtain the central Kerr identity of the present organization:

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2}L_H^{\text{Weyl}}. \quad (565)$$

Each equality has a different geometric origin:

$M - 2\Omega_H J$	Komar charge of the horizon generator,	
$\frac{\kappa_H A_H}{4\pi}$	horizon Killing-flux evaluation,	
$\sqrt{M^2 - a^2}$	Kerr nonextremality parameter,	(566)
$\frac{1}{2}L_H^{\text{Weyl}}$	Weyl horizon-rod half-length.	

The role of the response spectrum is to select Ω_H through the coalescence of the two null stationary roots.

6.12 The Kerr Smarr relation

Using

$$J = Ma \quad (567)$$

and

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad (568)$$

the Komar identity becomes

$$M = 2\Omega_H J + \frac{\kappa_H A_H}{4\pi}. \quad (569)$$

This decomposition admits a useful response interpretation.

The term

$$2\Omega_H J \quad (570)$$

contains the angular velocity selected by the horizon response double root, while

$$\frac{\kappa_H A_H}{4\pi} \quad (571)$$

contains the transverse first-jet scale of the same selected horizon generator.

Thus the two terms entering the stationary vacuum Smarr balance arise from two different aspects of the horizon response geometry:

$$\begin{aligned} \Omega_H & \text{ from spectral coalescence in } \Omega, \\ \kappa_H & \text{ from transverse differentiation in spacetime.} \end{aligned} \quad (572)$$

The angular momentum J and area A_H , however, are additional global geometric quantities. They are not reconstructed from the pointwise response polynomial alone.

6.13 Extremal Kerr

For extremal Kerr,

$$|a| = M, \quad (573)$$

so

$$r_+ = r_- = M. \quad (574)$$

The surface gravity vanishes:

$$\kappa_H = 0. \quad (575)$$

The horizon area remains finite:

$$A_H = 8\pi M^2. \quad (576)$$

The angular velocity is

$$\Omega_H = \frac{1}{2M} \quad (577)$$

for $a = M$, and

$$J = M^2. \quad (578)$$

Hence

$$2\Omega_H J = M, \quad (579)$$

and

$$M - 2\Omega_H J = 0. \quad (580)$$

At the same time,

$$L_H^{\text{Weyl}} = 0. \quad (581)$$

Thus the fourfold Kerr identity has the extremal limit

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}} = 0. \quad (582)$$

This does not mean that the extremal horizon itself disappears. Its area, angular momentum, and total mass remain nonzero. What vanishes is the specific horizon-generator Komar combination and the associated nonextremality scale.

The response interpretation is therefore

$$\text{extremality} \iff \begin{cases} \text{transverse first jet of } -\chi_H^2 \text{ vanishes,} \\ \text{horizon-generator Komar charge vanishes,} \\ \text{finite Weyl horizon rod collapses.} \end{cases} \quad (583)$$

The angular double-root condition itself remains present in both nonextremal and extremal cases.

6.14 Response spectrum, horizon geometry, and Smarr balance

We can now assemble the complete stationary chain.

The massive response spectrum determines the Killing Gram matrix:

$$\mathcal{R}_x(\Omega) \longrightarrow G_{AB}. \quad (584)$$

Its discriminant gives the Weyl radial invariant:

$$\frac{1}{4} [(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}''] = \rho^2. \quad (585)$$

Its maximum selects the locally nonrotating direction:

$$\arg \max \mathcal{R} = \Omega_Z. \quad (586)$$

At a regular non-axis horizon, the two null roots merge:

$$\mathcal{R}_H(\Omega_H) = \mathcal{R}'_H(\Omega_H) = 0. \quad (587)$$

The selected generator is

$$\chi_H = k + \Omega_H m. \quad (588)$$

Its normalized transverse response derivative gives

$$\kappa_H = -\frac{1}{2} n^\mu \nabla_\mu \mathcal{R}(\Omega_H) \Big|_{\mathcal{H}}. \quad (589)$$

Its global Komar charge is

$$M - 2\Omega_H J. \quad (590)$$

Vacuum transport to the horizon gives

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi}. \quad (591)$$

Thus the complete structure is

$$\begin{aligned}
& \text{massive response spectrum} \longrightarrow \text{Killing Gram geometry} \\
& \longrightarrow \text{Weyl discriminant and frame dragging} \\
& \longrightarrow \text{horizon response double root} \\
& \longrightarrow \Omega_H \\
& \longrightarrow \chi_H \\
& \xrightarrow{\text{transverse first jet}} \kappa_H \\
& \xrightarrow{\text{Komar transport}} M - 2\Omega_H J \\
& = \frac{\kappa_H A_H}{4\pi}.
\end{aligned} \tag{592}$$

For Kerr this chain collapses further to

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \tag{593}$$

6.15 What is standard and what is new in the present organization

The following ingredients are standard stationary black-hole geometry:

$$\chi_H = k + \Omega_H m, \tag{594}$$

the Killing-horizon definition of κ_H , the Komar mass and angular momentum, vacuum surface independence of Komar fluxes, the horizon identity

$$\mathcal{K}[\chi_H] = \frac{\kappa_H A_H}{4\pi}, \tag{595}$$

and the Smarr formula

$$M = 2\Omega_H J + \frac{\kappa_H A_H}{4\pi}. \tag{596}$$

The Kerr formulas

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad \kappa_H = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \tag{597}$$

and

$$A_H = 4\pi(r_+^2 + a^2) \tag{598}$$

are likewise classical.

The new structural content proposed here is their organization through the MS response hierarchy:

$$\begin{aligned}
\Omega_H : & \quad \text{double root of the reconstructed stationary response spectrum,} \\
\kappa_H : & \quad \text{normalized transverse first derivative of the selected horizon response gap,} \\
M - 2\Omega_H J : & \quad \text{Komar charge of the response-selected Killing generator.}
\end{aligned} \tag{599}$$

The Smarr identity then becomes the global vacuum closure of these local and differential response data.

No claim is made that the compact MS state relation independently derives the black-hole rigidity theorem, the constancy of surface gravity, the Einstein vacuum equations, or the Smarr formula.

6.16 Smarr without thermodynamic interpretation

It is useful to state one further limitation explicitly.

The relation

$$M = 2\Omega_H J + \frac{\kappa_H A_H}{4\pi} \quad (600)$$

is used here as a classical geometric identity among stationary vacuum charges and horizon data.

We do not identify

$$\kappa_H \quad (601)$$

with temperature, nor

$$A_H \quad (602)$$

with entropy, within the present MS construction.

Likewise, the differential first law

$$\delta M = \frac{\kappa_H}{8\pi} \delta A_H + \Omega_H \delta J \quad (603)$$

is not required for any result in this paper.

The Smarr identity is retained because it follows directly from the global Killing-charge structure studied here and closes the response–Komar chain without introducing quantum or thermodynamic assumptions.

6.17 Transition to the discussion

The stationary MS construction developed in this paper may now be summarized in two intertwined chains.

The local differential chain is

$$\mathcal{R}(\Omega) \longrightarrow G_{AB} \longrightarrow (f, \rho, \omega_P) \longrightarrow \varpi \longrightarrow \psi \longrightarrow \mathcal{E}. \quad (604)$$

The horizon-charge chain is

$$\mathcal{R}_H(\Omega) \longrightarrow \Omega_H \longrightarrow \chi_H \longrightarrow \kappa_H \longrightarrow \mathcal{K}[\chi_H] \longrightarrow \text{Smarr}. \quad (605)$$

Schwarzschild is the real, static branch in which the response center is zero, the stationary-limit and horizon conditions coincide, the Ernst potential is real, and the horizon rod has length $2M$.

Kerr is the genuinely stationary branch in which the response spectrum is asymmetric, its maximum is shifted by frame dragging, the ergosurface and horizon are spectrally distinct, the Ernst potential is complex, and the same nonextremality scale appears as

$$\sqrt{M^2 - a^2} = M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (606)$$

The final section discusses the scope of these relations, the precise boundary between MS-derived structure and standard stationary general relativity, and several extensions suggested by the response formulation, including electrovacuum Ernst systems, Kerr–Newman geometry, multipole spectroscopy, and more general horizon-response problems.

7 Discussion and outlook

The purpose of this paper has been to develop a stationary extension of the motion–staticity response programme and to test it against the classical geometry of Schwarzschild and Kerr.

The central object introduced here is the stationary response polynomial

$$\mathcal{R}_x(\Omega) := -g_x(k + \Omega m, k + \Omega m), \quad (607)$$

associated with a commuting stationary–axial Killing pair

$$(k, m). \quad (608)$$

On the interval for which

$$k + \Omega m \quad (609)$$

is future timelike, the preceding MS response-polar construction gives

$$\mathcal{R}_x(\Omega) = \mu_x(k + \Omega m)^2. \quad (610)$$

Once this timelike response data have reconstructed the quadratic polynomial, the same Lorentzian quadratic form extends algebraically to its null and spacelike Killing directions.

This one-parameter spectrum organizes a substantial part of stationary geometry.

Because

$$\mathcal{R}(\Omega) = -g_{tt} - 2\Omega g_{t\phi} - \Omega^2 g_{\phi\phi}, \quad (611)$$

the complete metric on the Killing plane is recovered from $\mathcal{R}$. In particular,

$$g_{tt} = -\mathcal{R}(0), \quad (612)$$

$$g_{t\phi} = -\frac{1}{2}\mathcal{R}'(0), \quad (613)$$

$$g_{\phi\phi} = -\frac{1}{2}\mathcal{R}''(0). \quad (614)$$

Equivalently, three distinct massive response values at three timelike angular velocities reconstruct the three independent components of the Killing Gram matrix.

This is the first level of the stationary response construction:

$$\text{massive response spectrum} \longrightarrow \text{Killing-plane Lorentzian geometry}. \quad (615)$$

A second invariant follows from the discriminant of the same quadratic:

$$\rho^2 = -\det G = \frac{1}{4}[(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}'']. \quad (616)$$

In the stationary axisymmetric vacuum sector this becomes the canonical Weyl–Papapetrou radial coordinate.

Thus a standard orbit-space invariant acquires the response interpretation

$$\rho^2 = \frac{1}{4} \text{disc}_\Omega \mathcal{R}. \quad (617)$$

This relation is entirely algebraic in the one-parameter Killing family. It does not require space-time derivatives or the vacuum field equations.

A further consequence is the spectral characterization of locally nonrotating observers. Away from the rotation axis,

$$g_{\phi\phi} > 0, \quad (618)$$

so $\mathcal{R}$ is strictly concave and has a unique maximum:

$$\Omega_Z = \arg \max_{\Omega} \mathcal{R}(\Omega) = -\frac{g_{t\phi}}{g_{\phi\phi}}. \quad (619)$$

The normalized Killing direction

$$k + \Omega_Z m \quad (620)$$

is orthogonal to the axial generator and gives the standard zero-angular-momentum observer direction.

Frame dragging is therefore encoded by displacement of the response maximum away from $\Omega = 0$.

The two null stationary angular velocities are the roots of

$$\mathcal{R}(\Omega) = 0. \quad (621)$$

Their midpoint is Ω_Z , while their separation is controlled by ρ :

$$\Omega_+ - \Omega_- = \frac{2\rho}{g_{\phi\phi}}. \quad (622)$$

Thus the center and width of the allowed stationary-observer interval have direct geometric meanings.

These observations are kinematical. No Einstein equation is required.

7.1 Static response geometry and the Weyl potential

The static sector provides the simplest differential completion.

For a hypersurface-orthogonal stationary Killing field in a static axisymmetric representation,

$$g(k, m) = 0, \quad (623)$$

and the response spectrum becomes even:

$$\mathcal{R}(\Omega) = f - g_{\phi\phi}\Omega^2, \quad f := -g(k, k). \quad (624)$$

Where k is future timelike,

$$\mu(k) = \sqrt{f}. \quad (625)$$

The static response potential introduced in Sec. 2 is

$$U = \ln \mu(k) = \frac{1}{2} \ln \mathcal{R}(0). \quad (626)$$

Only after the vacuum Einstein equations are imposed does this potential satisfy

$$\left(\partial_\rho^2 + \frac{1}{\rho} \partial_\rho + \partial_z^2 \right) U = 0. \quad (627)$$

This distinction is important.

The relation

$$U = \ln \mu(k) \quad (628)$$

belongs to the response interpretation developed here.

The harmonic equation

$$\Delta_{\text{ax}} U = 0 \quad (629)$$

belongs to standard static vacuum Einstein geometry.

For Schwarzschild these two structures combine into

$$\ln \mu_{\text{Schw}}(k) = -\frac{1}{2} \int_{-M}^M \frac{d\zeta}{\sqrt{\rho^2 + (z - \zeta)^2}}. \quad (630)$$

Thus the logarithm of the stationary response amplitude has the auxiliary Weyl potential of a finite rod of length

$$2M. \quad (631)$$

The rod is not a distribution of local massive MS generators. It is an auxiliary source representation of the harmonic potential arising in the Weyl reduction of the global vacuum geometry.

This distinction preserves the separation already required in the preceding asymptotic MS analysis between local massive generator scales and global gravitational mass.

7.2 Rotation as spectral asymmetry

Kerr makes the stationary response construction substantially richer.

Its exact response spectrum may be written

$$\mathcal{R}_{\text{Kerr}}(\Omega) = \alpha^2 - g_{\phi\phi} (\Omega - \omega)^2, \quad (632)$$

where

$$\omega = -\frac{g_{t\phi}}{g_{\phi\phi}} \quad (633)$$

is the locally nonrotating angular velocity and

$$\alpha \quad (634)$$

is the Kerr lapse.

Therefore

$$\omega = \arg \max_{\Omega} \mathcal{R}_{\text{Kerr}}, \quad (635)$$

and

$$\alpha^2 = \max_{\Omega} \mathcal{R}_{\text{Kerr}}. \quad (636)$$

This gives the standard Kerr lapse and frame-dragging angular velocity a common response-spectral interpretation.

It also separates two response gaps:

$$\mathcal{R}_{\text{stat}} = \mathcal{R}(0) = -g(k, k), \quad (637)$$

$$\mathcal{R}_{\text{opt}} = \max_{\Omega} \mathcal{R}(\Omega). \quad (638)$$

For Schwarzschild,

$$\mathcal{R}_{\text{stat}} = \mathcal{R}_{\text{opt}}, \quad (639)$$

because the response center is zero.

For Kerr,

$$\mathcal{R}_{\text{stat}} \neq \mathcal{R}_{\text{opt}}, \quad (640)$$

and their zeros define different surfaces.

The stationary-limit surface satisfies

$$\mathcal{R}(0) = 0, \quad \mathcal{R}_{\text{max}} > 0, \quad (641)$$

so the zero-angular-velocity stationary direction has become null while other co-rotating stationary timelike directions still exist.

By contrast, at a regular non-axis horizon,

$$\mathcal{R}_{\text{max}} = 0. \quad (642)$$

Equivalently, the two stationary null roots coalesce:

$$\mathcal{R}_H(\Omega_H) = \partial_\Omega \mathcal{R}_H(\Omega_H) = 0. \quad (643)$$

This yields a compact distinction:

$$\begin{aligned} \text{ergosurface} & : \text{one null root crosses } \Omega = 0, \\ \text{horizon} & : \text{the two null roots merge.} \end{aligned} \quad (644)$$

In Schwarzschild these phenomena coincide because $\Omega_Z = 0$. In Kerr rotation separates them.

7.3 From response geometry to the Ernst system

The response polynomial reconstructs the pointwise Killing Gram matrix, but stationary rotation also contains first-derivative information.

The stationary twist one-form is

$$\varpi = \star(k^b \wedge dk^b). \quad (645)$$

This is a first-jet quantity and cannot be recovered from the value of the Killing Gram matrix at one point alone.

In vacuum,

$$d\varpi = 0, \quad (646)$$

so locally

$$d\psi = \varpi. \quad (647)$$

The Ernst potential is then

$$\mathcal{E} = f + i\psi, \quad f = \mathcal{R}(0). \quad (648)$$

This gives what may be regarded as the differential completion of the stationary response spectrum:

$$\Re \mathcal{E} = \mathcal{R}(0), \quad d \Im \mathcal{E} = \star(k^b \wedge dk^b). \quad (649)$$

The two sides belong to different levels.

The real part is a zeroth-order response quantity.

The imaginary part requires the first spacetime jet and vacuum integrability.

In Weyl–Papapetrou variables, the metric coefficient ω_P is already determined algebraically by the response spectrum:

$$\omega_P = -\frac{\partial_\Omega \mathcal{R}(0)}{2\mathcal{R}(0)}. \quad (650)$$

Its spacetime derivatives determine the twist potential through the standard stationary vacuum equations.

Thus the stationary hierarchy is

$$\begin{aligned} \mathcal{R}(\Omega) &\longrightarrow G_{AB}, \\ G_{AB} &\longrightarrow (f, \rho, \omega_P), \\ \nabla G_{AB} &\longrightarrow \varpi, \\ R_{\mu\nu} = 0 &\longrightarrow d\varpi = 0, \\ \varpi &\longrightarrow \psi, \\ (f, \psi) &\longrightarrow \mathcal{E}. \end{aligned} \quad (651)$$

Schwarzschild occupies the real branch

$$\psi = 0, \quad (652)$$

whereas Kerr gives the complex branch

$$\mathcal{E}_{\text{Kerr}} = 1 - \frac{2M}{r - ia \cos \theta}. \quad (653)$$

The compactness of this expression makes the relation between mass and rotation particularly transparent, but it remains a classical exact solution of the vacuum Ernst equation rather than an MS derivation of Kerr.

7.4 Local response spectra and global Killing charges

The second major transition in the paper is from local stationary geometry to global conserved Killing charges.

For

$$\chi_\Omega = k + \Omega m, \quad (654)$$

the local response spectrum is

$$\mathcal{R}_x(\Omega) = -g_x(\chi_\Omega, \chi_\Omega), \quad (655)$$

while the Komar functional is

$$\mathcal{K}_S(\Omega) = -\frac{1}{8\pi} \int_S \star d\chi_\Omega^b = M_K[S] - 2\Omega J_K[S]. \quad (656)$$

The first function is quadratic in Ω :

$$\deg_\Omega \mathcal{R} = 2. \quad (657)$$

The second is affine:

$$\deg_\Omega \mathcal{K} = 1. \quad (658)$$

This gives a compact local–global duality of stationary Killing information:

local	global	
$\mathcal{R}_x(\Omega)$	$\mathcal{K}_S(\Omega)$	(659)
quadratic Killing norm	linear Killing charge	
causal spectrum	conserved flux	

In vacuum,

$$d \star d\xi^b = 0 \quad (660)$$

for every Killing field ξ , so the Komar functional is invariant under homologous deformation of the integration surface.

This mechanism is different from the canonical boundary completion used in the preceding MS asymptotic paper.

There,

$$\text{bulk constraint} + \text{functional differentiability} \quad (661)$$

produced an asymptotic Hamiltonian boundary charge.

Here,

$$\text{exact Killing symmetry} + \text{vacuum closed flux} \quad (662)$$

produces a Komar charge.

In the common stationary asymptotically flat Einstein sector, standard general relativity gives

$$M_K = M_{\text{ADM}}. \quad (663)$$

Combining this with the preceding MS–ADM comparison gives

$$M_K = M_{\text{ADM}} = M_{\infty}^{\text{MS}}. \quad (664)$$

This equality does not mean that the three constructions are identical. They arrive at the same stationary mass through different geometric and Hamiltonian mechanisms.

The angular momentum requires a further caution. The preceding MS asymptotic paper constructed only the translation sector. It did not construct the boundary generator of asymptotic rotations.

Therefore, although standard GR gives

$$J_K = J_{\text{ADM}} \quad (665)$$

under the appropriate stationary asymptotic hypotheses, no independent “MS rotational boundary charge” has been derived in the present sequence of papers.

7.5 The horizon as a meeting point

A regular stationary horizon provides the point where the local quadratic spectrum and global affine charge line meet.

The response double root selects

$$\Omega_H. \quad (666)$$

This defines

$$\chi_H = k + \Omega_H m. \quad (667)$$

The corresponding global Komar value is

$$\mathcal{K}(\Omega_H) = M - 2\Omega_H J. \quad (668)$$

Holding this Ω_H fixed, define the horizon response gap

$$\mathcal{G}_H(x) := -g_x(\chi_H, \chi_H) = \mathcal{R}_x(\Omega_H). \quad (669)$$

It vanishes on the horizon.

If n is a transverse null vector normalized by

$$g(\chi_H, n) = -1, \quad (670)$$

then the standard Killing-horizon identity gives

$$\kappa_H = -\frac{1}{2}n^\mu \nabla_\mu \mathcal{G}_H \Big|_{\mathcal{H}}. \quad (671)$$

The horizon therefore carries two distinct response derivatives:

$$\begin{aligned} \partial_\Omega \mathcal{R}_H &\longrightarrow \Omega_H, \\ \nabla_\perp \mathcal{R}(\Omega_H) &\longrightarrow \kappa_H. \end{aligned} \quad (672)$$

The first acts in Killing-parameter space.

The second acts in spacetime.

This separation is conceptually useful because surface gravity is not encoded in the zeroth-order double root alone. It belongs to the transverse first jet of the selected horizon Killing norm.

Vacuum Komar transport then gives

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi}, \quad (673)$$

or

$$M = 2\Omega_H J + \frac{\kappa_H A_H}{4\pi}. \quad (674)$$

This is the classical vacuum Smarr relation.

The new structural interpretation is that the Killing generator entering the Komar–Smarr identity is selected by the local response spectrum before the global flux calculation is performed.

7.6 The Kerr fourfold nonextremality identity

For Kerr,

$$J = Ma, \quad (675)$$

and

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \quad (676)$$

The horizon-generator Komar charge is

$$M - 2\Omega_H J = \sqrt{M^2 - a^2}. \quad (677)$$

The surface-gravity–area combination gives independently

$$\frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2}. \quad (678)$$

The Weyl horizon rod has length

$$L_H^{\text{Weyl}} = 2 \sqrt{M^2 - a^2}. \quad (679)$$

Thus the different structures developed in the paper converge on

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (680)$$

The four terms have different origins:

$$\begin{aligned} M - 2\Omega_H J & : \text{Komar charge of } \chi_H, \\ \kappa_H A_H / (4\pi) & : \text{horizon Killing-flux value,} \\ \sqrt{M^2 - a^2} & : \text{Kerr nonextremality parameter,} \\ L_H^{\text{Weyl}} / 2 & : \text{Weyl horizon-rod half-length.} \end{aligned} \quad (681)$$

The response spectrum supplies the common horizon angular velocity Ω_H through root coalescence and thereby selects the Komar combination that participates in this identity.

Equation (680) is not a new Kerr formula in the sense that each equality can be obtained within standard stationary black hole geometry. Its significance here is the unification of four descriptions through the response-selected horizon generator.

The extremal limit is especially transparent:

$$|a| \rightarrow M \implies M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \frac{1}{2} L_H^{\text{Weyl}} = 0. \quad (682)$$

The horizon itself does not disappear:

$$M \neq 0, \quad J \neq 0, \quad A_H \neq 0. \quad (683)$$

What vanishes is the common nonextremality scale.

In response terms, extremality adds a transverse degeneracy to the Killing-plane double root:

$$\begin{aligned} \mathcal{R}_H(\Omega_H) &= 0, \\ \partial_\Omega \mathcal{R}_H(\Omega_H) &= 0, \\ n^\mu \nabla_\mu \mathcal{R}(\Omega_H)|_H &= 0. \end{aligned} \quad (684)$$

The first two conditions also hold for a nonextremal stationary horizon; the third distinguishes the extremal case.

7.7 What is genuinely new in the present paper

The classical objects appearing in the paper should be separated carefully from the response organization introduced here.

The following are standard results of stationary general relativity:

- the Schwarzschild and Kerr metrics;
- Weyl–Papapetrou coordinates and the static Weyl reduction;
- the stationary twist one-form and the Ernst equation;
- the Kerr Ernst potential;
- the Komar mass and angular momentum;
- the equality of Komar and ADM mass in the stationary asymptotically flat vacuum sector;
- the Kerr ergosurface, horizon angular velocity, surface gravity, area, and Weyl rod structure;
- the Komar horizon identity and the Smarr relation.

The present paper does not claim these classical formulas as new general-relativistic results.

The MS-specific and current-paper structural contributions are instead:

1. the introduction of the stationary massive-response spectrum

$$\mathcal{R}_x(\Omega) = -g_x(k + \Omega m, k + \Omega m), \quad (685)$$

interpreted directly as μ_x^2 on its future-timelike interval;

2. reconstruction of the full Killing Gram matrix from this one-parameter response spectrum;
3. interpretation of

$$-\det G \quad (686)$$

as the discriminant of the response polynomial;

4. characterization of the locally nonrotating observer as

$$\arg \max_{\Omega} \mathcal{R}; \quad (687)$$

5. separation of the ergosurface and horizon as two different spectral phenomena,

$$\mathcal{R}(0) = 0 \quad (688)$$

versus

$$\mathcal{R}(\Omega_H) = \mathcal{R}'(\Omega_H) = 0; \quad (689)$$

6. interpretation of the static Weyl potential as

$$U = \ln \mu(k); \quad (690)$$

7. organization of the Ernst potential into the response hierarchy

$$\Re \mathcal{E} = \mathcal{R}(0), \quad d \Im \mathcal{E} = \star(k^b \wedge dk^b); \quad (691)$$

8. the pairing of the local quadratic response spectrum $\mathcal{R}_x(\Omega)$ with the global affine Komar profile

$$\mathcal{K}(\Omega) = M - 2\Omega J; \quad (692)$$

9. interpretation of Ω_H as an angular spectral degeneracy and κ_H as a transverse first-jet scale of the same selected Killing response;

10. the organization of the Kerr relation

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}} \quad (693)$$

as a common endpoint of the response, Komar, horizon, and Weyl descriptions.

These claims are organizational and reconstructive rather than claims of a new black-hole solution or new Einstein identity.

7.8 What has not been derived from MS

Several limitations are fundamental.

First, the stationary vacuum Einstein equations have not been derived in this paper from the compact MS state relation. The Lorentzian metric is taken from the preceding response-soldering construction, while the vacuum Einstein dynamics used in the Weyl and Ernst reductions remains the minimal metric gravitational branch already identified in the preceding MS work.

Second, the existence of the stationary and axial Killing fields is an additional symmetry assumption. Neither stationarity nor axisymmetry follows from the compact state relation.

Third, the rigidity theorem is not derived. The statement that a rotating stationary black-hole horizon is generated by

$$k + \Omega_H m \quad (694)$$

belongs to the standard stationary black-hole framework under its usual hypotheses.

Fourth, the zeroth law,

$$\kappa_H = \text{constant on } \mathcal{H}, \quad (695)$$

is not derived from response geometry alone. The response formula

$$\kappa_H = -\frac{1}{2} n^\mu \nabla_\mu \mathcal{R}(\Omega_H) \quad (696)$$

re-expresses the standard surface gravity once a regular Killing horizon has been supplied.

Fifth, the Kerr uniqueness theorem has not been proved or reformulated. Kerr is used as the canonical exact rotating solution on which the stationary response hierarchy can be evaluated.

Sixth, no microscopic origin of the gravitational coupling is supplied. As in the preceding asymptotic MS paper, the Einstein normalization remains an empirical gravitational input.

Seventh, no thermodynamic interpretation is assumed. In particular, this paper does not derive

$$T_H = \frac{\hbar \kappa_H}{2\pi c k_B} \quad (697)$$

or

$$S_{\text{BH}} = \frac{k_B c^3 A_H}{4G\hbar}. \quad (698)$$

The Smarr formula is used as a classical stationary Killing-charge identity, not as a derivation of black-hole thermodynamics.

Finally, the local one-particle MS shell scale Q should not be identified with any horizon rod density, Komar mass density, or black-hole mass. The same Lorentzian response geometry appears at different levels of the construction, but the associated physical quantities remain distinct.

7.9 Electrovacuum and Kerr–Newman

The most immediate extension is stationary axisymmetric electrovacuum.

There the gravitational Ernst potential couples to a complex electromagnetic potential. Schematically,

$$(\mathcal{E}, \Phi) \tag{699}$$

replaces the single vacuum Ernst field.

The response polynomial

$$\mathcal{R}(\Omega) \tag{700}$$

would continue to reconstruct the Killing-plane metric, while the electromagnetic field introduces additional local gauge-invariant response data and modifies both the Ernst equations and the global horizon balance.

For Kerr–Newman one expects the horizon relation

$$M = 2\Omega_H J + \Phi_H Q_e + \frac{\kappa_H A_H}{4\pi} \tag{701}$$

in geometrized units.

This would provide a useful test of whether the response hierarchy can accommodate simultaneously:

$$\text{kinematic Killing response} + \text{electromagnetic gauge response} + \text{horizon charge balance.} \tag{702}$$

Such an extension would be substantially stronger than simply inserting the Kerr–Newman metric into the formulas above.

7.10 Stationary multipole spectroscopy

The Ernst representation suggests a second natural direction.

For Kerr, the Cayley-transformed Ernst field

$$\xi = \frac{1 - \mathcal{E}}{1 + \mathcal{E}} \tag{703}$$

takes the simple axis form

$$\xi_{\text{Kerr}}^{\text{axis}}(z) = \frac{M}{z - ia}. \tag{704}$$

Its asymptotic expansion is

$$\xi_{\text{Kerr}}^{\text{axis}}(z) = \frac{M}{z} + \frac{iMa}{z^2} - \frac{Ma^2}{z^3} - \frac{iMa^3}{z^4} + \dots \tag{705}$$

Within the standard Geroch–Hansen formalism this structure is related to the Kerr multipole hierarchy

$$M_\ell + iS_\ell = M(ia)^\ell. \tag{706}$$

A future response formulation could ask whether these multipoles can be reconstructed directly from the asymptotic angular and spacetime jets of $\mathcal{R}_x(\Omega)$ and its twist completion.

The natural hierarchy would be

stationary response spectrum $\longrightarrow$ asymptotic response jets $\longrightarrow$ Ernst axis data $\longrightarrow$ mass-current multipoles. (707)

Such a construction would provide a genuine “stationary response spectroscopy” beyond the mass and angular momentum sectors studied here.

7.11 Solution-generating symmetries

The Ernst equations possess a rich nonlinear solution-generating structure. Fractional-linear transformations, Ehlers transformations, Bäcklund methods, inverse-scattering techniques, and the Geroch-group viewpoint connect families of stationary solutions.

The response representation raises a natural question:

How do the nonlinear symmetries of the Ernst system act directly on the stationary response polynomial $\mathcal{R}(\Omega)$ and its first-jet twist completion?

The zeroth-order Killing-plane response is quadratic in Ω , while the Ernst field transforms nonlinearly. Understanding the induced action on

$$(\mathcal{R}, d\psi) \tag{708}$$

could provide a useful response-theoretic formulation of stationary solution generation.

No such construction is attempted here.

7.12 Extremal and near-horizon response geometry

The extremal limit suggests another direction.

For nonextremal Kerr,

$$L_H^{\text{Weyl}} > 0, \quad \kappa_H > 0. \tag{709}$$

For extremal Kerr,

$$L_H^{\text{Weyl}} = 0, \quad \kappa_H = 0. \tag{710}$$

The horizon response retains its angular double root but acquires an additional transverse first-jet degeneracy.

This suggests organizing extremality by the order of vanishing of the horizon response gap

$$\mathcal{G}_H = -\chi_H^2. \tag{711}$$

A systematic near-horizon analysis could ask whether the near-horizon extremal Kerr geometry can be characterized by a universal scaling limit of the response spectrum and its transverse derivatives.

Such a construction may provide a cleaner response-theoretic distinction between ordinary horizon degeneracy and extremal throat formation.

7.13 Beyond orthogonal transitivity

The Weyl–Papapetrou and Ernst analysis used here assumes the standard orthogonally transitive stationary axisymmetric setting.

More general two-Killing geometries need not admit the same reduction. It would therefore be useful to determine how much of the stationary response construction survives without circularity.

The algebraic results

$$\mathcal{R}(\Omega) \longrightarrow G_{AB} \quad (712)$$

and

$$\rho^2 = -\det G \quad (713)$$

do not depend on orthogonal transitivity.

By contrast, the canonical Weyl–Papapetrou form and scalar Ernst description do.

This separates a potentially broader MS stationary geometry from the special integrable Einstein sector:

$$\text{Killing response geometry} \supset \text{Weyl–Ernst geometry}. \quad (714)$$

Exploring this larger class may be more informative than restricting future work entirely to known exact black-hole solutions.

7.14 Possible relation to the full asymptotic Poincaré sector

There is also a direct connection to the unresolved problem in the preceding asymptotic MS paper.

That work constructed only the translation boundary charge. The present stationary paper uses the exact axial Killing generator m and its standard Komar angular momentum, but it does not derive the corresponding MS rotational Hamiltonian generator.

A future canonical analysis should therefore allow asymptotic rotation and boost smearings and construct the complete Poincaré boundary sector:

$$\text{translations} + \text{rotations} + \text{boosts}. \quad (715)$$

Such a construction would make it possible to derive, within one MS Hamiltonian framework,

$$E, \quad P_i, \quad J_i, \quad C_i, \quad (716)$$

rather than importing the angular-momentum sector from standard stationary GR.

It would also provide the proper setting in which to compare the canonical rotational boundary charge directly with the Komar angular momentum used in the present paper.

7.15 Final synthesis

The stationary extension developed here can be summarized by two interlocking chains.

The first is local and differential:

$$\begin{aligned} \text{massive MS response} &\longrightarrow \mathcal{R}_x(\Omega) \\ &\longrightarrow G_{AB} \\ &\longrightarrow (f, \rho, \omega_P) \\ &\longrightarrow \varpi \\ &\xrightarrow{R_{\mu\nu}=0} \psi \\ &\longrightarrow \mathcal{E}. \end{aligned} \quad (717)$$

The second is global and horizon based:

$$\begin{aligned}
\mathcal{R}_H(\Omega) &\longrightarrow \Omega_H \\
&\longrightarrow \chi_H = k + \Omega_H m \\
&\xrightarrow{\nabla_\perp} \kappa_H \\
&\xrightarrow{\text{Komar flux}} M - 2\Omega_H J \\
&= \frac{\kappa_H A_H}{4\pi}.
\end{aligned} \tag{718}$$

Schwarzschild is the static branch:

$$\Omega_Z = \Omega_H = 0, \quad \psi = 0, \tag{719}$$

with

$$M = \frac{\kappa_H A_H}{4\pi} = \frac{1}{2} L_H^{\text{Weyl}}. \tag{720}$$

Kerr is the rotating complex branch:

$$\Omega_Z \neq 0, \quad \Omega_H \neq 0, \quad \psi \neq 0, \tag{721}$$

with the characteristic synthesis

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \tag{722}$$

The compact MS relation alone does not produce these stationary black-hole structures. Their appearance requires the complete sequence

MS massive-generator response

$$\begin{aligned}
&\longrightarrow \text{soldered Lorentzian geometry} \\
&\longrightarrow \text{stationary Killing reduction} \\
&\longrightarrow \text{vacuum Weyl–Ernst dynamics} \\
&\longrightarrow \text{Komar conservation} \\
&\longrightarrow \text{horizon geometry and Smarr balance.}
\end{aligned} \tag{723}$$

Within that chain, the distinctive stationary MS contribution is the response spectrum

$$\mathcal{R}_x(\Omega) = -g_x(k + \Omega m, k + \Omega m), \tag{724}$$

which turns redshift, dragging, the stationary causal interval, Weyl discriminant, ergosurface, and horizon generator into different features of one Lorentzian response profile.

The resulting picture may be stated succinctly:

Schwarzschild and Kerr differ not by the existence of a stationary response spectrum, but by its displacement, asymptotically (725)

This provides the stationary counterpart of the preceding MS asymptotic translation analysis and opens a natural route from local response geometry toward stationary multipoles, electrovacuum black holes, and the full asymptotic rotational sector.

A Algebra of the stationary response polynomial

This appendix collects the algebraic identities associated with the stationary response polynomial

$$\mathcal{R}_x(\Omega) := -g_x(k + \Omega m, k + \Omega m), \quad (726)$$

used throughout the main text.

No field equation is required in this appendix. The results follow only from the Lorentzian geometry of the two-dimensional Killing plane

$$\mathcal{K}_x := \text{span}\{k_x, m_x\}. \quad (727)$$

Away from the rotation axis we assume

$$g(m, m) > 0. \quad (728)$$

A.1 Quadratic form and Gram reconstruction

Define

$$A := g(k, k), \quad B := g(k, m), \quad C := g(m, m). \quad (729)$$

Then the Killing Gram matrix is

$$G = \begin{pmatrix} A & B \\ B & C \end{pmatrix}. \quad (730)$$

The stationary response polynomial is

$$\mathcal{R}(\Omega) = -A - 2B\Omega - C\Omega^2. \quad (731)$$

Consequently,

$$\mathcal{R}'(\Omega) = -2B - 2C\Omega, \quad (732)$$

$$\mathcal{R}''(\Omega) = -2C. \quad (733)$$

Thus

$$A = -\mathcal{R}(0), \quad (734)$$

$$B = -\frac{1}{2}\mathcal{R}'(0), \quad (735)$$

$$C = -\frac{1}{2}\mathcal{R}''. \quad (736)$$

Hence the full Killing Gram matrix can be written directly as

$$G = \begin{pmatrix} -\mathcal{R}(0) & -\frac{1}{2}\mathcal{R}'(0) \\ -\frac{1}{2}\mathcal{R}'(0) & -\frac{1}{2}\mathcal{R}'' \end{pmatrix}. \quad (737)$$

This proves the derivative form of the reconstruction theorem stated in Sec. ??.

A.2 Reconstruction from three response values

Differentiability with respect to Ω is not required operationally. Three distinct timelike response measurements suffice.

Let

$$\Omega_1, \quad \Omega_2, \quad \Omega_3 \quad (738)$$

be distinct values for which

$$\chi_{\Omega_a} = k + \Omega_a m \quad (739)$$

are future timelike, and define

$$R_a := \mu(\chi_{\Omega_a})^2 = \mathcal{R}(\Omega_a). \quad (740)$$

The unique quadratic interpolation is

$$\mathcal{R}(\Omega) = \sum_{a=1}^3 R_a L_a(\Omega), \quad (741)$$

where

$$L_a(\Omega) := \prod_{\substack{b=1 \\ b \neq a}}^3 \frac{\Omega - \Omega_b}{\Omega_a - \Omega_b}. \quad (742)$$

Writing

$$\mathcal{R}(\Omega) = q_0 + q_1 \Omega + q_2 \Omega^2, \quad (743)$$

one has

$$g(k, k) = -q_0, \quad g(k, m) = -\frac{1}{2}q_1, \quad g(m, m) = -q_2. \quad (744)$$

Thus three finite massive-response values reconstruct the complete two-dimensional Killing metric without requiring access to a null or spacelike Killing direction.

A.3 Response discriminant

The Gram determinant is

$$\det G = AC - B^2. \quad (745)$$

Define

$$\rho^2 := -\det G = B^2 - AC. \quad (746)$$

Using Eqs. (731)–(733),

$$\begin{aligned} (\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}'' &= 4(B + C\Omega)^2 \\ &\quad + 4C(-A - 2B\Omega - C\Omega^2) \\ &= 4(B^2 - AC). \end{aligned} \quad (747)$$

Hence

$$\rho^2 = \frac{1}{4} [(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}'']. \quad (748)$$

The derivative of the right-hand side vanishes identically:

$$\begin{aligned} \frac{d}{d\Omega} \left[(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}'' \right] &= 2\mathcal{R}'\mathcal{R}'' - 2\mathcal{R}'\mathcal{R}'' - 2\mathcal{R}\mathcal{R}''' \\ &= 0, \end{aligned} \quad (749)$$

because

$$\mathcal{R}''' = 0. \quad (750)$$

Thus the response discriminant is independent of where it is evaluated in the angular-velocity family.

A.4 Null roots

The stationary-null directions satisfy

$$\mathcal{R}(\Omega) = 0. \quad (751)$$

Equivalently,

$$C\Omega^2 + 2B\Omega + A = 0. \quad (752)$$

Hence

$$\Omega_{\pm} = \frac{-B \pm \rho}{C}. \quad (753)$$

The midpoint is

$$\frac{\Omega_+ + \Omega_-}{2} = -\frac{B}{C}. \quad (754)$$

The separation is

$$\Omega_+ - \Omega_- = \frac{2\rho}{C}. \quad (755)$$

The product is

$$\Omega_+\Omega_- = \frac{A}{C}. \quad (756)$$

The polynomial factorizes as

$$\mathcal{R}(\Omega) = C(\Omega_+ - \Omega)(\Omega - \Omega_-). \quad (757)$$

Because $C > 0$,

$$\mathcal{R}(\Omega) > 0 \iff \Omega_- < \Omega < \Omega_+. \quad (758)$$

Thus the massive stationary response domain is exactly the interval between the two null Killing directions.

A.5 Vertex form and the response maximum

Since

$$\mathcal{R}'' = -2C < 0, \quad (759)$$

the polynomial is strictly concave.

Its unique stationary point is determined by

$$\mathcal{R}'(\Omega_Z) = 0, \quad (760)$$

which gives

$$\Omega_Z = -\frac{B}{C} = -\frac{g(k, m)}{g(m, m)}. \quad (761)$$

Hence

$$\Omega_Z = \frac{\Omega_+ + \Omega_-}{2}. \quad (762)$$

Completing the square gives

$$\begin{aligned} \mathcal{R}(\Omega) &= -A - 2B\Omega - C\Omega^2 \\ &= \frac{B^2 - AC}{C} - C\left(\Omega + \frac{B}{C}\right)^2. \end{aligned} \quad (763)$$

Therefore

$$\mathcal{R}(\Omega) = \frac{\rho^2}{C} - C(\Omega - \Omega_Z)^2. \quad (764)$$

The maximum value is

$$\mathcal{R}_{\max} = \frac{\rho^2}{g(m, m)}. \quad (765)$$

Moreover,

$$\begin{aligned} g(k + \Omega_Z m, m) &= B + \Omega_Z C \\ &= 0. \end{aligned} \quad (766)$$

Thus the response maximum selects the Killing direction orthogonal to the axial generator.

A.6 Diagonalization of the Killing two-metric

Define

$$\widetilde{k} := k + \Omega_Z m. \quad (767)$$

Then

$$g(\widetilde{k}, m) = 0 \quad (768)$$

and

$$g(\widetilde{k}, \widetilde{k}) = -\frac{\rho^2}{C}. \quad (769)$$

Thus in the basis $(\widetilde{k}, m)$,

$$G \sim \begin{pmatrix} -\rho^2/C & 0 \\ 0 & C \end{pmatrix}. \quad (770)$$

In coordinates adapted to (k, m) , the Killing two-metric can therefore be written

$$ds_{\mathcal{K}}^2 = -\frac{\rho^2}{g_{\phi\phi}} dt^2 + g_{\phi\phi} (d\phi - \Omega_Z dt)^2. \quad (771)$$

Expanding gives

$$g_{t\phi} = -\Omega_Z g_{\phi\phi}, \quad (772)$$

$$g_{tt} = -\frac{\rho^2}{g_{\phi\phi}} + \Omega_Z^2 g_{\phi\phi}. \quad (773)$$

Thus

$$(g_{\phi\phi}, \Omega_Z, \rho) \quad (774)$$

provide an equivalent parametrization of the local Killing-plane geometry.

A.7 Stationary-limit and double-root conditions

The stationary Killing field k is null when

$$\mathcal{R}(0) = 0. \quad (775)$$

Since

$$\mathcal{R}(0) = C\Omega_+(-\Omega_-), \quad (776)$$

this means that one of the two null roots passes through

$$\Omega = 0. \quad (777)$$

By contrast, a double root occurs precisely when

$$\rho^2 = 0. \quad (778)$$

Indeed,

$$\rho^2 = 0 \iff \Omega_+ = \Omega_-. \quad (779)$$

At the repeated root

$$\Omega_H = -\frac{B}{C}, \quad (780)$$

one has

$$\mathcal{R}(\Omega_H) = 0, \quad \mathcal{R}'(\Omega_H) = 0. \quad (781)$$

The converse also holds: if

$$\mathcal{R}(\Omega_0) = \mathcal{R}'(\Omega_0) = 0, \quad (782)$$

then

$$\rho^2 = 0. \quad (783)$$

Thus

$$\mathcal{R}(0) = 0 \quad (784)$$

and

$$\mathcal{R}(\Omega_H) = \mathcal{R}'(\Omega_H) = 0 \quad (785)$$

are algebraically different phenomena.

The first makes one prescribed Killing direction null.

The second makes the complete Killing plane degenerate.

Geometric identification of the latter with a regular horizon still requires the additional non-axis and horizon hypotheses stated in the main text.

A.8 Static response symmetry

The response polynomial is even if and only if

$$B = g(k, m) = 0. \quad (786)$$

Indeed,

$$\mathcal{R}(\Omega) - \mathcal{R}(-\Omega) = -4B\Omega. \quad (787)$$

Hence

$$\mathcal{R}(\Omega) = \mathcal{R}(-\Omega) \quad \forall \Omega \iff g(k, m) = 0. \quad (788)$$

This is only a pointwise algebraic statement. It should not be confused with the invariant differential condition for staticity,

$$k^b \wedge dk^b = 0. \quad (789)$$

A.9 Change of stationary Killing representative

Let

$$k' := k + \Omega_0 m, \quad m' := m, \quad (790)$$

where Ω_0 is constant.

Then

$$k' + \Omega' m' = k + (\Omega_0 + \Omega') m. \quad (791)$$

Therefore

$$\mathcal{R}'(\Omega') = \mathcal{R}(\Omega_0 + \Omega'). \quad (792)$$

The roots transform as

$$\Omega'_\pm = \Omega_\pm - \Omega_0, \quad (793)$$

and the response center transforms as

$$\Omega'_Z = \Omega_Z - \Omega_0. \quad (794)$$

However,

$$\Omega'_+ - \Omega'_- = \Omega_+ - \Omega_-, \quad (795)$$

$$\mathcal{R}'_{\max} = \mathcal{R}_{\max}, \quad (796)$$

$$\rho'^2 = \rho^2. \quad (797)$$

This confirms that the position of the response spectrum along the Ω -axis depends on the chosen stationary Killing representative, whereas its width, maximum height, and Gram determinant do not.

In an asymptotically flat spacetime, the ambiguity is removed by fixing k to be the asymptotically nonrotating unit time translation.

A.10 Summary

The entire local Killing-plane geometry can be organized in the following equivalent data:

$$G_{AB} \iff \mathcal{R}(\Omega) \iff (g_{\phi\phi}, \Omega_Z, \rho). \quad (798)$$

The basic identities are

$$\mathcal{R}(\Omega) = -g_{tt} - 2\Omega g_{t\phi} - \Omega^2 g_{\phi\phi}, \quad (799)$$

$$\rho^2 = \frac{1}{4} [(\mathcal{R}')^2 - 2\mathcal{R}\mathcal{R}''], \quad (800)$$

$$\Omega_Z = \frac{\Omega_+ + \Omega_-}{2} = -\frac{g_{t\phi}}{g_{\phi\phi}}, \quad (801)$$

and

$$\mathcal{R}(\Omega) = \frac{\rho^2}{g_{\phi\phi}} - g_{\phi\phi} (\Omega - \Omega_Z)^2. \quad (802)$$

These identities contain no Einstein dynamics. They are purely Lorentzian properties of the response-reconstructed stationary Killing plane.

B Schwarzschild and Kerr coordinate identities

This appendix verifies the coordinate and algebraic identities used in the Schwarzschild and Kerr sections. All calculations are standard properties of the corresponding exact metrics; they are collected here to keep the main text focused on the response interpretation.

Throughout this appendix

$$G = c = 1. \quad (803)$$

B.1 Schwarzschild Weyl coordinates

The Schwarzschild exterior metric is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (804)$$

Introduce

$$\rho = \sqrt{r(r - 2M)} \sin \theta, \quad (805)$$

$$z = (r - M) \cos \theta. \quad (806)$$

The stationary and axial Killing coefficients are

$$g_{tt} = -\left(1 - \frac{2M}{r}\right), \quad (807)$$

$$g_{t\phi} = 0, \quad (808)$$

$$g_{\phi\phi} = r^2 \sin^2 \theta. \quad (809)$$

Therefore

$$\begin{aligned}
-\det G &= \left(1 - \frac{2M}{r}\right) r^2 \sin^2 \theta \\
&= r(r - 2M) \sin^2 \theta \\
&= \rho^2.
\end{aligned} \tag{810}$$

Thus the Weyl radial coordinate agrees exactly with the response discriminant.
Define

$$R_{\pm} := \sqrt{\rho^2 + (z \pm M)^2}. \tag{811}$$

Substitution of Eqs. (805) and (806) gives

$$\begin{aligned}
R_+^2 &= r(r - 2M) \sin^2 \theta + [(r - M) \cos \theta + M]^2 \\
&= [r - M + M \cos \theta]^2,
\end{aligned} \tag{812}$$

and therefore, in the exterior region,

$$R_+ = r - M + M \cos \theta. \tag{813}$$

Similarly,

$$R_- = r - M - M \cos \theta. \tag{814}$$

Hence

$$R_+ + R_- = 2(r - M), \tag{815}$$

$$R_+ - R_- = 2M \cos \theta. \tag{816}$$

The inverse transformation is therefore

$$r = M + \frac{R_+ + R_-}{2}, \tag{817}$$

$$\cos \theta = \frac{R_+ - R_-}{2M}. \tag{818}$$

B.2 Schwarzschild Weyl potential

The static response coefficient is

$$f = 1 - \frac{2M}{r}. \tag{819}$$

Using Eq. (817),

$$\begin{aligned}
f &= \frac{r - 2M}{r} \\
&= \frac{R_+ + R_- - 2M}{R_+ + R_- + 2M}.
\end{aligned} \tag{820}$$

Thus

$$f_{\text{Schw}} = \frac{R_+ + R_- - 2M}{R_+ + R_- + 2M}. \tag{821}$$

The Weyl potential is

$$U_{\text{Schw}} = \frac{1}{2} \ln \left(\frac{R_+ + R_- - 2M}{R_+ + R_- + 2M} \right). \quad (822)$$

The second Weyl metric function is

$$e^{2\gamma_{\text{Schw}}} = \frac{(R_+ + R_-)^2 - 4M^2}{4R_+R_-}. \quad (823)$$

B.3 Finite-rod integral

Consider

$$I(\rho, z) := \int_{-M}^M \frac{d\zeta}{\sqrt{\rho^2 + (z - \zeta)^2}}. \quad (824)$$

Set

$$y = z - \zeta. \quad (825)$$

Then

$$d\zeta = -dy \quad (826)$$

and

$$\begin{aligned} I &= \int_{z-M}^{z+M} \frac{dy}{\sqrt{\rho^2 + y^2}} \\ &= \left[\operatorname{arsinh} \left(\frac{y}{\rho} \right) \right]_{z-M}^{z+M}. \end{aligned} \quad (827)$$

Using the standard logarithmic form of arsinh and the definitions of $R_{\pm}$, one obtains

$$I = \ln \left(\frac{R_+ + R_- + 2M}{R_+ + R_- - 2M} \right). \quad (828)$$

Hence

$$U_{\text{Schw}} = -\frac{1}{2} \int_{-M}^M \frac{d\zeta}{\sqrt{\rho^2 + (z - \zeta)^2}}. \quad (829)$$

Thus the auxiliary Weyl source occupies the segment

$$\rho = 0, \quad -M \leq z \leq M. \quad (830)$$

B.4 Near-rod expansion

Fix

$$|z| < M \quad (831)$$

and let

$$\rho \rightarrow 0. \quad (832)$$

Then

$$R_+ = M + z + \frac{\rho^2}{2(M+z)} + O(\rho^4), \quad (833)$$

$$R_- = M - z + \frac{\rho^2}{2(M-z)} + O(\rho^4). \quad (834)$$

Therefore

$$R_+ + R_- = 2M + \frac{M\rho^2}{M^2 - z^2} + O(\rho^4). \quad (835)$$

Substitution into Eq. (821) gives

$$f_{\text{Schw}} = \frac{\rho^2}{4(M^2 - z^2)} + O(\rho^4). \quad (836)$$

Hence

$$\mu_{\text{Schw}}(k) = \frac{\rho}{2\sqrt{M^2 - z^2}} + O(\rho^3). \quad (837)$$

Since

$$g_{\phi\phi} = \frac{\rho^2}{f_{\text{Schw}}}, \quad (838)$$

we obtain

$$g_{\phi\phi} \longrightarrow 4(M^2 - z^2) \quad (839)$$

on the interior of the rod.

The response polynomial therefore approaches

$$\mathcal{R}_{\text{Schw}}(\Omega) \longrightarrow -4(M^2 - z^2)\Omega^2 \quad (840)$$

for $|z| < M$, demonstrating explicitly the double root

$$\Omega_H = 0. \quad (841)$$

B.5 Kerr Killing determinant

For Kerr define

$$\Sigma := r^2 + a^2 \cos^2 \theta, \quad (842)$$

$$\Delta := r^2 - 2Mr + a^2, \quad (843)$$

$$\mathcal{A} := (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta. \quad (844)$$

The Killing coefficients are

$$g_{tt} = -\frac{\Delta - a^2 \sin^2 \theta}{\Sigma}, \quad (845)$$

$$g_{t\phi} = -\frac{2Mar \sin^2 \theta}{\Sigma}, \quad (846)$$

$$g_{\phi\phi} = \frac{\mathcal{A} \sin^2 \theta}{\Sigma}. \quad (847)$$

We calculate

$$\begin{aligned}
-\det G &= g_{t\phi}^2 - g_{tt}g_{\phi\phi} \\
&= \frac{4M^2a^2r^2 \sin^4 \theta}{\Sigma^2} \\
&\quad + \frac{(\Delta - a^2 \sin^2 \theta)\mathcal{A} \sin^2 \theta}{\Sigma^2}.
\end{aligned} \tag{848}$$

Using

$$\mathcal{A} = (r^2 + a^2)\Sigma + 2Ma^2r \sin^2 \theta, \tag{849}$$

and

$$r^2 + a^2 - \Delta = 2Mr, \tag{850}$$

the numerator simplifies to

$$\Delta \Sigma^2 \sin^2 \theta. \tag{851}$$

Therefore

$$-\det G = \Delta \sin^2 \theta. \tag{852}$$

Thus the canonical Weyl radial invariant is

$$\rho = \sqrt{\Delta} \sin \theta \tag{853}$$

in the exterior $\Delta > 0$.

B.6 Kerr response vertex form

Define

$$\alpha^2 := \frac{\Delta \Sigma}{\mathcal{A}}, \tag{854}$$

$$\omega := \frac{2Mar}{\mathcal{A}}. \tag{855}$$

Then

$$g_{t\phi} = -\omega g_{\phi\phi}. \tag{856}$$

Since

$$-\det G = \alpha^2 g_{\phi\phi}, \tag{857}$$

we also have

$$g_{tt} = -\alpha^2 + \omega^2 g_{\phi\phi}. \tag{858}$$

Therefore

$$\begin{aligned}
\mathcal{R}_{\text{Kerr}}(\Omega) &= -g_{tt} - 2\Omega g_{t\phi} - \Omega^2 g_{\phi\phi} \\
&= \alpha^2 - g_{\phi\phi} (\Omega - \omega)^2.
\end{aligned} \tag{859}$$

Thus

$$\mathcal{R}_{\text{Kerr}}(\Omega) = \alpha^2 - g_{\phi\phi} (\Omega - \omega)^2. \tag{860}$$

Immediately,

$$\arg \max_{\Omega} \mathcal{R}_{\text{Kerr}} = \omega, \tag{861}$$

and

$$\max_{\Omega} \mathcal{R}_{\text{Kerr}} = \alpha^2. \tag{862}$$

B.7 Kerr null angular velocities

From Eq. (860), the null roots are

$$\Omega_{\pm} = \omega \pm \frac{\alpha}{\sqrt{g_{\phi\phi}}}. \quad (863)$$

Since

$$g_{\phi\phi} = \frac{\mathcal{A} \sin^2 \theta}{\Sigma}, \quad (864)$$

we obtain

$$\frac{\alpha}{\sqrt{g_{\phi\phi}}} = \frac{\Sigma \sqrt{\Delta}}{\mathcal{A} \sin \theta} \quad (865)$$

for $0 < \theta < \pi$.

Hence

$$\Omega_{\pm} = \frac{2Mar}{\mathcal{A}} \pm \frac{\Sigma \sqrt{\Delta}}{\mathcal{A} \sin \theta}. \quad (866)$$

Their separation is

$$\Omega_+ - \Omega_- = \frac{2\Sigma \sqrt{\Delta}}{\mathcal{A} \sin \theta}. \quad (867)$$

Their midpoint is

$$\frac{\Omega_+ + \Omega_-}{2} = \omega. \quad (868)$$

B.8 Ergosurface

At $\Omega = 0$,

$$\mathcal{R}_{\text{Kerr}}(0) = -g_{tt} = 1 - \frac{2Mr}{\Sigma}. \quad (869)$$

Thus the stationary-limit equation is

$$\Sigma - 2Mr = 0, \quad (870)$$

or

$$r^2 - 2Mr + a^2 \cos^2 \theta = 0. \quad (871)$$

Hence

$$r_{\text{E}\pm} = M \pm \sqrt{M^2 - a^2 \cos^2 \theta}. \quad (872)$$

At a generic non-axis point of the outer ergosurface,

$$\alpha^2 > 0. \quad (873)$$

Therefore the vanishing of $\mathcal{R}(0)$ does not imply collapse of the full stationary timelike interval.

B.9 Outer horizon and response-root merger

Let

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}. \quad (874)$$

Then

$$\Delta = (r - r_+)(r - r_-). \quad (875)$$

At

$$r = r_+, \quad (876)$$

we have

$$\Delta = 0 \quad (877)$$

and

$$r_+^2 + a^2 = 2Mr_+. \quad (878)$$

Moreover,

$$\mathcal{A}_+ = (r_+^2 + a^2)^2. \quad (879)$$

Therefore

$$\begin{aligned} \omega_H &= \frac{2Mar_+}{(r_+^2 + a^2)^2} \\ &= \frac{a}{r_+^2 + a^2}. \end{aligned} \quad (880)$$

Thus

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \quad (881)$$

Since

$$\alpha^2 = \frac{\Delta\Sigma}{\mathcal{A}}, \quad (882)$$

we have

$$\alpha^2 \longrightarrow 0 \quad (883)$$

at the outer horizon.

The two null roots therefore satisfy

$$\Omega_{\pm} \longrightarrow \Omega_H. \quad (884)$$

On the horizon,

$$g_{\phi\phi}^H = \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+}, \quad (885)$$

where

$$\Sigma_+ = r_+^2 + a^2 \cos^2 \theta. \quad (886)$$

Hence the exact horizon response polynomial is

$$\mathcal{R}_H(\Omega) = -\frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+} (\Omega - \Omega_H)^2. \quad (887)$$

Therefore, at every non-axis horizon point,

$$\mathcal{R}_H(\Omega_H) = 0, \quad \mathcal{R}'_H(\Omega_H) = 0. \quad (888)$$

B.10 Kerr Weyl coordinates

For subextremal Kerr define

$$\ell := \sqrt{M^2 - a^2}. \quad (889)$$

Then

$$r_{\pm} = M \pm \ell. \quad (890)$$

Introduce Weyl coordinates

$$\rho = \sqrt{\Delta} \sin \theta, \quad (891)$$

$$z = (r - M) \cos \theta. \quad (892)$$

Define focal distances

$$R_{\pm} := \sqrt{\rho^2 + (z \pm \ell)^2}. \quad (893)$$

Using

$$\Delta = (r - M)^2 - \ell^2, \quad (894)$$

one finds

$$\begin{aligned} R_{\pm}^2 &= \left[(r - M)^2 - \ell^2 \right] \sin^2 \theta \\ &\quad + [(r - M) \cos \theta \pm \ell]^2 \\ &= [r - M \pm \ell \cos \theta]^2. \end{aligned} \quad (895)$$

Hence in the exterior

$$R_{\pm} = r - M \pm \ell \cos \theta. \quad (896)$$

Therefore

$$r - M = \frac{R_+ + R_-}{2}, \quad (897)$$

$$\cos \theta = \frac{R_+ - R_-}{2\ell}. \quad (898)$$

B.11 Prolate spheroidal variables

Define

$$x := \frac{r - M}{\ell}, \quad y := \cos \theta. \quad (899)$$

Then

$$R_{\pm} = \ell(x \pm y), \quad (900)$$

$$\rho = \ell \sqrt{x^2 - 1} \sqrt{1 - y^2}, \quad (901)$$

$$z = \ell xy. \quad (902)$$

The exterior corresponds to

$$x > 1, \quad -1 \leq y \leq 1. \quad (903)$$

The outer horizon

$$r = r_+ = M + \ell \quad (904)$$

is

$$x = 1. \quad (905)$$

Hence

$$\rho = 0, \quad z = \ell y. \quad (906)$$

Therefore the horizon maps to

$$\rho = 0, \quad -\ell \leq z \leq \ell. \quad (907)$$

Its Weyl coordinate length is

$$L_H^{\text{Weyl}} = 2\ell = 2\sqrt{M^2 - a^2}. \quad (908)$$

The exterior parts of the rotation axis correspond to

$$\rho = 0, \quad |z| > \ell. \quad (909)$$

Thus the response discriminant vanishes on both the horizon rod and the axis, while the complete response polynomial distinguishes the two.

B.12 Schwarzschild limit

Take

$$a \rightarrow 0. \quad (910)$$

Then

$$\ell \rightarrow M, \quad (911)$$

and

$$\rho \rightarrow \sqrt{r(r - 2M)} \sin \theta, \quad (912)$$

$$z \rightarrow (r - M) \cos \theta. \quad (913)$$

Thus the Kerr Weyl transformation reduces to the Schwarzschild transformation.

The horizon rod becomes

$$-M \leq z \leq M, \quad (914)$$

with length

$$L_H^{\text{Weyl}} \rightarrow 2M. \quad (915)$$

Likewise,

$$\omega \rightarrow 0 \quad (916)$$

and

$$\alpha^2 \rightarrow 1 - \frac{2M}{r}. \quad (917)$$

The Kerr response polynomial reduces to

$$\mathcal{R}_{\text{Kerr}}(\Omega) \rightarrow 1 - \frac{2M}{r} - r^2 \sin^2 \theta \Omega^2. \quad (918)$$

B.13 Extremal limit

For

$$|a| \rightarrow M, \quad (919)$$

we have

$$\ell = \sqrt{M^2 - a^2} \rightarrow 0. \quad (920)$$

Therefore

$$L_H^{\text{Weyl}} \rightarrow 0. \quad (921)$$

The prolate coordinate system based on a nonzero focal separation 2ℓ degenerates in this limit. This is a coordinate degeneration of the subextremal Weyl representation, not the disappearance of the extremal horizon.

The Killing-plane response remains well defined. The horizon still satisfies

$$\mathcal{R}_H(\Omega_H) = \mathcal{R}'_H(\Omega_H) = 0, \quad (922)$$

while the additional transverse first-jet degeneracy

$$\kappa_H = 0 \quad (923)$$

distinguishes extremality, as discussed in Sec. 6.

B.14 Kerr nonextremality identities

The horizon radii satisfy

$$r_+ + r_- = 2M, \quad (924)$$

$$r_+ r_- = a^2, \quad (925)$$

$$r_+ - r_- = 2\ell. \quad (926)$$

The horizon angular velocity is

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \quad (927)$$

Since

$$J = Ma \quad (928)$$

and

$$r_+^2 + a^2 = 2Mr_+, \quad (929)$$

we have

$$\begin{aligned} M - 2\Omega_H J &= M - \frac{2Ma^2}{r_+^2 + a^2} \\ &= M - \frac{a^2}{r_+} \\ &= M - r_- \\ &= \frac{r_+ - r_-}{2} \\ &= \ell. \end{aligned} \quad (930)$$

Thus

$$M - 2\Omega_H J = \ell = \frac{1}{2} L_H^{\text{Weyl}}. \quad (931)$$

The Kerr surface gravity is

$$\kappa_H = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \quad (932)$$

and the horizon area is

$$A_H = 4\pi(r_+^2 + a^2). \quad (933)$$

Hence

$$\begin{aligned} \frac{\kappa_H A_H}{4\pi} &= \frac{r_+ - r_-}{2} \\ &= \ell. \end{aligned} \quad (934)$$

Therefore

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \ell = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (935)$$

This identity combines several classical Kerr formulas. The role of the main text is to show how the horizon angular velocity entering the first term is selected by the double root of the stationary MS response spectrum.

C Twist, Ernst, orientation, and sign conventions

This appendix fixes the orientation and sign conventions used for the stationary twist, the Weyl–Papapetrou rotational coefficient, and the Ernst potential.

These conventions are not unique in the literature. In particular, a reversal of spacetime orientation changes the sign of the twist potential and therefore complex conjugates the Ernst potential, while leaving the spacetime metric unchanged.

Throughout this appendix we use geometrized units

$$G = c = 1 \quad (936)$$

and Lorentzian signature

$$(-, +, +, +). \quad (937)$$

C.1 Stationary Killing field and twist

Let

$$k \quad (938)$$

be the stationary Killing field and define

$$k^\flat := g(k, \cdot). \quad (939)$$

The stationary norm potential is

$$f := -g(k, k). \quad (940)$$

Where k is timelike,

$$f > 0 \quad (941)$$

and, in the MS response interpretation,

$$f = \mathcal{R}(0) = \mu(k)^2. \quad (942)$$

The twist one-form is defined by

$$\varpi := \star (k^b \wedge dk^b). \quad (943)$$

Equivalently,

$$\varpi_\mu = \epsilon_{\mu\nu\rho\sigma} k^\nu \nabla^\rho k^\sigma. \quad (944)$$

The spacetime orientation is chosen so that the Weyl–Papapetrou formulas below take the signs

$$\psi_{, \rho} = \frac{f^2}{\rho} \omega_{\text{P}, z}, \quad (945)$$

$$\psi_{, z} = -\frac{f^2}{\rho} \omega_{\text{P}, \rho}. \quad (946)$$

With this orientation, the Kerr twist potential used in the main text is

$$\psi_{\text{Kerr}} = -\frac{2Ma \cos \theta}{r^2 + a^2 \cos^2 \theta}. \quad (947)$$

If the spacetime orientation is reversed, then

$$\star \mapsto -\star, \quad (948)$$

and therefore

$$\varpi \mapsto -\varpi, \quad \psi \mapsto -\psi. \quad (949)$$

All metric quantities, including

$$f, \quad \rho, \quad \mathcal{R}(\Omega), \quad \Omega_Z, \quad (950)$$

are unchanged.

C.2 Staticity

The Frobenius condition for local hypersurface orthogonality of the stationary Killing field is

$$k^b \wedge dk^b = 0. \quad (951)$$

Since the Hodge star is invertible,

$$k^b \wedge dk^b = 0 \iff \varpi = 0. \quad (952)$$

Thus vanishing twist is the invariant local staticity condition.

This should be distinguished from

$$g(k, m) = 0, \quad (953)$$

which is only a pointwise algebraic statement about the chosen stationary and axial Killing basis.

C.3 Vacuum integrability

For a Killing vector field, the Killing and Ricci identities imply

$$\nabla_{[\mu}\varpi_{\nu]} = \epsilon_{\mu\nu\rho\sigma}k^\rho R^\sigma{}_\lambda k^\lambda. \quad (954)$$

Therefore in vacuum,

$$R_{\mu\nu} = 0, \quad (955)$$

the twist is closed:

$$d\varpi = 0. \quad (956)$$

On a simply connected stationary vacuum region, there consequently exists a twist potential

$$d\psi = \varpi. \quad (957)$$

The potential is defined up to a constant. In an asymptotically flat stationary region we impose

$$\psi \longrightarrow 0 \quad \text{at spatial infinity.} \quad (958)$$

The stationary Killing field is independently normalized by

$$g(k, k) \longrightarrow -1, \quad (959)$$

so

$$f \longrightarrow 1. \quad (960)$$

C.4 Weyl–Papapetrou convention

In an orthogonally transitive stationary axisymmetric region, write

$$ds^2 = -f(dt - \omega_P d\phi)^2 + f^{-1} \left[e^{2\gamma} (d\rho^2 + dz^2) + \rho^2 d\phi^2 \right]. \quad (961)$$

The subscript in ω_P is used deliberately. It denotes the Weyl–Papapetrou metric coefficient and not the angular velocity of the locally nonrotating observer.

Expanding the Killing part gives

$$g_{tt} = -f, \quad (962)$$

$$g_{t\phi} = f\omega_P, \quad (963)$$

$$g_{\phi\phi} = f^{-1}\rho^2 - f\omega_P^2. \quad (964)$$

Consequently,

$$-\det G = \rho^2. \quad (965)$$

Since

$$f = \mathcal{R}(0) \quad (966)$$

and

$$g_{t\phi} = -\frac{1}{2}\partial_\Omega \mathcal{R}(0), \quad (967)$$

we obtain

$$\omega_{\text{P}} = -\frac{\partial_{\Omega}\mathcal{R}(0)}{2\mathcal{R}(0)}. \quad (968)$$

This formula is valid wherever

$$f = \mathcal{R}(0) \neq 0. \quad (969)$$

At a stationary-limit surface $f = 0$, the particular Weyl–Papapetrou parametrization based on division by f becomes singular even though the underlying Killing Gram matrix and response polynomial remain geometrically meaningful.

C.5 Papapetrou coefficient versus locally nonrotating angular velocity

The response maximum occurs at

$$\Omega_{\text{Z}} = -\frac{g_{t\phi}}{g_{\phi\phi}}. \quad (970)$$

Using Eqs. (963) and (964),

$$\Omega_{\text{Z}} = -\frac{f\omega_{\text{P}}}{f^{-1}\rho^2 - f\omega_{\text{P}}^2}. \quad (971)$$

Therefore

$$\Omega_{\text{Z}} \neq \omega_{\text{P}} \quad (972)$$

in general.

Their roles are different:

$$\begin{aligned} \omega_{\text{P}} &: \text{metric coefficient in the Papapetrou parametrization,} \\ \Omega_{\text{Z}} &: \text{angular velocity of the response-maximizing ZAMO direction.} \end{aligned} \quad (973)$$

This distinction is especially important in Kerr.

C.6 Derivation of the twist equations

For

$$k = \partial_t, \quad (974)$$

the dual one-form obtained from Eq. (961) is

$$k^{\flat} = -f(dt - \omega_{\text{P}}d\phi). \quad (975)$$

Differentiation gives

$$dk^{\flat} = -df \wedge (dt - \omega_{\text{P}}d\phi) + f d\omega_{\text{P}} \wedge d\phi. \quad (976)$$

Hence

$$\begin{aligned} k^{\flat} \wedge dk^{\flat} &= -f^2 (dt - \omega_{\text{P}}d\phi) \wedge d\omega_{\text{P}} \wedge d\phi \\ &= -f^2 dt \wedge d\omega_{\text{P}} \wedge d\phi. \end{aligned} \quad (977)$$

With the orientation chosen in this appendix, taking the Hodge dual yields

$$\varpi = \frac{f^2}{\rho} [\omega_{\text{P},z} d\rho - \omega_{\text{P},\rho} dz]. \quad (978)$$

Since

$$d\psi = \varpi, \quad (979)$$

we recover

$$\psi_{,\rho} = \frac{f^2}{\rho} \omega_{\text{P},z}, \quad (980)$$

$$\psi_{,z} = -\frac{f^2}{\rho} \omega_{\text{P},\rho}. \quad (981)$$

Under reversal of orientation both right-hand sides change sign, which is equivalent to

$$\psi \mapsto -\psi. \quad (982)$$

C.7 Twist directly from response data

Substituting

$$f = \mathcal{R}(0) \quad (983)$$

and

$$\omega_{\text{P}} = -\frac{\mathcal{R}_{\Omega}(0)}{2\mathcal{R}(0)} \quad (984)$$

into Eqs. (980) and (981) gives

$$\psi_{,\rho} = \frac{\mathcal{R}(0)^2}{\rho} \partial_z \left[-\frac{\mathcal{R}_{\Omega}(0)}{2\mathcal{R}(0)} \right], \quad (985)$$

$$\psi_{,z} = -\frac{\mathcal{R}(0)^2}{\rho} \partial_{\rho} \left[-\frac{\mathcal{R}_{\Omega}(0)}{2\mathcal{R}(0)} \right]. \quad (986)$$

Thus the pointwise response spectrum determines

$$f, \quad \omega_{\text{P}}, \quad \rho, \quad (987)$$

while its first spacetime jet determines the twist one-form.

The logical hierarchy is

$$\mathcal{R} \longrightarrow (f, \omega_{\text{P}}, \rho) \longrightarrow d\psi. \quad (988)$$

Only after imposing vacuum integrability may $d\psi$ be globally integrated on a simply connected region to a single-valued twist potential.

C.8 Ernst potential

With the above orientation convention, define

$$\mathcal{E} := f + i\psi. \quad (989)$$

Therefore

$$\Re \mathcal{E} = f = \mathcal{R}(0), \quad (990)$$

and

$$d \Im \mathcal{E} = \star (k^{\flat} \wedge dk^{\flat}). \quad (991)$$

Under reversal of spacetime orientation,

$$\psi \mapsto -\psi, \quad (992)$$

so

$$\mathcal{E} \mapsto \bar{\mathcal{E}}. \quad (993)$$

Thus two common sign conventions for the Kerr Ernst potential differ only by complex conjugation.

C.9 Ernst equation

The stationary axisymmetric vacuum Einstein equations imply

$$(\Re \mathcal{E}) \left(\mathcal{E}_{,\rho\rho} + \frac{1}{\rho} \mathcal{E}_{,\rho} + \mathcal{E}_{,zz} \right) = \mathcal{E}_{,\rho}^2 + \mathcal{E}_{,z}^2. \quad (994)$$

Equivalently, for

$$\mathcal{E} = f + i\psi, \quad (995)$$

$$f \left(f_{,\rho\rho} + \frac{1}{\rho} f_{,\rho} + f_{,zz} \right) = f_{,\rho}^2 + f_{,z}^2 - \psi_{,\rho}^2 - \psi_{,z}^2, \quad (996)$$

$$f \left(\psi_{,\rho\rho} + \frac{1}{\rho} \psi_{,\rho} + \psi_{,zz} \right) = 2 \left(f_{,\rho} \psi_{,\rho} + f_{,z} \psi_{,z} \right). \quad (997)$$

These are standard stationary vacuum Einstein equations [?]. Their role in the present paper is not to follow from the MS response polynomial alone, but to provide the vacuum dynamical closure of the reconstructed stationary response geometry.

C.10 Static limit

If

$$\varpi = 0, \quad (998)$$

then with $\psi \rightarrow 0$ at infinity,

$$\psi = 0. \quad (999)$$

Therefore

$$\mathcal{E} = f. \quad (1000)$$

The Ernst equation becomes

$$f \Delta_{\text{ax}} f = |\nabla f|^2, \quad (1001)$$

so

$$\Delta_{\text{ax}} \ln f = 0. \quad (1002)$$

Writing

$$f = e^{2U}, \quad (1003)$$

one obtains

$$\Delta_{\text{ax}} U = 0. \quad (1004)$$

Since

$$U = \ln \mu(k) \quad (1005)$$

in the timelike static region, this reproduces the static response–Weyl relation of Sec. 2.

C.11 Kerr Ernst potential

For Kerr define

$$\Sigma := r^2 + a^2 \cos^2 \theta. \quad (1006)$$

With the orientation convention of this appendix, the Ernst potential is

$$\mathcal{E}_{\text{Kerr}} = 1 - \frac{2M}{r - ia \cos \theta}. \quad (1007)$$

Multiplying numerator and denominator by

$$r + ia \cos \theta \quad (1008)$$

gives

$$\begin{aligned} \mathcal{E}_{\text{Kerr}} &= 1 - \frac{2M(r + ia \cos \theta)}{\Sigma} \\ &= 1 - \frac{2Mr}{\Sigma} - i \frac{2Ma \cos \theta}{\Sigma}. \end{aligned} \quad (1009)$$

Hence

$$\Re \mathcal{E}_{\text{Kerr}} = 1 - \frac{2Mr}{\Sigma} = -g_{tt} = \mathcal{R}_{\text{Kerr}}(0), \quad (1010)$$

and

$$\Im \mathcal{E}_{\text{Kerr}} = -\frac{2Ma \cos \theta}{\Sigma} = \psi_{\text{Kerr}}. \quad (1011)$$

The opposite spacetime orientation gives instead

$$\overline{\mathcal{E}_{\text{Kerr}}} = 1 - \frac{2M}{r + ia \cos \theta}. \quad (1012)$$

Both conventions describe the same Kerr metric.

C.12 Asymptotic expansion

At large r ,

$$\frac{1}{r - ia \cos \theta} = \frac{1}{r} + i \frac{a \cos \theta}{r^2} - \frac{a^2 \cos^2 \theta}{r^3} + O(r^{-4}). \quad (1013)$$

Therefore

$$\begin{aligned} \mathcal{E}_{\text{Kerr}} &= 1 - \frac{2M}{r} - i \frac{2Ma \cos \theta}{r^2} \\ &\quad + \frac{2Ma^2 \cos^2 \theta}{r^3} + O(r^{-4}). \end{aligned} \quad (1014)$$

Since

$$J = Ma, \quad (1015)$$

the leading twist term is

$$\psi_{\text{Kerr}} = -\frac{2J \cos \theta}{r^2} + O(r^{-4}). \quad (1016)$$

The leading real and imaginary sectors therefore encode respectively the mass monopole and the current dipole.

C.13 Cayley-transformed potential

Define

$$\xi := \frac{1 - \mathcal{E}}{1 + \mathcal{E}}. \quad (1017)$$

For Eq. (1007),

$$1 - \mathcal{E}_{\text{Kerr}} = \frac{2M}{r - ia \cos \theta}, \quad (1018)$$

$$1 + \mathcal{E}_{\text{Kerr}} = \frac{2(r - M - ia \cos \theta)}{r - ia \cos \theta}. \quad (1019)$$

Hence

$$\xi_{\text{Kerr}} = \frac{M}{r - M - ia \cos \theta}. \quad (1020)$$

For subextremal Kerr, let

$$\ell = \sqrt{M^2 - a^2}, \quad x = \frac{r - M}{\ell}, \quad y = \cos \theta. \quad (1021)$$

Then

$$\xi_{\text{Kerr}} = \frac{M}{\ell x - iay}. \quad (1022)$$

On the north exterior symmetry axis,

$$y = 1, \quad z = r - M, \quad (1023)$$

so

$$\xi_{\text{Kerr}}^{\text{axis}}(z) = \frac{M}{z - ia}. \quad (1024)$$

The asymptotic expansion is

$$\xi_{\text{Kerr}}^{\text{axis}} = \frac{M}{z} + \frac{iMa}{z^2} - \frac{Ma^2}{z^3} - \frac{iMa^3}{z^4} + O(z^{-5}). \quad (1025)$$

This alternating real–imaginary structure anticipates the standard Kerr mass-current multipole hierarchy. A derivation of the Geroch–Hansen moments requires additional conformal-completion machinery and is not part of the present paper.

C.14 Summary of convention dependence

With the orientation used in this paper,

$$\begin{aligned} \varpi &= \star(k^b \wedge dk^b), \\ d\psi &= \varpi, \\ \mathcal{E} &= f + i\psi, \\ \mathcal{E}_{\text{Kerr}} &= 1 - \frac{2M}{r - ia \cos \theta}. \end{aligned} \quad (1026)$$

Under orientation reversal,

$$\varpi \rightarrow -\varpi, \quad \psi \rightarrow -\psi, \quad \mathcal{E} \rightarrow \overline{\mathcal{E}}. \quad (1027)$$

The stationary response polynomial $\mathcal{R}(\Omega)$, the Killing Gram matrix, the Weyl discriminant, the horizon angular velocity, and all metric observables are invariant under this convention change.

D Komar normalization, horizon evaluation, and Smarr factors

This appendix fixes the normalization and orientation conventions entering the Komar mass, angular momentum, horizon-generator charge, and Smarr identity.

The main purpose is to make explicit the relative factor of two between the stationary and axial Komar charges and the factor $1/(4\pi)$ appearing in the horizon contribution.

Throughout the principal derivation we use geometrized units

$$G = c = 1 \quad (1028)$$

and signature

$$(-, +, +, +). \quad (1029)$$

D.1 Komar functional

Let

$$\xi \quad (1030)$$

be a Killing field and

$$\xi^b := g(\xi, \cdot). \quad (1031)$$

For an oriented closed two-surface S , define

$$\mathcal{K}_S[\xi] := -\frac{1}{8\pi} \int_S \star d\xi^b. \quad (1032)$$

The functional is linear:

$$\mathcal{K}_S[a\xi + b\zeta] = a\mathcal{K}_S[\xi] + b\mathcal{K}_S[\zeta] \quad (1033)$$

for constant a, b .

Let k be the stationary Killing field normalized by

$$g(k, k) \rightarrow -1 \quad (1034)$$

at spatial infinity. Define the Komar mass by

$$M_K[S] := \mathcal{K}_S[k] = -\frac{1}{8\pi} \int_S \star dk^b. \quad (1035)$$

Let m be the axial Killing field normalized so that regular orbits have period 2π . Define the Komar angular momentum by

$$J_K[S] := \frac{1}{16\pi} \int_S \star dm^b. \quad (1036)$$

Comparison with Eq. (1032) immediately gives

$$\mathcal{K}_S[m] = -2J_K[S]. \quad (1037)$$

The relative factor of two is therefore not an additional horizon assumption; it is already fixed by the standard Komar normalization of angular momentum.

D.2 The affine Komar profile

For the stationary Killing family

$$\chi_\Omega = k + \Omega m \quad (1038)$$

with constant Ω , linearity gives

$$\begin{aligned} \mathcal{K}_S[\chi_\Omega] &= \mathcal{K}_S[k] + \Omega \mathcal{K}_S[m] \\ &= M_K[S] - 2\Omega J_K[S]. \end{aligned} \quad (1039)$$

Thus

$$\mathcal{K}_S(\Omega) := \mathcal{K}_S[k + \Omega m] = M_K[S] - 2\Omega J_K[S]. \quad (1040)$$

The Komar profile is affine in Ω , whereas the local response spectrum

$$\mathcal{R}_x(\Omega) = -g_x(k + \Omega m, k + \Omega m) \quad (1041)$$

is quadratic.

D.3 Vacuum closure

The Killing identity is

$$\nabla_\mu \nabla^\mu \xi_\nu = -R_{\nu\mu} \xi^\mu. \quad (1042)$$

In vacuum,

$$R_{\mu\nu} = 0, \quad (1043)$$

this implies that the Komar two-form has vanishing exterior derivative:

$$d \star d\xi^b = 0. \quad (1044)$$

Let S_1 and S_2 be homologous closed two-surfaces bounding a three-region $\mathcal{V}$ contained entirely in the vacuum domain. Then

$$\begin{aligned} \mathcal{K}_{S_2}[\xi] - \mathcal{K}_{S_1}[\xi] &= -\frac{1}{8\pi} \int_{\mathcal{V}} d \star d\xi^b \\ &= 0. \end{aligned} \quad (1045)$$

Therefore

$$\mathcal{K}_{S_2}[\xi] = \mathcal{K}_{S_1}[\xi]. \quad (1046)$$

In particular,

$$M_K, \quad J_K, \quad \mathcal{K}(\Omega) \quad (1047)$$

are independent of the homologous integration surface in stationary vacuum.

D.4 Orientation on a horizon cross-section

Let $\mathcal{B}$ be a spacelike bifurcation two-surface of a regular nonextremal Killing horizon.

Choose an orthonormal normal pair

$$u^\mu, \quad s^\mu, \quad (1048)$$

where u^μ is future timelike and s^μ is outward spacelike:

$$g(u, u) = -1, \quad g(s, s) = 1, \quad g(u, s) = 0. \quad (1049)$$

Define the normal binormal two-form

$$\epsilon^\perp := u^\flat \wedge s^\flat. \quad (1050)$$

Its normalization is

$$\epsilon_{\mu\nu}^\perp \epsilon^{\mu\nu} = -2. \quad (1051)$$

Let $\epsilon_{\mathcal{B}}$ be the positively oriented area two-form on $\mathcal{B}$. We choose the spacetime orientation so that

$$\star \epsilon^\perp = -\epsilon_{\mathcal{B}}. \quad (1052)$$

This orientation is compatible with the positive Schwarzschild Komar mass under Eq. (1035).

D.5 Derivative of the horizon generator

Let

$$\chi_H = k + \Omega_H m \quad (1053)$$

be the Killing field generating the horizon.

On the bifurcation surface,

$$\chi_H = 0. \quad (1054)$$

Its derivative is purely normal and satisfies

$$\nabla_\mu \chi_\nu^H = \kappa_H \epsilon_{\mu\nu}^\perp \quad (1055)$$

with the orientation of $\epsilon^\perp$ chosen consistently with $\kappa_H > 0$ for an ordinary future outer horizon.

Because χ_H is Killing,

$$\nabla_\mu \chi_\nu^H = -\nabla_\nu \chi_\mu^H. \quad (1056)$$

Therefore

$$\begin{aligned} (d\chi_H^\flat)_{\mu\nu} &= 2\nabla_{[\mu} \chi_{\nu]}^H \\ &= 2\nabla_\mu \chi_\nu^H. \end{aligned} \quad (1057)$$

Hence on $\mathcal{B}$,

$$d\chi_H^\flat = 2\kappa_H \epsilon^\perp. \quad (1058)$$

Taking the Hodge dual and using Eq. (1052),

$$\star d\chi_H^\flat = -2\kappa_H \epsilon_{\mathcal{B}}. \quad (1059)$$

D.6 Horizon Komar integral

Substitution into the Komar functional gives

$$\begin{aligned}\mathcal{K}_{\mathcal{B}}[\chi_H] &= -\frac{1}{8\pi} \int_{\mathcal{B}} \star d\chi_H^\flat \\ &= -\frac{1}{8\pi} \int_{\mathcal{B}} (-2\kappa_H \epsilon_{\mathcal{B}}) \\ &= \frac{\kappa_H}{4\pi} \int_{\mathcal{B}} \epsilon_{\mathcal{B}}.\end{aligned}\tag{1060}$$

Since

$$A_H := \int_{\mathcal{B}} \epsilon_{\mathcal{B}},\tag{1061}$$

we obtain

$$\mathcal{K}_{\mathcal{B}}[\chi_H] = \frac{\kappa_H A_H}{4\pi}.\tag{1062}$$

This explicitly fixes the factor $1/(4\pi)$.

The factor arises from three ingredients:

$$\underbrace{\frac{1}{8\pi}}_{\text{Komar normalization}} \times \underbrace{2}_{d\chi=2\nabla\chi} \times \underbrace{\kappa_H A_H}_{\text{horizon geometry}}.\tag{1063}$$

D.7 Transport to spatial infinity

Assume that the exterior region between $\mathcal{B}$ and spatial infinity is vacuum.

Since

$$d \star d\chi_H^\flat = 0,\tag{1064}$$

surface independence gives

$$\mathcal{K}_{S_\infty}[\chi_H] = \mathcal{K}_{\mathcal{B}}[\chi_H].\tag{1065}$$

At infinity,

$$\begin{aligned}\mathcal{K}_{S_\infty}[\chi_H] &= \mathcal{K}[k] + \Omega_H \mathcal{K}[m] \\ &= M_K - 2\Omega_H J_K.\end{aligned}\tag{1066}$$

At the bifurcation surface,

$$\mathcal{K}_{\mathcal{B}}[\chi_H] = \frac{\kappa_H A_H}{4\pi}.\tag{1067}$$

Therefore

$$M_K - 2\Omega_H J_K = \frac{\kappa_H A_H}{4\pi}.\tag{1068}$$

Equivalently,

$$M_K = 2\Omega_H J_K + \frac{\kappa_H A_H}{4\pi}.\tag{1069}$$

This is the vacuum Smarr relation [?].

D.8 How the response spectrum enters

The Komar derivation above does not require MS geometry. The response construction enters by selecting the horizon Killing combination.

At a regular non-axis horizon,

$$\mathcal{R}_H(\Omega_H) = 0, \quad \partial_\Omega \mathcal{R}_H(\Omega_H) = 0. \quad (1070)$$

This determines

$$\chi_H = k + \Omega_H m. \quad (1071)$$

The Komar functional is then evaluated on precisely that generator:

$$\mathcal{K}[\chi_H] = M_K - 2\Omega_H J_K. \quad (1072)$$

Thus the logical relation is

$$\mathcal{R}_H \xrightarrow{\text{double root}} \Omega_H \longrightarrow \chi_H \xrightarrow{\text{Komar}} M - 2\Omega_H J. \quad (1073)$$

The equality of this quantity with $\kappa_H A_H / (4\pi)$ is supplied by standard horizon geometry and vacuum Komar transport.

D.9 Surface gravity from the response gap

Holding Ω_H fixed, define

$$\mathcal{G}_H := -\chi_H^2 = \mathcal{R}(\Omega_H). \quad (1074)$$

On a Killing horizon,

$$\nabla_\mu (\chi_H^2) = -2\kappa_H \chi_\mu^H. \quad (1075)$$

Therefore

$$\nabla_\mu \mathcal{G}_H = 2\kappa_H \chi_\mu^H. \quad (1076)$$

Let n^μ be an auxiliary null transverse vector satisfying

$$g(\chi_H, n) = -1. \quad (1077)$$

Then

$$\kappa_H = -\frac{1}{2} n^\mu \nabla_\mu \mathcal{G}_H \Big|_{\mathcal{H}}. \quad (1078)$$

Thus the two response derivatives entering the horizon problem are

$$\begin{aligned} \partial_\Omega \mathcal{R}_H &\longrightarrow \Omega_H, \\ n^\mu \nabla_\mu \mathcal{R}(\Omega_H) &\longrightarrow \kappa_H. \end{aligned} \quad (1079)$$

The first acts in Killing-parameter space, while the second acts transversely in spacetime.

D.10 Schwarzschild normalization check

For Schwarzschild,

$$k = \partial_t \quad (1080)$$

and

$$g_{tt} = -1 + \frac{2M}{r} + O(r^{-2}). \quad (1081)$$

Thus

$$k^b = \left(-1 + \frac{2M}{r} + O(r^{-2}) \right) dt. \quad (1082)$$

Hence

$$dk^b = \frac{2M}{r^2} dt \wedge dr + O(r^{-3}). \quad (1083)$$

With the orientation used here,

$$\star(dt \wedge dr) = -r^2 \sin \theta d\theta \wedge d\phi + O(r). \quad (1084)$$

Therefore

$$\begin{aligned} \int_{S_\infty} \star dk^b &= -2M \int_{S^2} \sin \theta d\theta d\phi \\ &= -8\pi M. \end{aligned} \quad (1085)$$

Hence

$$M_K = -\frac{1}{8\pi} \int_{S_\infty} \star dk^b = M. \quad (1086)$$

This verifies the mass sign convention directly.

D.11 Kerr horizon evaluation

For Kerr,

$$M_K = M, \quad J_K = J = Ma. \quad (1087)$$

The horizon angular velocity is

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad (1088)$$

with

$$r_\pm = M \pm \sqrt{M^2 - a^2}. \quad (1089)$$

The horizon-generator Komar charge is

$$\begin{aligned} \mathcal{K}[\chi_H] &= M - 2\Omega_H J \\ &= M - \frac{2Ma^2}{r_+^2 + a^2}. \end{aligned} \quad (1090)$$

Since

$$r_+^2 + a^2 = 2Mr_+, \quad (1091)$$

$$\begin{aligned}
\mathcal{K}[\chi_H] &= M - \frac{a^2}{r_+} \\
&= M - r_- \\
&= \frac{r_+ - r_-}{2}.
\end{aligned} \tag{1092}$$

Therefore

$$\mathcal{K}[\chi_H] = \sqrt{M^2 - a^2}. \tag{1093}$$

The surface gravity is

$$\kappa_H = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \tag{1094}$$

and the area is

$$A_H = 4\pi(r_+^2 + a^2). \tag{1095}$$

Hence

$$\begin{aligned}
\frac{\kappa_H A_H}{4\pi} &= \frac{r_+ - r_-}{2} \\
&= \sqrt{M^2 - a^2}.
\end{aligned} \tag{1096}$$

Finally, Appendix B gave

$$L_H^{\text{Weyl}} = 2\sqrt{M^2 - a^2}. \tag{1097}$$

Thus

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2}L_H^{\text{Weyl}}. \tag{1098}$$

D.12 Nonextremal versus extremal horizons

The bifurcation-surface derivation above assumes a nonextremal horizon.

For

$$\kappa_H \neq 0, \tag{1099}$$

the maximally extended stationary black-hole geometry contains a bifurcation surface on which

$$\chi_H = 0. \tag{1100}$$

The derivation of Eq. (1062) then applies directly.

An extremal horizon has

$$\kappa_H = 0 \tag{1101}$$

and has no bifurcation two-surface of the same type.

Therefore one should not justify the extremal formula by inserting an extremal horizon directly into the bifurcation-surface proof.

For Kerr, the extremal result follows unambiguously by taking the continuous limit of the subextremal family:

$$|a| \rightarrow M. \tag{1102}$$

Then

$$r_+ - r_- \rightarrow 0, \tag{1103}$$

so

$$\kappa_H \rightarrow 0, \quad (1104)$$

$$M - 2\Omega_H J \rightarrow 0, \quad (1105)$$

and

$$L_H^{\text{Weyl}} \rightarrow 0. \quad (1106)$$

The area remains finite:

$$A_H \rightarrow 8\pi M^2 \quad (1107)$$

for extremal Kerr.

Thus the vanishing quantity is the common nonextremality scale, not the horizon itself.

D.13 Relation to ADM and MS asymptotic mass

The Komar and Hamiltonian asymptotic constructions are logically distinct.

The Komar mass requires an exact stationary Killing field and vacuum closedness:

$$\text{Killing symmetry} + R_{\mu\nu} = 0 \longrightarrow M_K. \quad (1108)$$

The preceding MS asymptotic construction instead used differentiability of the canonical generator:

$$\text{bulk constraint generator} + \text{boundary completion} \longrightarrow P_\infty^{\text{MS}}. \quad (1109)$$

For a stationary asymptotically flat vacuum spacetime, standard general relativity gives

$$M_K = M_{\text{ADM}}. \quad (1110)$$

After the Einstein normalization used in the preceding MS paper,

$$M_{\text{ADM}} = M_\infty^{\text{MS}}. \quad (1111)$$

Hence in the common stationary rest-frame sector,

$$M_K = M_{\text{ADM}} = M_\infty^{\text{MS}}. \quad (1112)$$

This equality does not identify the underlying constructions.

Moreover, the preceding MS boundary paper did not construct an asymptotic rotation generator.

Therefore the present use of

$$J_K \quad (1113)$$

does not constitute a derivation of an MS rotational boundary charge.

D.14 Restoring physical units

The formulas above use geometrized quantities.

Let

$$M_{\text{geom}} = \frac{GM_{\text{phys}}}{c^2}, \quad (1114)$$

$$J_{\text{geom}} = \frac{GJ_{\text{phys}}}{c^3}, \quad (1115)$$

and

$$\Omega_{\text{geom}} = \frac{\Omega_{\text{phys}}}{c}. \quad (1116)$$

If κ_{phys} denotes surface gravity with acceleration dimension, then

$$\kappa_{\text{geom}} = \frac{\kappa_{\text{phys}}}{c^2}. \quad (1117)$$

Multiplying the geometrized Smarr relation by

$$\frac{c^4}{G} \quad (1118)$$

gives the physical-energy form

$$M_{\text{phys}}c^2 = 2\Omega_H J_{\text{phys}} + \frac{\kappa_{\text{phys}}c^2 A_H}{4\pi G}. \quad (1119)$$

Thus

$$M_{\text{phys}}c^2 - 2\Omega_H J_{\text{phys}} = \frac{\kappa_{\text{phys}}c^2 A_H}{4\pi G}. \quad (1120)$$

For Kerr, if L_H^{Weyl} is expressed as a physical length, then

$$M_{\text{phys}}c^2 - 2\Omega_H J_{\text{phys}} = \frac{c^4}{2G} L_H^{\text{Weyl}}. \quad (1121)$$

This is the physical-unit version of

$$M - 2\Omega_H J = \frac{1}{2} L_H^{\text{Weyl}} \quad (1122)$$

in geometrized units.

D.15 Final normalization summary

The convention chain used in the paper is

$$\begin{aligned} M_K &= -\frac{1}{8\pi} \int \star dk^b, \\ J_K &= \frac{1}{16\pi} \int \star dm^b, \\ \mathcal{K}[k + \Omega m] &= M_K - 2\Omega J_K, \\ \mathcal{K}[\chi_H] &= \frac{\kappa_H A_H}{4\pi}, \\ M_K &= 2\Omega_H J_K + \frac{\kappa_H A_H}{4\pi}. \end{aligned} \quad (1123)$$

For Kerr this becomes

$$M - 2\Omega_H J = \frac{\kappa_H A_H}{4\pi} = \sqrt{M^2 - a^2} = \frac{1}{2} L_H^{\text{Weyl}}. \quad (1124)$$

The response-specific input is the selection of Ω_H and hence of χ_H from the local stationary response double root. The Komar normalization, vacuum transport, horizon evaluation, and Smarr identity are standard general-relativistic structures.